UNIVERSIDADE D
COIMBRA

Hugo Jorge da Cunha Fernandes

Oscillatory Sign-Switching Dark Energy

Dissertação no âmbito do Mestrado em Física, ramo de Física Nuclear e de Partículas, orientada pela Doutora Mariam Bouhmadi López e Doutor Professor João Pedro Trancoso Gomes Rosa e apresentada ao Departamento de Física da Faculdade de Ciências e Tecnologia da Universidade de Coimbra.

Setembro de 2026

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Hugo Jorge da Cunha Fernandes

Supervised by Prof. Dr. Mariam Bouhmadi López and Prof. Dr. João Pedro Trancoso Gomes Rosa.

University of Coimbra

Dissertation submitted in partial fulfillment for the degree of

Master of Science in Physics

September 2026

Abstract

The ΛCDM model, combined with inflation and formulated within the framework of general relativity, remains the simplest model that fits remarkably well the majority of cosmological observations.However, the increasing precision of cosmological measurements has revealed persistent discrepancies between early- and late-Universe determinations of the Hubble constant H_0, with the tension between the Planck and SH0ES collaborations at the approximately 5σ level .

Among the proposed late-time extensions of ΛCDM, sign-switching dark energy (DE) models, in which the cosmological constant evolves from negative to positive values at low redshift, have attracted growing attention. In this thesis, we introduce a new class of smooth, oscillatory sign-switching DE models, organized into two families: $\Lambda_{\rm osc}$CDM and $\Lambda_{\rm OI}$CDM. Both feature a continuous transition from a negative to a positive DE density, but differ in where the oscillatory behavior appears relative to the transition. We constrain the model parameters using Markov Chain Monte Carlo (MCMC) methods, combining different cosmological datasets, including the recent Baryon Acoustic Oscillation (BAO) measurements from DESI DR2. The statistical comparison with ΛCDM is carried out using the AIC, $\rm AIC_c$, and BIC information criteria. We implement as well a cosmographic analysis,i.e. a kinematics-based approach that assumes only the FLRW metric and study the behavior the models can exhibit through the deceleration parameter and higher-order quantities. We also carry out a linear perturbation analysis of matter and radiation on the background expansion of the sign-switching models, computing the evolution of the matter density contrast, the gravitational potential, the matter power spectrum, and the growth rate. We compare these results with the corresponding ΛCDM predictions, while $f\sigma_8$ is also compared with available observational data.

Resumo

O modelo ΛCDM, combinado com inflação e formulado no contexto da relatividade geral, continua a ser o modelo mais simples a ajustar-se notavelmente bem à maioria das observações cosmológicas. No entanto, a crescente precisão das medições cosmológicas revelou discrepâncias persistentes entre as determinações da constante de Hubble H_0 obtidas a partir de sondas do Universo primordial e do Universo tardio, com a tensão entre as colaborações Planck e SH0ES atualmente ao nível de aproximadamente 5σ.

Entre as extensões tardias propostas ao ΛCDM, os modelos de Energia Escura (DE) com mudança de sinal, nos quais a constante cosmológica evolui de valores negativos para positivos a baixo desvio para o vermelho, têm atraído uma atenção crescente. Nesta tese, introduzimos uma nova classe de modelos suaves e oscilatórios de DE com mudança de sinal, organizados em duas famílias: $\Lambda_{\rm osc}$CDM e $\Lambda_{\rm OI}$CDM. Ambas apresentam uma transição contínua de uma densidade de DE negativa para positiva, mas diferem na forma como o comportamento oscilatório surge relativamente à transição.

Restringimos os parâmetros dos modelos utilizando métodos de Monte Carlo com Cadeias de Markov (MCMC), combinando diferentes conjuntos de dados cosmológicos, incluindo as recentes medições de Oscilações Acústicas de Bariões (BAO) da segunda divulgação de dados do DESI. A comparação estatística com o ΛCDM é realizada através dos critérios de informação AIC, $\rm AIC_c$ e BIC. Implementamos também uma análise cosmográfica, isto é, uma abordagem cinemática que assume apenas a métrica FLRW, para estudar o comportamento que os modelos podem exibir através do parâmetro de desaceleração e de quantidades de ordem superior. Realizamos também uma análise de perturbações lineares da matéria e da radiação sobre a expansão de fundo dos modelos com mudança de sinal, calculando a evolução do contraste de densidade da matéria, do potencial gravitacional, do espectro de potência da matéria e da taxa de crescimento. Comparamos estes resultados com as previsões correspondentes do ΛCDM, enquanto $f\sigma_8$ é também comparado com os dados observacionais disponíveis.

Acknowledgements

Como já estudo física há alguns anos, é sempre complicado agradecer a todas as pessoas que, direta ou indiretamente, me ajudaram a tornar esta jornada, por vezes complicada, muito mais calma e divertida. Aqui vai a minha melhor tentativa.

Primeiro, gostava de expressar a minha gratidão aos meus orientadores, Professor Doutor João Pedro Trancoso Gomes Rosa e Professora Doutora Mariam Bouhmadi López. O Professor João teve uma grande influência na minha decisão de seguir a área da cosmologia, desde as incríveis aulas de T.F.T.G ou de R.G. e Cosmologia até ao facto de ter estabelecido uma ponte entre mim e a Professora Mariam. Dizer obrigado é o mínimo! Também gostava de agradecer profundamente à Professora Mariam, que me acompanhou durante este ano com carinho e atenção. Não importava se as minhas perguntas eram "tontas" ou se eu demorava mais tempo a pensar sobre uma tarefa, a professora tentou sempre manter-me num caminho em que aprender e ser produtivo andavam de mãos dadas.

In the same spirit, I would like to thank the rest of the research group: Hsu-Wen, Carlos and Javier, for always being available to help me whenever I was struggling to understand how Cobaya worked or simply had a question. A special thank you also goes to Beñat for all his help and friendship, and for the few months during which we shared an office, where he always brought a welcome sense of calm.

Fora da Universidade, morar em Bilbao trouxe-me surpresas magníficas, principalmente pelas amizades que fiz pelo caminho. A special thank you goes to my group of friends from the first semester, *En el festin*, especially to Valerio, Marghe and Bea, my favourite Italians. Como bom português que sou, no segundo semestre tive a grande sorte de conhecer 3 seres humanos espetaculares, que me ensinaram, entre muitas coisas, que começar a levar a vida com leveza ontem já é tarde. Muito obrigado à Mafaldinha, à Clarinha e, especialmente, ao João, por estarem sempre lá para mim com um pintxo, uma caña e o jogo dos gatinhos. Já tenho saudades das nossas noites passadas em Casco Viejo.

Voltando às origens, a minha jornada em Coimbra permitiu-me conhecer diversas pessoas incríveis, desde professores a amigos. Gostava de agradecer a todos os professores que me deram aulas, especialmente aos Professores João Moreira e Fernando Nogueira, que sempre deram excelentes aulas com uma paixão contagiante por ensinar. Também gostava de agradecer aos meus amigos seniores, que sempre serviram de exemplo e me deram lições valiosas pelo caminho. Obrigado, Faria, Carvalho, Valéria

e Brito, por terem sempre as palavras certas nos momentos certos. Ao melhor que Coimbra me deu, sou eternamente grato por vos ter conhecido, Alex, Maria e maninha Ana. Todas as saídas, gargalhadas e fofoquices desnecessárias que tivemos foram incríveis e sinceramente repetia tudo. O vosso coração é gigante e tenho muito orgulho de poder chamar-vos de amigos. Como a nossa geração diz, um grande shoutout também aos meus parceiros das semanas de exames: à Lilics, ao Faus, ao Dudu, aos meus dois Henriques, ao Manel e ao Cartaxo. Não havia nada como chegar aos últimos dias dos exames e, durante as nossas pausas dinâmicas, ouvir um de vocês soltar uma piada sem sentido nenhum. Como nós dizíamos, antes rir do que chorar. Por último, um obrigado especial também à Carol, à Hermanita, ao PS, à Maria e à Mar, pela amizade que construímos para além destas equações e teorias complicadas do universo.

Aos meus amigos de Felgueiras, que gostam de me chamar emigrante, obrigado por ajudarem a manter a nossa amizade viva, mesmo depois de tantos anos fora. Podíamos passar meses sem nos ver, mas bastava um minuto juntos para voltarmos a falar como se eu nunca tivesse ido embora. Obrigado, Mariana, Files, Guigo e, especialmente, Carlitos. Vocês ainda são dos poucos motivos pelos quais volto a essa terra com um genuíno sorriso na cara.

Antes de falar da família, quero deixar um agradecimento especial ao professor do secundário que me influenciou a seguir física: muito obrigado, Professor Valdemar, pela exemplar capacidade de ensinar matemática e por sempre acreditar no poder transformador do ensino.

Finalmente, gostava de agradecer à minha família, especialmente à minha mãe, ao meu pai, ao Ricardo, ao meu mano Afonso e à Bela. Mesmo sem entenderem totalmente porque eu fazia turnos à noite para estudar ou porque quis seguir este caminho, vocês sempre me encorajaram e esperaram o melhor de mim. Não há nada mais reconfortante do que saber que tenho sempre um lugar seguro e feliz ao qual voltar caso o meu plano Z não funcione. Por último, escrevo para ti, avó Mila, a quem dedico esta tese. És o ser humano mais resiliente que conheço, e estarás para sempre no meu coração. Do teu neto,

Huguinho.

"Se durante o processo de construção
O autor vos remeter para uma estrutura quase demente
É porque foi usado mais por demais coração
Condição absoluta no limiar do obstinadamente
Esculpindo frases ao sabor da inspiração
Revestidas por minimal batida envolvente
Construindo fortalezas muralhadas de emoção
Provocando aqui e ali a moralidade digente
E se, de repente, o sonho ganha asas de vulcão
Derramando palavras e punhais principalmente
Requerendo seguramente mais que uma audição
O que é, por vezes, não existe, mas sim aquilo que se pressente
Após longas e repetidas noites de agitação
O resultado surge... emergente
Esvai-se a permanente inquietação...
Entre(tanto), Sobre(tudo),
Pratica(mente)"

Napoleão Mira

x

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List of Abbreviations

ΛCDM . . . . Lambda Cold Dark Matter
AIC . . . . . Akaike Information Criterion
AdS . . . . . Anti-de Sitter
BAO . . . . . Baryon Acoustic Oscillation
BIC . . . . . Bayesian Information Criterion
BBN . . . . . Big Bang Nucleosynthesis
CC . . . . . Cosmic Chronometers
CMB . . . . . Cosmic Microwave Background
DE . . . . . Dark Energy
DESI . . . . . Dark Energy Spectroscopic Instrument
DM . . . . . Dark Matter
DR . . . . . Data Release
dS . . . . . de Sitter
EoS . . . . . Equation of State
FLRW . . . . . Friedman-Lemaître-Robertson-Walker
GR . . . . . General Relativity
MCMC . . . . Markov Chain Monte Carlo
PDF . . . . . Probability Distribution Function
SH0ES . . . . Supernovae H0 for the Equation of State of Dark Energy
SNe Ia . . . . . Supernovae of Type Ia

I get ideas in fragments... It's as if in the other room there's a puzzle, and all the pieces are together.. But in my room , they just flip one piece at a time into me. And the first piece I get is just a fragment of the whole puzzle... and it holds a promise for more, and I keep it, I write it down. And then I say that having the fragment is more bait on the hook ... the more it come in, the more faster the rest come in.

— David Lynch

1 Introduction

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1.1 Historical Review

This is the only personal remark I will make in this thesis. To me, Lynch's quote captures, in an intuitive way, what studying cosmology feels like. What are the pieces of our puzzle? They might be an additional parameter that explains a property of some astrophysical object, a different theory of gravity, or a new method to infer cosmological parameters. Our goal as cosmologists, I believe, is to discover new pieces and connect them with those already in place — testing our hypotheses and checking their consistency with what is already known.

Metaphors aside, we can say that modern cosmology is defined as the field of science that studies the origin, evolution, structure and history of the entire Universe. This current notion was built thanks to the efforts of thousands of scientists who, over the millennia dedicated their lives to studying the cosmos.

The foundation of modern cosmology can be traced back to the early 20th century. Inspired by the work of physicists such as James Maxwell [1] , Einstein created his theory of Special Relativity [2] following two principles:

- The speed of light in vacuum is the same for all inertial observers, independently of the motion of the source or the observer.
- The laws of physics take the same form in all inertial frames of reference.

This gave birth to his famous theory of General Relativity (GR), in which Einstein presented a framework relating gravity to the geometry of spacetime. Two years later he published an article entitled Cosmological Considerations in the General Theory of Gravity [3], in which he applied the theoretical framework of GR to the Universe as a whole .

Using Einstein's work as a basis, the mathematicians Willem de Sitter [4–6] and Alexander Friedmann [7, 8] and the physicists Georges Lemaitre [9] and Karl Schwarzschild [10, 11] contributed by developing cosmological models. By finding specific solutions to Einstein field equations, Friedmann showed that an expanding Universe was a possible solution. On the observational side, before Einstein's publication, Henrietta Leavitt [12] noticed the relationship between the period and luminosity of astrophysical objects known as Cepheids which became the first known standard candles.

Using Leavitt's work, Hubble produced his redshift–distance relation in 1929 [13], showing that more distant galaxies were generally receding at higher velocities. This result provided strong evidence that the Universe is expanding and played an important role in establishing the expanding-Universe picture within the scientific community.

The work of Howard P. Robertson [14–16] and Arthur G. Walker [17] established the most general form of a spacetime compatible with homogeneity and isotropy. Together with the dynamical solutions developed by Friedmann and Lemaitre, this led to the formulation of the Friedmann-Lemaître-Robertson-Walker (FLRW) metric.

During the same period, evidence also began to emerge for a form of matter that neither emits nor absorbs electromagnetic radiation. Measurements of galactic rotation curves [18] later lent further credibility to what we now call dark matter (DM).

In parallel, Hot Big Bang model was developed, predicting a hot and dense early Universe. Later, the discovery in 1965 [19] of a thermal radiation background, now known as the Cosmic Microwave Background (CMB), provided strong observational support for this model.

Then, in 1998, an unexpected result was discovered by two separate teams: the Supernova Cosmology Project [20] and the High-z Supernova Search Team [21], the Universe was expanding not only, but its expansion had also been accelerating at recent cosmological times. The simplest explanation within GR is the cosmological constant, Λ.

We arrive at the current standard model of cosmology, also known as the ΛCDM model. Adopting GR as the theory of gravity and the FLRW metric, it describes, on large scales, a Universe with the following constituents:

- Matter from the Standard Model of Particle Physics. Its contribution must be included to accurately describe the expansion history of the Universe.
- Cold Dark Matter (CDM): a type of matter needed to accurately describe the dynamics of formation of large-scale structure, such as galaxy clusters.
- Λ: a constituent of the Universe that contributes to its accelerated expansion.

Our cosmological system evolves dynamically and, therefore, we must specify suitable initial conditions. By introducing an early period of accelerated cosmic

expansion, the ΛCDM model incorporates inflation as a mechanism that helps explain these initial conditions.

This framework is remarkably simple, with only six main cosmological parameters, yet it provides an excellent description of a wide range of cosmological data, including observations of the CMB, large-scale structure, and other astrophysical probes.

Despite these successes, the model faces theoretical shortcomings such as the cosmological constant problem [22] and the cosmic coincidence problem [23]. In addition to these theoretical issues, we now live in an era of major advances in both the quantity and quality of observations. As the number of surveys and observational techniques increases, different probes can reveal apparently irreconcilable predictions, usually referred to as discrepancies or tensions.

Among these, the tension that has received the most attention is related to the Hubble constant, H_0. The most commonly cited discrepancy is the 5.0σ disagreement between the value inferred from the Planck collaboration [24], $H_0^{\rm Planck} = (67.27 \pm 0.60)$ km/s/Mpc at 68% confidence level (CL) assuming ΛCDM and the local distance-ladder measurement from the Supernovae H0 for the Equation of State of Dark Energy (SH0ES) collaboration [25] $H_0^{\rm SH0ES} = (73.04 \pm 1.04)$ km/s/Mpc at 68% CL

For a comprehensive review of the different cosmological discrepancies and proposed solutions, see Ref. [26].

1.2 Motivation

Since the H_0 tension persists across different measurements and cosmological probes, there is growing interest in the possibility of new physics beyond the standard cosmological model. For a comprehensive review of the current proposals, see Ref. [26].

Among these proposals, one of the most widely explored avenues consists modifying the standard dark sector by considering models beyond cold dark matter and the cosmological constant Λ. These proposals can broadly be divided into two classes: models that modify the physics before recombination, usually referred to as early-time solutions, and models that modify the physics after recombination, known as late-time solutions. Within the latter class, increasing attention has recently been devoted to scenarios in which the dark energy (DE) density changes sign during the late-time evolution of the Universe. Such an evolution can be interpreted phenomenologically as a transition from an anti-de Sitter (AdS)-like negative-energy phase to a positive-energy de Sitter (dS)-like phase characterizing the present accelerated expansion.

Inspired by the graduated DE model [27], the $\Lambda_{\rm s}$CDM model was introduced in Ref. [28]. With only one additional parameter relative to ΛCDM , it allows the effective cosmological constant to take negative values in the past before transitioning to a positive value at low redshift.

At the observational level, $\Lambda_{\rm s}$CDM has been shown to alleviate several cosmological discrepancies for particular combinations of data [28].

In its minimal formulation, the sign change is implemented through an abrupt transition. This leads to a Type-II (sudden) singularity [29] at the transition point, where the scale factor and the total energy density remain finite, while the total pressure diverges. This has motivated the study of smooth sign-switching extensions,

such as those proposed in Refs. [30–34].

Additional motivation comes from recent baryon acoustic oscillation (BAO) measurements. When data from the Dark Energy Spectroscopic Instrument (DESI) [35] are combined with other cosmological probes, some analyses report a mild preference for a dynamical dark energy (DDE) equation of state that evolves towards, or even crosses, the cosmological-constant value $w = -1$ at late times. Although these indications are not yet statistically conclusive and remain compatible with ΛCDM within current uncertainties, they further motivate the exploration of late-time extensions in which the DE sector is dynamical.

In this spirit, we present a new class of smooth sign-switching DE models which, during the late-time evolution of the Universe, can also exhibit oscillatory behavior. We divide them into two families, the $\Lambda_{\rm osc}$CDM and $\Lambda_{\rm OI}$CDM models, whose phenomenological properties and observational constraints are discussed in Chapter 5.

1.3 Conventions and Units

Conventions:

- Einstein's summation convention is always assumed, unless stated otherwise.
- Greek indices, μ, ν, ... run from 0 to 3, while Latin indices, i, j, ... stand for spatial indices.
- The Minkowski spacetime metric is written as $\eta_{\mu\nu}$, and non-Minkowski metrics are written as $g_{\mu\nu}$. We use the signature $(-1, +1, +1, +1)$.
- General model parameters will be denoted θ while Λ will refer to the cosmological parameters from ΛCDM model and Ξ to additional parameters in extensions of the standard models of cosmology.

Units:

- Unless stated otherwise, throughout this thesis we adopt natural units such that $c = \hbar = k_B = 1$. Consequently, energy, mass and temperature have units of inverse of length and reverse time:

$$[\text{Energy}] = [\text{Mass}] = [\text{Temperature}] = [\text{Length}]^{-1} = [\text{Time}]^{-1} \tag{1.1}$$

Esse é o verdadeiro Poder, o Poder da Luz
Muita gente pensa que dinheiro é poder.
Mas pessoas que verdadeiramente conseguem influenciar
E guiar outras pessoas são os génios, os filantropos, os criadores de Luz

— Valete

2

Brief Introduction to Gravity and Cosmology

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2.1 General Relativity

This section provides a brief review of the theory of gravity used throughout this thesis. For more extensive discussions of general relativity, we refer the reader to Refs [36–38].

A fundamental quantity in the theory is the metric g, which allows us to calculate the space-time distance between two neighboring points, corresponding to the infinitesimal line element:

$$\mathrm{ds}^2 = g_{\mu\nu} dx^\mu dx^\nu. \tag{2.1}$$

The metric is a symmetric 4×4 tensor and is therefore specified by 10 components. In flat Minkowski spacetime, it reduces to diag $(-1, +1, +1, +1)$.

Pairs of points with $ds^2 < 0$ are said to be timelike separated while pair with $ds^2 > 0$ are spacelike separated. When $ds^2 = 0$ we say that they are lightlike separated, or null.

Now, one might ask: what is the shortest path between two points? A natural answer would be a straight line. In Euclidean space, this is correct, but in GR we are dealing with curved spacetime. The corresponding generalization of a straight line is a geodesic, obtained by maximizing the spacetime interval. For timelike trajectories, this is equivalent to extremizing the proper time.

For a given path $x^\mu(\sigma)$ between point A and B, the proper time is defined as:

$$\Delta\tau = \int_{\sigma_A}^{\sigma_B} d\sigma \sqrt{-g_{\mu\nu}(x)\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}} \tag{2.2}$$

Massive freely falling particles moving in a curved spacetime follow a geodesic trajectory, which are described by:

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0. \tag{2.3}$$

Physical laws must be independent of the coordinate system used to describe them. Tensors are geometrical objects with well-defined transformation rules between different coordinate systems. The quantity $\Gamma^\alpha_{\mu\nu}$ is known as the Christoffel symbol and is defined as:

$$\Gamma^\alpha_{\mu\nu} \equiv \frac{1}{2}g^{\alpha\lambda}\left[\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}\right], \tag{2.4}$$

where $g^{\mu\nu}$ denotes the inverse metric, satisfying $g^{\mu\sigma}g_{\sigma\nu} = \delta^\mu{}_\nu$.

While tensors have well-defined transformation properties between coordinate systems, the ordinary partial derivative of a tensor does not, in general, transform as a tensor. This motivates the introduction of a new derivative operator with the appropriate tensorial transformation properties [36]:

$$\nabla_\nu v^\mu = \partial_\nu v^\mu + \Gamma^\mu_{\nu\rho}v^\rho \tag{2.5}$$

For simplicity, it is also common to denote partial derivatives by a comma($v^\mu{}_{,\nu}$), instead of $\partial_\nu v^\mu$, and covariant derivatives by a semicolon ($v^\mu{}_{;\nu}$), instead of $\nabla_\nu v^\mu$. Einstein's theory connects gravity with the geometry of the spacetime, whose curvature is described by the Riemann tensor:

$$R^\alpha_{\beta\mu\nu} \equiv \partial_\mu\Gamma^\alpha_{\nu\beta} - \partial_\nu\Gamma^\alpha_{\mu\beta} + \Gamma^\sigma_{\beta\nu}\Gamma^\alpha_{\mu\sigma} - \Gamma^\sigma_{\beta\mu}\Gamma^\alpha_{\nu\sigma}. \tag{2.6}$$

Another tensor that can be defined is the Ricci tensor. It is obtained by contracting the upper index of the Riemann tensor with its second lower spacetime index:

$$R^\alpha_{\mu\alpha\nu} = R_{\mu\nu} = \partial_\alpha\Gamma^\alpha_{\mu\nu} - \partial_\nu\Gamma^\alpha_{\mu\alpha} + \Gamma^\alpha_{\beta\alpha}\Gamma^\beta_{\mu\nu} - \Gamma^\alpha_{\beta\nu}\Gamma^\beta_{\mu\alpha}. \tag{2.7}$$

Its trace, known as the Ricci scalar, $R \equiv g^{\mu\nu}R_{\mu\nu}$, is a scalar quantity that characterizes the curvature of spacetime and is invariant under coordinate transformations.

Now we want to establish a connection between the tensors previously defined and their dynamics. We can achieve that with the action:

$$S = \int d^4x\sqrt{-g}\left(\frac{R}{16\pi G} + \mathcal{L}_{\rm M}\right), \tag{2.8}$$

where the full action of the theory is given by the sum of Einstein-Hilbert term (the first one) and $\mathcal{L}_{\rm M}$, the Lagrangian of the matter field. G is Newton's gravitational constant.

Applying the principle of least action and the action in Eq. (2.8) with respect to the metric $g_{\mu\nu}$, we obtain the Einstein field equations:

$$\delta S = 0 \quad \longrightarrow G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G\, T_{\mu\nu}. \tag{2.9}$$

We finally arrive at the equations of motion of GR, known as Einstein's field equations. In the words of physicist John Wheeler:

"Spacetime tells matter how to move; matter tells spacetime how to curve"

In Eq. (2.9), the left-hand-side (LHS) encodes the geometrical properties of spacetime, while the right-hand-side (RHS) contains the matter contribution through the energy-momentum tensor $T_{\mu\nu}$.

2.2 Background Cosmology

2.2.1 The Universe is expanding

At the beginning of the 20th century, astronomers were still uncertain about the size of the Milky Way and the extent of the observable Universe. Reliable distance measurements were limited, making it difficult to determine whether the spiral nebulae observed in the sky were nearby objects within the Milky Way or distant systems lying far beyond it.

During the same period, an important historical discussion took place between Shapley and Curtis, known as the "Great Debate", which focused on two competing views regarding the size and structure of the Universe. Shapley argued that the nebulae observed were relatively small and lay within the outskirts of the Milky Way Galaxy, while Curtis argued that they were in fact independent galaxies, implying that they were large and distant.

There was no immediate agreement on which view was correct. A year later, both scientists published independent papers under the title “The Scale of the Universe”, in which they defended their own interpretations and presented counterarguments aimed at challenging the competing view.

An important step towards resolving this debate came with Hubble's famous redshift–distance plot, shown in Fig. 2.1, which revealed that more distant galaxies were, on average, receding at higher velocities.

The vertical axis, representing the recession velocity of the galaxies, was made possible largely through the work of Vesto Slipher [39]. By measuring the redshift of spectral lines in the light emitted by galaxies, Slipher was able to infer their recession velocities through the Doppler effect. His observations revealed that many of these objects exhibited substantial redshifts, indicating that they were receding from us.

The horizontal axis, representing the distance to the galaxies, required an independent method for determining astronomical distances. Henrietta Leavitt [12] established the relation between the pulsation period and intrinsic luminosity of Cepheid variable stars. Once appropriately calibrated, this relation allowed distances to be inferred from their observed brightness. Hubble applied this method to Cepheids observed in nearby galaxies, allowing him to estimate their distances.

Combining these distance estimates with measurements of recession velocities led to the relation known as the Hubble–Lemaître law:

$$v = H_0 D, \tag{2.10}$$

where v is the recessional velocity of the galaxy, measured in km/s and D is its distance, measured in megaparsec (1 Mpc $= 3.2615 \times 10^6$ light-years $= 3.0856 \times 10^{24}$cm). The slope, known as the Hubble constant H_0 corresponds the present-day expansion rate of the Universe.

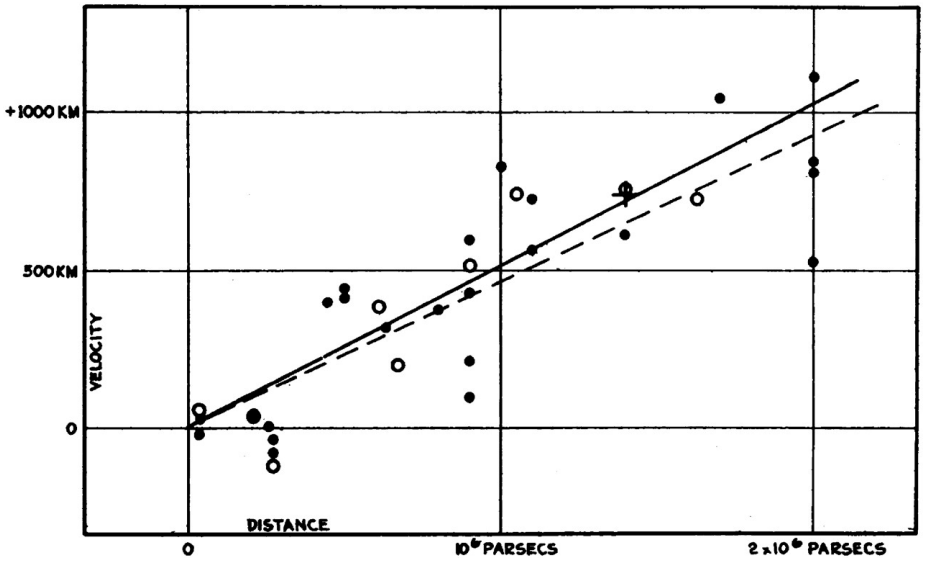

Figure 2.1: Original plot obtained by Hubble of the velocity-distance relation among extragalatic nebulae [13].The open circles and the dashed line correspond to grouped galaxies and their corresponding Hubble constant. The black dots and the black solid line correspond to individual galaxies and their average slope, the Hubble constant.

Since we are dealing with an expanding universe, we must distinguish between two types of coordinates. The coordinates $x^i \equiv (x^1, x^2, x^3)$ are called comoving coordinates and x^i_{phys} are known as physical coordinates. The former move along with expansion and are related to the latter by $x^i_{phys} = a(t)x^i$, where $a(t)$ is the quantity that relates the two coordinate systems, known as the scale factor, describing the relative size of the Universe (see Fig. 2.2). The physical velocity of an object is given by:

$$v^i_{phys} \equiv \frac{\mathrm{d}x^i_{phys}}{\mathrm{d}t} = a(t)\frac{\mathrm{d}x^i}{\mathrm{d}t} + \frac{\mathrm{d}a}{\mathrm{d}t}x^i \equiv v^i_{pec} + Hx^i_{phys}, \tag{2.11}$$

where we see two contributions: one from the peculiar velocity $v^i_{pec} = a(t)\frac{\mathrm{d}x^i}{\mathrm{d}t}$ and

the other from the Hubble flow[1].

The second contribution contains an important quantity used to describe the time evolution of the scale factor, known as the Hubble parameter:

$$H(t) = \frac{\dot{a}(t)}{a(t)}, \tag{2.12}$$

where $\dot{a} = \mathrm{d}a/\mathrm{d}t$ is the time derivative with respect to cosmic time of the scale factor. The Hubble constant and the Hubble parameter are related by $H_0 = H(t = t_0)$, where H_0 denotes the present-day value of the Hubble parameter.

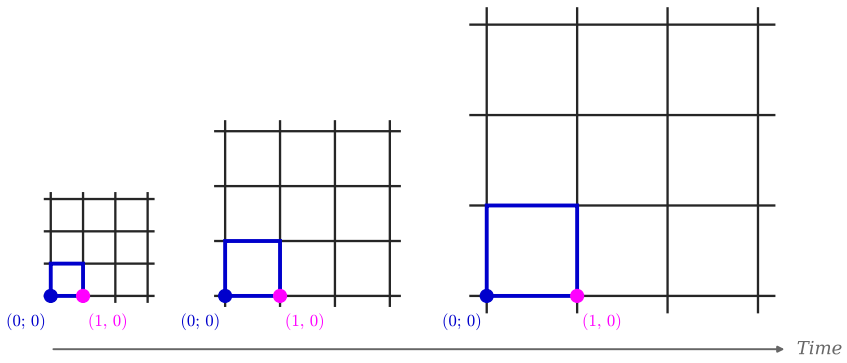

Figure 2.2: Visual relation between the physical distance and the comoving distance in an expanding Universe.

For galaxies whose peculiar velocities are negligible with the velocity associated with the expansion, the first term on the RHS of Eq. (2.11) can be neglected recovering the Hubble-Lemaître introduced previously $v^i_{phys} \approx H x^i_{phys}$.

A consequence of an expanding Universe is that the wavelength of light is stretched. Suppose that light from a distant galaxy travels through space. If the respective photons were emitted at a wavelength λ_e at a time t_e, we observe them with a larger wavelength λ_o at time t_o. Both wavelengths can be related by the scale factor and the cosmological redshift as:

$$z \equiv \frac{\lambda_o - \lambda_e}{\lambda_e}, \quad \lambda_o = \frac{a(t_o)}{a(t_e)}\lambda_e \implies \frac{1}{a} = 1 + z. \tag{2.13}$$

In the last equality we adopt the conventional normalization $a(t_o) = 1$ since the observer is located at Earth at the present time.

The last equality gives a relation between the time-dependent scale factor and the redshift parameter, which is also frequently used to specify events in the history of the Universe.

The Hubble parameter is also commonly expressed in terms of the dimensionless quantity h:

[1]The motions of objects or structures can be influenced by effects other than the expansion of the Universe, commonly referred to as the Hubble flow. Objects within gravitationally bound systems can exhibit deviations from the pure Hubble flow, known as peculiar motions.

$$H_0 = 100h \text{ km/s/Mpc}. \tag{2.14}$$

Another useful quantity is the Hubble rate which defines a Hubble time:

$$t_H = \frac{1}{H_0} = 9.78\text{h}^{-1} \text{ Gyr}. \tag{2.15}$$

2.2.2 Homogeneity and Isotropy

As physicist, when we study an system, we often make use of symmetry arguments, since they can provide conserved quantities that simplify our computations. When Einstein applied his theory of general relativity to the Universe as a whole, he made an important assumption. He argued that, on sufficiently large scales, the properties of the Universe should be the same regardless of our location or the direction in which we observe. This culminates in what we know today as the Cosmological Principle : on sufficiently large scales, the Universe is statistically homogeneous and isotropic.

An important consequence of a such principle is that we can then work and test a theory that describes the Universe as a whole, while knowing that the laws of physics are the same everywhere. Such an assumption reduces the number of possible cosmological models and ensures that there are no privileged positions or directions in the Universe, which is particularly useful observationally, since our observations are made from Earth.

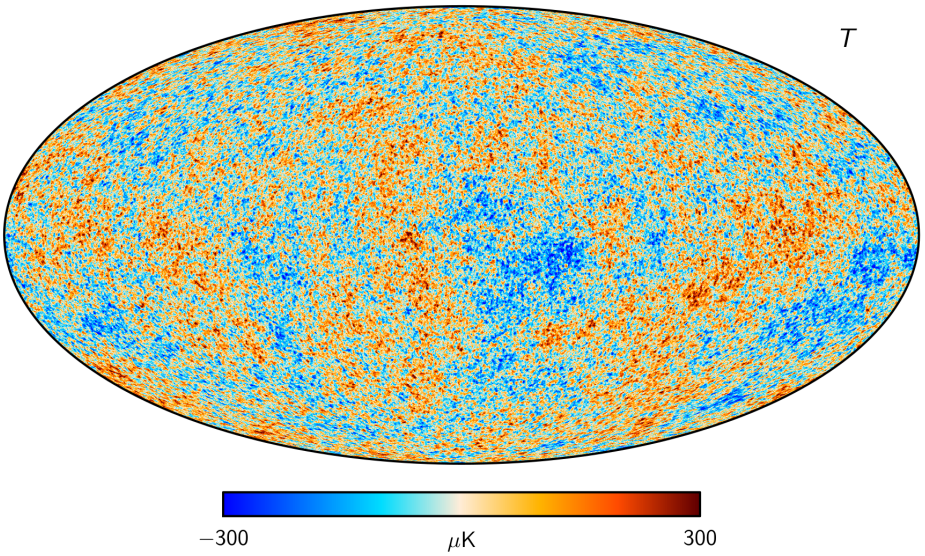

Figure 2.3: The sky map of temperature anisotropies in the CMB as measured by the Planck Collaboration [40].

Initially, this concept emerged from a philosophical perspective, reflecting the intuition behind Einstein's assumption. However, like any other assumption, it must be supported by obervational evidence to retain its honorable status as a cornerstone of modern cosmology.

If isotropy is preserved around every point, this implies homogeneity [41]. Isotropy is supported on large scales by observations such as the small CMB temperature

fluctuations, $\delta T/T \sim 10^{-5}$, which indicate a high isotropic Universe at the time the radiation was emitted. This, in turn, implies that the early Universe was extremely homogeneous and isotropic.

Figure 2.3 shows a map of the CMB temperature fluctuations measured by the Planck satellite.

2.2.3 Friedmann Equations

The most general spacetime that is maximally symmetric in the spatial coordinates while also allowing for expansion is described by the FLRW metric, which, in spherical comoving coordinates (t, r, θ, ϕ) can be written as:

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\left[\frac{\mathrm{d}r^2}{1-kr^2} + r^2(\mathrm{d}\theta^2 + \sin^2(\theta)\mathrm{d}\phi^2)\right], \tag{2.16}$$

where k refers to the curvature of the Universe and takes the values $k = \{-1, 0, +1\}$ and t corresponds to the cosmic time.

The spatial curvature parameter is defined as:

$$\Omega_{\mathrm{K}} \equiv -\frac{k}{a^2H^2}. \tag{2.17}$$

When Planck 2018 data are combined with BAO measurements [24], the present value of the curvature parameter is constrained to $\Omega_{\mathrm{K},0} = 0.001 \pm 0.002$, which is consistent with a spatially flat Universe. The subscript 0 denotes the present-day value of a given quantity. Motivated by these constraints, throughout this thesis we assume a spatially flat Universe, $k = 0$, which simplifies the computation of the relevant quantities.

For a homogeneous and isotropic Universe, the simplest assumption we can make is that the energy-momentum tensor should be that of a perfect fluid:

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu}, \tag{2.18}$$

where ρ is known as the energy density and p as the pressure in the rest frame of the fluid. u^μ corresponds to the velocity 4-vector $u^\mu(\vec{x}, t)$ such that $u^\mu u_\nu = -1$.

Substituting the metric into the Einstein field equations (2.9), one obtains the Friedmann and Raychaudhuri equations, respectively:

$$\begin{aligned}\left(\frac{\dot{a}}{a}\right)^2 &= H^2 = \frac{8\pi G}{3}\rho, \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}(\rho + 3p).\end{aligned} \tag{2.19}$$

2.2.4 Cosmic Inventory

The energy-momentum tensor is obtained from the variation of the matter action with respect to the metric:

$$T^{\mu\nu} \equiv \frac{2}{\sqrt{-g}}\frac{\delta S_{\mathrm{M}}}{\delta g_{\mu\nu}}. \tag{2.20}$$

Diffeomorphism invariance of the matter action implies the covariant conservation of the energy-momentum tensor:

$$\nabla_\mu T^{\mu\nu} = 0. \tag{2.21}$$

For the $\nu = 0$ component, this conservation law leads to the continuity equation:

$$\dot{\rho}_f + 3H(\rho_f + p_f) = 0 \quad \xrightarrow{\text{integrating over t}} \quad \rho_f \propto a^{-3(1+w_f)}, \tag{2.22}$$

where we assume that the fluid f is separately conserved and has a constant equation of state (EoS) w_f, such that $p_f = w_f \rho_f$.

From Eq. (2.19), we can define the critical density as:

$$\rho_c(t) \equiv \frac{3H^2(t)}{8\pi G}, \tag{2.23}$$

which corresponds to the total energy density required for a spatially flat Universe. Since throughout this thesis we assume a flat geometry, the total energy density satisfies:

$$\rho_{\rm tot}(t) = \sum_f \rho_f(t) = \rho_c(t). \tag{2.24}$$

This allows us to express the energy density of each fluid component f in dimensionless form through the density parameter:

$$\Omega_f(t) \equiv \frac{\rho_f(t)}{\rho_c(t)} = \frac{8\pi G\, \rho_{f,0}}{3H_0^2} \frac{\varrho_f(t)}{E^2(t)} = \Omega_{f,0} \frac{\varrho_f(t)}{E^2(t)}. \tag{2.25}$$

Here, we have written the energy density of each fluid as $\rho_f(t) = \rho_{f,0}\varrho_f(t)$ and introduced the dimensionless Hubble expansion rate, $E(t) \equiv H(t)/H_0$. As stated earlier, the subscript 0 denotes the present-day value of a given quantity. For a spatially flat Universe, the density parameters therefore satisfy:

$$\sum_f \Omega_f(t) = 1. \tag{2.26}$$

We can categorize the energy content of the Universe according to the properties of its different components, in particular their pressure and equation of state. We describe their main properties and cosmological evolution below.

Matter

By matter, we mean both baryonic and non-baryonic matter. The former consists of non-relativistic particles described by the Standard Model of particle physics. In an astrophysical context, this includes, for example, stars, planets, and clouds of cold gas.

Most of the matter content of the Universe, however, comes from the latter component, also known as dark matter, which is approximately five times more abundant than baryonic matter. Its exact nature remains unknown, but in the standard cosmological model it is assumed to behave as cold dark matter (CDM).

The pressure in these fluids compared with their energy density is negligible, such that $p \ll \rho \implies w \approx 0$ resulting in the contribution to the relative energy density:

$$\Omega_{\mathrm{m}}(a) = \Omega_{\mathrm{m},0}\frac{a^{-3}}{E^2(a)} \xrightarrow{a=\frac{1}{1+z}} \Omega_{\mathrm{m}}(z) = \Omega_{\mathrm{m},0}\frac{(1+z)^3}{E^2(z)}. \tag{2.27}$$

Radiation

By radiation we mean the relativistic components of the Universe, primarily photons and neutrinos while they are relativistic. Although neutrinos are known to have non-zero masses, in the early Universe their thermal energies were much larger than their rest masses, allowing them to be treated effectively as relativistic species.

The matter density dilutes as a^{-3} because the universe's volume increases by that factor. Radiation dilutes faster than matter by an additional factor a^{-1}, because the expansion of space not only causes light to dilute away in three dimensions but also stretches its wavelength.

The pressure in this fluid is related to the energy density as $p = \rho/3 \implies w = 1/3$ which results in the following contribution for the relative energy density:

$$\Omega_{\mathrm{r}}(a) = \Omega_{\mathrm{r},0}\frac{a^{-4}}{E^2(a)} \xrightarrow{a=\frac{1}{1+z}} \Omega_{\mathrm{r}}(z) = \Omega_{\mathrm{r},0}\frac{(1+z)^4}{E^2(z)}. \tag{2.28}$$

Dark energy

To take into account the cosmic acceleration observed today [20], the Raychaudhuri equation Eq. (2.19) requires a component with negative pressure:

$$\ddot{a} > 0 \implies p_{\mathrm{tot}} < -\rho_{\mathrm{tot}}/3 \implies w_{\mathrm{tot}} < -\frac{1}{3}. \tag{2.29}$$

For $w = -1$, corresponds to the cosmological constant Λ. Initially used by Einstein to create a static Universe, the cosmological constant Λ is the simplest candidate for driving accelerated expansion, causing gravity to be repulsive.

These three components–radiation, matter, and a cosmological constant — constitute the standard model of cosmology, whose strength lies in its simplicity and observational success.

The corresponding Friedmann equation can then be written as:

$$E^2(z) = \frac{H^2}{H_0^2} = \Omega_{\mathrm{r},0}(1+z)^4 + \Omega_{\mathrm{m},0}(1+z)^3 + \Omega_{\Lambda,0}. \tag{2.30}$$

Despite the observational success of ΛCDM, the cosmological constant itself faces well-known theoretical difficulties, including the cosmological constant problem [22] and the cosmic coincidence problem [23]. While these issues are not directly addressed in this work, they form part of the broader motivation to replace the cosmological constant with a dynamical dark energy (DDE) component whose properties evolve throughout cosmic history. Various theoretical frameworks have been proposed for dynamical dark energy, from canonical quintessence [42] to non-canonical k-essence [43]. In this thesis, however, we take a purely phenomenological approach and parametrize the DE density directly as a function of redshift.

As a function redshift, the energy density can be obtained by integrating Eq. 2.22:

$$\rho_{\rm de}(z) = \rho_{\rm de,0} \exp\left(3\int_0^z \frac{1+w_{\rm de}(z')}{1+z'}{\rm d}z'\right). \tag{2.31}$$

Conversely, given a specific DE density, the corresponding EoS can be obtained from the continuity equation:

$$w_{\rm de} = -1 + \frac{1+z}{3\rho_{\rm de}(z)}\frac{{\rm d}\rho_{\rm de}(z)}{{\rm d}z}. \tag{2.32}$$

We note that if the DE density crosses zero, Eq. (2.32) diverges. Models in which this occurs will be the focus of Chapter 5.

The corresponding Friedmann equation for a dynamical DE component is obtained by the substitution $\Omega_{\Lambda,0} \to \Omega_{\rm de,0}\varrho_{\rm de}(z)$ in Eq. (2.30):

$$E^2(z) = \Omega_{\rm r,0}(1+z)^4 + \Omega_{\rm m,0}(1+z)^3 + \Omega_{\rm de,0}\varrho_{\rm de}(z), \tag{2.33}$$

Evaluating Eq. 2.30 at $z = 0$ gives the following constraint:

$$\Omega_{\rm r,0} + \Omega_{\rm m,0} + \Omega_{\rm de,0} = 1, \tag{2.34}$$

which allows us to express $\Omega_{\rm de,0}$ as a derived quantity, $\Omega_{\rm de,0} = 1 - \Omega_{\rm m,0} - \Omega_{\rm r,0}$.

2.2.5 Distances

To estimate the parameters of our models, we must make measurements. In studies of large scale structure, this information usually comes from distant astronomical sources. Thus, it is useful to understand how we can measure the separation between events along radial null trajectories, i.e., photon trajectories that terminate at the observer. As stated in Section 2.1 these geodesics satisfy the relation ${\rm d}s^2 = 0$.

Since these distances will be used in the cosmological probes introduced in Section 4.3, we restore the speed of light c explicitly in this section, rather than setting $c = 1$. Considering an FLRW metric with flat spatial geometry and radial geodesics (${\rm d}\theta = {\rm d}\phi = 0$), Eq. (2.16) reduces to:

$${\rm d}r^2 = \frac{(c\ {\rm d}t)^2}{a(t)^2}. \tag{2.35}$$

Now, we can integrate the distance D_M traveled by the photon in the interval $[t_i, t_f]$:

$$D_M(t) = \int_{t_f}^{t_i} \frac{c\ {\rm d}t}{a(t)}. \tag{2.36}$$

Making the variable substitution ${\rm d}t = -\frac{{\rm d}z}{(1+z)H}$, we obtain comoving distance in terms of redshift[2]:

[2]As previously mentioned, we are assume a spatially flat universe ($k = 0$). Thus, we do not distinguish the line-of-sight comoving distance $D_{\rm C}$ and the transverse comoving distance $D_{\rm M}$, since $D_{\rm M} = D_{\rm H}\frac{1}{\sqrt{\Omega_{\rm K}}}\sinh\left[\sqrt{\Omega_{\rm K}}\, D_{\rm C}/D_{\rm H}\right]$ for $\Omega_{\rm K} > 0$; $D_{\rm M} = D_{\rm C}$ for $\Omega_{\rm K} = 0$; and $D_{\rm M} = D_{\rm H}\frac{1}{\sqrt{|\Omega_{\rm K}|}}\sin\left[\sqrt{|\Omega_{\rm K}|}\, D_{\rm C}/D_{\rm H}\right]$ for $\Omega_{\rm K} < 0$. [44]

$$D_M(z) = \int_{z_i}^{z_f} \frac{c\,\mathrm{d}z}{H(z)} = \frac{c}{H_0}\int_0^z \frac{\mathrm{d}z'}{E(z')} \quad , \tag{2.37}$$

where we assume that the observer is located at Earth, thus $z_i = 0$.

In Euclidean geometry, if an object of known physical size l, oriented perpendicular to the line of sight, is observed to subtend a small angle $\Delta\beta$ perpendicular to the comoving distance D_M, we can relate the three quantities by:

$$D_A = \frac{l}{\Delta\beta}, \tag{2.38}$$

where D_A is known as the angular diameter distance, and we assume that $\Delta\beta \ll 1$. By adding a factor of $(1 + z)$ to account for the expansion of the Universe, we obtain the angular diameter distance:

$$D_A(z) = \frac{D_M(z)}{1+z} = \frac{c}{H_0(1+z)}\int_0^z \frac{\mathrm{d}z'}{E(z')} \quad . \tag{2.39}$$

Another important observable is the luminosity distance D_L, defined by the inverse-square law:

$$D_L \equiv \sqrt{\frac{L}{4\pi S}}, \tag{2.40}$$

where L is the intrinsic luminosity of the source and S is the observed flux. Suppose we have a source at comoving distance D_M that emits isotropically with a luminosity L. If we were in a static universe, the photons would spread over a sphere with area $4\pi D_M^2$, giving the inverse-square law:

$$S_{\text{static}} = \frac{L}{4\pi D_M^2}. \tag{2.41}$$

However, since we are dealing with an expanding Universe, we must add two factors of $(1 + z)$, one because each photon loses energy (the wavelength gets stretched) and another because photons arrive less frequently. Thus, in an expanding universe we can relate D_L and S by the expression:

$$S = \frac{L}{4\pi D_M^2(1+z)^2} \xrightarrow{D_L=\sqrt{\frac{L}{4\pi S}}} D_L = (1+z)D_M. \tag{2.42}$$

Explicitly, the luminosity distance is defined as:

$$D_L(z) = (1+z)D_M(z) = (1+z)^2 D_A(z) = \frac{c}{H_0}(1+z)\int_0^z \frac{\mathrm{d}z'}{E(z')}. \tag{2.43}$$

These three distances are related through the Etherington reciprocity relation, $D_L = (1 + z)^2 D_A = (1 + z)D_M$ which holds in any metric theory of gravity where photons propagate along null geodesics and their number is conserved. Throughout this work, we assume these conditions are satisfied.

2.2.6 Cosmography

In a physical theory, we can extract different types of information from a system. If we are studying a time-varying phenomenon in which motion, forces and other distinct physical properties play a role, we are usually studying the dynamics of a system. A different approach is kinematics, which focuses on the study of motion without invoking field equations such as Einstein Eqs. (2.9).

In cosmology, the kinematic framework based on the Cosmological Principle, without specifying the underlying dynamics, is known as cosmography. The term was introduced in this context by Weinberg [45] and subsequently developed by Visser [46].

We can write the Taylor expansion of the scale factor around the present-day value a_0 as:

$$\frac{a(t)}{a_0} = 1 + \sum_{n=1}^{\infty} \frac{A_n(t_0)}{n!}[H_0(t-t_0)^n], \tag{2.44}$$

where we can define the cosmographic parameters $A_n = \frac{a^{(n)}}{aH^n}$, with $a^{(n)}$ being the n^{th} derivative of the scale factor with respect to cosmic time. Therefore, we define the first four cosmography parameters : the deceleration parameter $q = -A_2$, the jerk $j = A_3$, the snap $s = A_4$ and the lerk $l = A_5$.

We can write them explicitly as [46]:

$$A_2 = \frac{1}{a}\left[\frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t}\right]^{-2}\frac{\mathrm{d}^2a}{\mathrm{d}t^2} = \frac{a^{(ii)}}{aH^2}, \tag{2.45}$$

$$A_3 = \frac{1}{a}\left[\frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t}\right]^{-3}\frac{\mathrm{d}^3a}{\mathrm{d}t^3} = \frac{a^{(iii)}}{aH^3}, \tag{2.46}$$

$$A_4 = \frac{1}{a}\left[\frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t}\right]^{-4}\frac{\mathrm{d}^4a}{\mathrm{d}t^4} = \frac{a^{(iv)}}{aH^4}, \tag{2.47}$$

$$A_5 = \frac{1}{a}\left[\frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t}\right]^{-5}\frac{\mathrm{d}^5a}{\mathrm{d}t^5} = \frac{a^{(v)}}{aH^5}. \tag{2.48}$$

Since the expansion of the Universe is accelerating, q_0 takes a negative value. Within the ΛCDM model, $q_0 = -0.55$.

For the discussion in Section 5.6 it is convenient to express the cosmographic parameters directly as functions of redshift:

$$q(z) = -1 + \frac{1+z}{H(z)}\frac{\mathrm{d}H(z)}{\mathrm{d}z}, \tag{2.49}$$

$$j(z) = -q(z)\left[1 - 2\left(1+q(z)\right)\right] + (1+z)\frac{\mathrm{d}q(z)}{\mathrm{d}z}, \tag{2.50}$$

$$s(z) = j(z)\left[1 - 3\left(1+q(z)\right)\right] - (1+z)\frac{\mathrm{d}j(z)}{\mathrm{d}z}, \tag{2.51}$$

$$l(z) = s(z)\left[1 - 4\left(1+q(z)\right)\right] - (1+z)\frac{\mathrm{d}s(z)}{\mathrm{d}z}, \tag{2.52}$$

2.3 Linear Perturbations in Cosmology

Up to this point we have assumed a perfectly homogeneous Universe, but studying cosmology solely at the background level has its limitations.

To study large-scale structure, we must introduce inhomogeneities and describe how they evolve. In particular, we can expand the Einstein equations order by order in perturbations to the metric and the stress-energy tensor, making the resulting system of coupled PDEs tractable.

As mentioned previously, different time parameterizations can play different roles depending on when where they are implemented. A convenient time coordinate that simplifies computations and is also useful for studying of photon propagation is as conformal time, defined as:

$$\mathrm{d}\tau = \frac{\mathrm{dt}}{\mathrm{a(t)}}, \tag{2.53}$$

where $c = 1$ since we are working with natural units. Earlier, in Eq. (2.13), we defined the relation between the scale factor and redshift. A third variable that will be useful for working with perturbation theory is:

$$x \equiv \ln(a). \tag{2.54}$$

Different time variables lead to different expressions for time derivatives, which we summarize below.

Derivatives with respect to cosmic and conformal time are related by:

$$\{\dot{}\} = \frac{1}{a}\{\ \}' \quad , \tag{2.55}$$

where a prime represents a derivative with respect to conformal time and a dot represents a derivative with respect to cosmic time.

This change of variables allows us to express the Hubble parameter defined in Eq. (2.12) in terms of the conformal Hubble parameter, $\mathcal{H} \equiv a'/a$ and its derivative $\mathcal{H}'$ as:

$$H = \frac{1}{a}, \quad \dot{H} = \frac{1}{a^2}(\mathcal{H}' - \mathcal{H}^2) \tag{2.56}$$

We can also introduce a transformation rule between conformal time and the new variable x:

$$\{\ \}' = \{\}_x \mathcal{H}, \quad \{\ \}'' = \{\}_{xx}\mathcal{H}^2 + \{\}_x \mathcal{H}' \quad , \tag{2.57}$$

where a subscript x denotes a differentiation with respect to x. We therefore have five variables– t, τ, a, z and x– that can be used to parametrize cosmic evolution.

With these variables in hand, we can now formulate the perturbation equations. Assuming that deviations from homogeneity remain small, we decompose the metric into a background and a perturbed part:

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g_{\mu\nu}, \quad \delta g_{\mu\nu} \ll 1, \tag{2.58}$$

where we require that the corresponding first and second partial derivatives of $\delta g_{\mu\nu}$ are also small.

Quantities with the overbar are associated with the background spacetime, while quantities with a delta are associated with the perturbed spacetime[3].

An important consequence of linear perturbation theory is that the correspondence between the background and perturbed spacetimes is not unique. Given a coordinate system on the background spacetime, different nearby coordinate systems can be chosen for the perturbed spacetime while Eq. (2.58) remains valid. This ambiguity is known as gauge freedom.

At linear order, scalar, vector, and tensor perturbations decouple and can be treated independently [37, 47]. In this work, we retain only scalar perturbations and fix the gauge freedom by working in the Newtonian gauge, in which the perturbed FLRW line element reads:

$$\mathrm{d}s^2 = a^2\Big[-(1+2\Phi)\,\mathrm{d}\tau^2 + (1-2\Psi)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\Big], \tag{2.59}$$

where Φ and Ψ are known as the Bardeen potentials [48]. The Newtonian gauge is commonly used for studying structure formation mainly for two reasons:

- the metric $g_{\mu\nu}$ is diagonal, simplifying the calculation.
- Ψ plays the role of the gravitational potential in the Newtonian limit and thus has a simple physical interpretation.

Our goal is to compute the perturbed Einstein equations:

$$\delta G^{\mu}{}_{\nu} = 8\pi G \delta T^{\mu}{}_{\nu}. \tag{2.60}$$

Thus, we must know how the components defined in previous sections for the background spacetime are expressed in the perturbed space. Below, we collect the expressions needed to arrive at Eq. (2.60); the full derivations can be found in Refs. [37, 47].

We therefore present the final expression for the relevant components and provide additional comments where necessary. If we want to compute the perturbed Einstein equations, we must first write the metric.

Metric

Looking at Eq. (2.59), we can write the metric as:

$$g_{\mu\nu} = a^2\begin{bmatrix}-(1+2\Phi) & 0\\ 0 & (1-2\Psi)\delta_{ij}\end{bmatrix} \tag{2.61}$$

Inverse Metric

We can write the inverse metric as:

[3]Since we work within linear perturbation theory, the homogeneous FLRW spacetime is treated as the zeroth-order background and we ignore the effect of backreaction.

$$g^{\mu\nu} = a^{-2}\begin{bmatrix}-1 & 0\\ 0 & \delta_{ij}\end{bmatrix} + a^{-2}\begin{bmatrix}2\Phi & 0\\ 0 & 2\Psi\delta_{ij}\end{bmatrix} = a^{-2}\begin{bmatrix}-1+2\Phi & 0\\ 0 & (1+2\Psi)\delta_{ij}\end{bmatrix} \tag{2.62}$$

Connections

As with previous quantities, we can separate the connections into background and perturbed parts:

$$\Gamma^{\alpha}_{\beta\gamma} = \bar{\Gamma}^{\alpha}_{\beta\gamma} + \delta\Gamma^{\alpha}_{\beta\gamma}, \tag{2.63}$$

where:

$$\begin{aligned}
&\delta\Gamma^0_{00} = \Phi' \qquad \delta\Gamma^0_{0k} = \Phi_{,k} \qquad \delta\Gamma^0_{ij} = -\Big[2\mathcal{H}(\Phi+\Psi) + \Psi'\Big]\delta_{ij}\\
&\delta\Gamma^i_{00} = \Phi_{,i} \qquad \delta\Gamma^i_{0j} = -\Psi'\delta^i_j \qquad \delta\Gamma^i_{kl} = -\Big(\Psi_{,l}\delta^i_k + \Psi_{,k}\delta^i_l\Big) + \Psi_{,i}\delta_{kl}
\end{aligned} \tag{2.64}$$

Ricci tensor and scalar

The perturbed Ricci tensor can be computed as:

$$R^{\mu}{}_{\nu} = g^{\mu\alpha}R_{\alpha\nu} = \bar{R}^{\mu}{}_{\nu} + \bar{g}^{\mu\alpha}\delta R_{\alpha\nu} + \delta g^{\mu\alpha}\bar{R}_{\alpha\nu}, \tag{2.65}$$

where $\delta R_{\alpha\nu}$ is obtained by substituting the perturbed connections from Eq. (2.63) into the definition of the Ricci tensor.

Finally, the corresponding scalar is obtained as:

$$\delta R = \delta g^{\mu\lambda}R_{\lambda\mu} + g^{\mu\lambda}\delta R_{\lambda\mu}. \tag{2.66}$$

Einstein tensor

We now have all the components needed to write the left-hand-side (LHS) of Eq. (2.60):

$$\delta G_{\mu\nu} = \delta R_{\mu\nu} - \frac{1}{2}\delta g_{\mu\nu}R - \frac{1}{2}g_{\mu\nu}\delta R \implies \delta G^{\mu}{}_{\nu} = \delta g^{\mu\lambda}G_{\lambda\nu} + g^{\mu\lambda}\delta G_{\lambda\nu}. \tag{2.67}$$

Writing the perturbed term explicitly with one upper index, we obtain:

$$\delta G^0{}_0 = 2a^{-2}\left[3\mathcal{H}(\mathcal{H}\Phi + \Psi') - \nabla^2\Psi\right], \tag{2.68}$$

$$\delta G^0{}_i = -2a^{-2}\left(\Psi' + \mathcal{H}\Phi\right)_{,i}, \tag{2.69}$$

$$\delta G^i{}_j = 2a^{-2}\Big[\left(\mathcal{H}^2 + 2\mathcal{H}'\right)\Phi + \mathcal{H}\Phi' + \Psi'' + 2\mathcal{H}\Psi'\Big]\delta^i_j + 2a^{-2}\Big[\nabla^2(\Phi-\Psi)\delta^i_j + (\Psi-\Phi)_{,ij}\Big]. \tag{2.70}$$

Energy-momentum tensor and matter

As with previous quantities, we decompose the energy-momentum tensor as:

$$T^{\mu}{}_{\nu} = \bar{T}^{\mu}{}_{\nu} + \delta T^{\mu}{}_{\nu}. \tag{2.71}$$

The perturbed quantity $\delta T^{\mu}{}_{\nu}$ should be written to keep $T^{\mu}{}_{\nu}$ in the perfect fluid form:

$$T^{\mu}{}_{\nu} = (\rho + p)u^{\mu}u_{\nu} + p\delta^{\mu}{}_{\nu}. \tag{2.72}$$

Applying the scalar-only decomposition to the energy-momentum tensor in Newtonian gauge, one obtains [47]:

$$\delta T^{\mu}{}_{\nu} = \begin{bmatrix} -\delta\rho & -(\bar{\rho}+\bar{p})v_{,i} \\ (\bar{\rho}+\bar{p})v_{,i} & \delta p\delta^{i}{}_{j} + \bar{p}(\Pi_{ij} - \frac{1}{3}\delta_{ij}\nabla^2\Pi), \end{bmatrix}. \tag{2.73}$$

$\delta\rho$ and δp denote the total energy density and pressure perturbations, respectively, while Π_{ij} denotes the anisotropic stress tensor . The quantity δv denotes the peculiar velocity perturbation, and is from the velocity 4-vector, under the assumption that $T^{\mu}{}_{\nu}$ retains the perfect fluid form.

As a first approximation, we assume that the fluid perturbations do not introduce any anisotropy, which leads to $\Pi_{ij} = 0$. From the off-diagonal terms of the perturbed Einstein Eqs. (2.60), we get:

$$(\Psi - \Phi)_{ij} = 8\pi G a^2 \bar{p}\ \Pi_{ij} \xrightarrow{\Pi_{ij}=0} \Phi = \Psi, \tag{2.74}$$

we then conclude that in the absence of the anisotropic stress tensor, the gravitational potentials are identical.

Following the multi-fluid approach used in Refs [49, 50] , the total energy perturbation, the total pressure perturbation , and the total peculiar velocity perturbation can be expressed in terms of the contributions from each individual fluid f:

$$\delta\rho = \sum_f \delta\rho_f, \quad \delta p = \sum_f \delta p_f \quad v = \sum_f \frac{1+w_f}{1+w_{\rm tot}}\Omega_f v_f, \tag{2.75}$$

, where we defined the total EoS parameter as:

$$w_{\rm tot} = \sum_f \Omega_f w_f. \tag{2.76}$$

Einstein and evolution equations

Given that we have $\Phi = \Psi$, we can combine together the perturbed Einstein and energy-momentum equations:

$$3\mathcal{H}\left(\mathcal{H}\Psi + \Psi'\right) - \nabla^2\Psi = -4\pi G a^2\,\delta\rho, \tag{2.77}$$

$$\nabla^2\left(\mathcal{H}\Psi + \Psi'\right) = -4\pi G a^2(\bar{\rho}+\bar{p})\nabla^2 v, \tag{2.78}$$

$$\Psi'' + 3\mathcal{H}\Psi' + \Psi\left(2\mathcal{H}' + \mathcal{H}^2\right) = 4\pi G a^2\,\delta p. \tag{2.79}$$

The perturbed evolution of the fractional energy density perturbation $\delta_f = \delta\rho_f/\bar{\rho}$ and the peculiar velocity potential can be obtained from the corresponding conservation relation of the energy-momentum tensor:

$$\nabla_\mu \delta T^\mu_{A\nu} + \delta\Gamma^\mu_{\mu\alpha}\bar{T}^\alpha_{A\nu} - \delta\Gamma^\alpha_{\mu\nu}\bar{T}^\mu_{A\alpha} = 0, \tag{2.80}$$

thus obtaining:

$$\delta'_A = 3\left(w_A - c^2_{sA}\right)\delta_A + 3(1+w_A)\Psi' + (1+w_A)\left[9\mathcal{H}\left(c^2_{sA} - c^2_{aA}\right) - \frac{\nabla^2}{\mathcal{H}}\right]v_A, \tag{2.81}$$

$$v'_A = \left(3c^2_{sA} - 1\right)v_A - \frac{c^2_{sA}}{\mathcal{H}(1+w_A)}\delta_A - \frac{\Psi}{\mathcal{H}} \tag{2.82}$$

where we introduce the adiabatic speed, c^2_{aA}, and the rest frame sound speed, c^2_{sA} which are defined, respectively, by:

$$c^2_{aA} = \frac{\bar{p}'_f}{\bar{\rho}'_f}, \quad c^2_{sA} = \left.\frac{\delta p_f}{\delta\rho_f}\right|_{r.f}. \tag{2.83}$$

Fourier Transformation

Eqs. (2.81) are coupled linear partial differential equations, that depend on both space and time. Since solving these in real space is not trivial, we decompose the function $\psi(\tau, x^i)$ into its Fourier components $\psi_k(\tau)$:

$$\psi(\tau, \vec{x}) = \frac{1}{(2\pi)^{3/2}}\int e^{-i\vec{k}\cdot\vec{x}}\psi_k(\tau)\, d^3k, \tag{2.84}$$

where $\vec{k}$ is a comoving wave vector. The comoving wave number $k \equiv |\vec{k}|$ and wavelength $\lambda = \frac{2\pi}{k}$ are related to the physical wavelength and wavenumber of the Fourier mode by:

$$k_{\rm phys} = \frac{2\pi}{\lambda_{\rm phys}} = \frac{2\pi}{a\lambda} = \frac{k}{a}. \tag{2.85}$$

Then, the equations for each mode are obtained after the substitution $\nabla^2 \to -k^2$ in Eqs. (2.81) for the different components of the universe (radiation, dust and DE) [51]:

$$(\delta_{\rm r})_x = \frac{4}{3}\left(\frac{k^2}{\mathcal{H}}v_{\rm r} + 3\Psi_x\right), \tag{2.86}$$

$$(v_{\rm r})_x = -\frac{1}{\mathcal{H}}\left(\frac{1}{4}\delta_{\rm r} + \Psi\right), \tag{2.87}$$

$$(\delta_{\rm m})_x = \left(\frac{k^2}{\mathcal{H}}v_{\rm m} + 3\Psi_x\right), \tag{2.88}$$

$$(v_{\rm m})_x = -\left(v_{\rm m} + \frac{\Psi}{\mathcal{H}}\right), \tag{2.89}$$

$$(\delta_{\rm de})_x = (1+w_{\rm de})\left\{\left[\frac{k^2}{\mathcal{H}} + 9\mathcal{H}\left(c^2_{\rm sd} - c^2_{\rm ad}\right)\right]v_{\rm de} + 3\Psi_x\right\} + 3\left(w_{\rm de} - c^2_{\rm sd}\right)\delta_{\rm de}, \tag{2.90}$$

$$(v_{\rm de})_x = -\frac{1}{\mathcal{H}}\left(\frac{c^2_{\rm sd}}{1+w_{\rm de}}\delta_{\rm de} + \Psi\right) + \left(3c^2_{\rm sd} - 1\right)v_{\rm de}, \tag{2.91}$$

and for the metric potential:

$$\Psi_x + \Psi\left(1 + \frac{k^2}{3\mathcal{H}^2}\right) = -\frac{1}{2}\delta, \tag{2.92}$$

$$\Psi_x + \Psi = -\frac{3}{2}\mathcal{H}v(1 + w_{\rm tot}), \tag{2.93}$$

$$\Psi_{xx} + \left[3 - \frac{1}{2}(1 + 3w_{\rm tot})\right]\Psi_x - 3w_{\rm tot}\Psi = \frac{3}{2}\frac{\delta p}{\rho}. \tag{2.94}$$

Eqs. (2.86) and (2.92) form a closed system of ODEs describing the evolution of density perturbations δ and velocity perturbations v for each fluid component–radiation, matter, and DE–coupled through the gravitational potential Ψ.

As stated earlier, each perturbation mode is characterized by a comoving wavenumber k. The behavior of cosmological perturbations depends on their wavelength relative to the characteristic scales of the problem. In our case, the relevant scale is the comoving Hubble radius, $\mathcal{H}^{-1} = (aH)^{-1}$.

When $k/\mathcal{H} \ll 1$, the characteristic comoving scale of the perturbation is much larger than the comoving Hubble radius. Such modes are referred to as super-Hubble modes. In this regime, the perturbations remain approximately constant until the mode crosses the Hubble radius, $k \sim \mathcal{H}$. By contrast, when $k/\mathcal{H} \gg 1$, the mode lies well inside the Hubble radius and is referred to as sub-Hubble.

2.3.1 Initial Conditions

Having obtained the system of ODEs, we must specify the initial conditions for each mode. Inflation, a short period of accelerated expansion in the very early Universe, provides a framework for establishing the primordial perturbations that later seed the formation of cosmic structure. We set the initial conditions during the radiation-dominated epoch, at approximately $z \sim 10^6$. At this stage, all relevant modes in the linear regime are super-Hubble, with wavenumbers satisfying $k \ll \mathcal{H} \implies \frac{k^2}{\mathcal{H}^2} \ll 1$, and thus lie outside the horizon. Under these approximations, combining the first and third equations of Eq. (2.92), one can show that:

$$\Psi_{xx} + 3\Psi_x \simeq 0. \tag{2.95}$$

The dominant solution of this equation is constant, $\Psi_{\rm ini} = \Psi(x_{\rm ini})$. Using this result in the same system of equations from which the previous relation was obtained, we find:

$$\Psi_{\rm ini} = -\frac{\delta_{\rm ini}}{2\left[1 + k^2/(3\mathcal{H}_{\rm ini}^2)\right]}, \tag{2.96}$$

and

$$\Psi_{\rm ini} = -\frac{3}{2}\mathcal{H}_{\rm ini}v_{\rm ini}\left(1 + w_{\rm ini}\right). \tag{2.97}$$

In addition, motivated by single scalar-field models of inflation [52], we assume adiabatic initial conditions:

$$\frac{\delta_{\rm r}}{1+w_{\rm r}} = \frac{\delta_{\rm m}}{1+w_{\rm m}} = \frac{\delta}{1+w_{\rm tot}}, \qquad v_{\rm r,ini} = v_{\rm m,ini} = v_{\rm ini}. \tag{2.98}$$

Combining these relations with the expressions obtained above, the initial conditions for the perturbations and their corresponding velocities can then be written as:

$$\frac{3}{4}\delta_{\rm r,ini} = \delta_{\rm m,ini} \simeq \frac{3}{4}\delta_{\rm ini} = -\frac{2\left[1+k^2/(3\mathcal{H}_{\rm ini}^2)\right]\Psi_{\rm ini}}{1+w_{\rm ini}}, \tag{2.99}$$

and

$$v_{\rm r,ini} = v_{\rm m,ini} = -\frac{2\Psi_{\rm ini}}{3\mathcal{H}_{\rm ini}(1+w_{\rm ini})}. \tag{2.100}$$

After performing the Fourier decomposition, the linear perturbation equations can be evolved independently for each mode. Taking advantage of the linearity of the system, we first compute the evolution of the perturbation variables using the initial conditions above and setting $\Psi_{\rm ini} = 1$. In the super-Hubble limit, from Eq. (2.96) we can obtain:

$$\Psi_{\rm ini} \simeq -\frac{\delta_{\rm ini}}{2} \rightarrow \delta_{\rm ini} = -2\Psi_{\rm ini} = -2, \tag{2.101}$$

the resulting solutions are then rescaled by the physical value of $\delta_{\rm phys}(k)$ [51]. This quantity is taken from the Planck observational fit to single-field inflation [24]:

$$\delta_{\rm phys}(k) = \frac{8\pi}{3}\sqrt{2A_s}\left(\frac{k}{k_{\rm pivot}}\right)^{\frac{n_s-1}{2}} k^{-\frac{3}{2}}. \tag{2.102}$$

Here, A_s and n_s denote the amplitude and spectral index of the primordial inflationary power spectrum, respectively, evaluated at the chosen pivot scale $k_{\rm pivot}$. In this work, we adopt $k_{\rm pivot} = 0.05\,h\,{\rm Mpc}^{-1}$, $A_s = 2.101 \times 10^{-9}$, and $n_s = 0.9665$, consistent with the Planck observational data [24].

2.3.2 Cosmological Observables

Having described the evolution of the linear perturbations, we now introduce the growth rate of matter perturbations and observables that will be used to characterize structure formation and to compare the predictions of the models in Chapter 5 .

Growth rate

A possible method to constrain our models is through the linear growth rate, a quantity that quantifies the growth of matter perturbations as:

$$f \equiv \frac{{\rm d}(\ln\delta_{\rm m})}{{\rm d}(\ln a)}, \tag{2.103}$$

where δ_m is the matter density contrast. During matter domination, and for modes well inside the Hubble radius, its evolution can be approximately written as:

$$f \simeq \Omega_{\rm m}^{\gamma}, \tag{2.104}$$

where γ is known as the growth index and takes a value close to $\gamma \simeq 0.55$ for the ΛCDM model.

Matter power spectrum

The matter power spectrum describes the distribution of matter fluctuations across different scales. Following the Newtonian-gauge variables introduced above, we define the comoving matter density perturbation as:

$$\Delta_{\rm m} \equiv \delta_{\rm m} - 3\mathcal{H}v_{\rm m}, \tag{2.105}$$

such that the matter power spectrum is given by:

$$P(k) \equiv |\Delta_{\rm m}|^2 = |\delta_{\rm m} - 3\mathcal{H}v_{\rm m}|^2 . \tag{2.106}$$

The dependence on the comoving wavenumber k allows us to study the amplitude of matter fluctuations at different cosmological scales.

$f\sigma_8$ distribution

Observational constraints on structure growth are commonly expressed in terms of the combination $f\sigma_8$. The quantity σ_8 is the root-mean-square of the matter density fluctuations within spheres of radius $8\,h^{-1}$Mpc, serving as a normalization for the power spectrum.

Its evolution can be written as:

$$\sigma_8(z) = \sigma_{8,0}(0, k_{\sigma_8})\frac{\delta_{\rm m}(z, k_{\sigma_8})}{\delta_{\rm m}(0, k_{\sigma_8})}, \tag{2.107}$$

where:

$$k_{\sigma_8} \equiv 0.125\,h\,{\rm Mpc}^{-1}, \tag{2.108}$$

denotes the wavenumber corresponding to a physical scale of $8{\rm h}^{-1}$Mpc and $\sigma_{8,0}$ denotes its present-day value.

The observable is therefore computed as:

$$f\sigma_8(z) = f(z)\sigma_8(z). \tag{2.109}$$

An advantage of this combination is that it reduces the degeneracy between σ_8 and the linear galaxy bias, which relates the DM perturbations to the galaxy density perturbations.

For all the models considered in the analysis, we adopt the Planck 2018 value $\sigma_8(0, k_{\sigma 8})$ [40].

. . . se me vês acenar a mano
Diz ao meu Porto que eu o amo
Diz ao meu país que eu o amo

— Keso

3 The Hubble Tension

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3.1 Introduction

Often referred as the standard model of cosmology, the ΛCDM model provides a remarkably successful description of observations spanning very different epochs and physical scales using only six physical parameters, including the anisotropies of the cosmic microwave background (CMB), the large-scale distribution of matter, baryon acoustic oscillations (BAO), the luminosity distances of Type Ia supernovae (SNe Ia), and the abundances of light elements produced during Big Bang nucleosynthesis (BBN).

However,with the increase precision and volume of cosmological meaurements, different probes have revealed apparently incompatible results. These descrepancies,also referred to as tensions, usually arise between from observations probing the early and stages of the Universe. In this context, by early times we mean epochs before recombination, while later times refer to the subsequent evolution after photon decoupling. This distinction applies both to observations and theoretical models.

Since the observational analysis presented in this thesis is performed at the background level, we focus on the tension most directly constrained by background observables : the Hubble constant H_0. For a comprehensive review of other cosmological tensions, including those related to the growth of structure, see Ref. [26].

A possible way of framing, the Hubble tension is as a disagreement between early- and late-Universe measurements. As discussed in Chapter 1, the most commonly

cited manifestation of this tension is the 5.0σ disagreement between the value inferred by Planck Collaboration [24] and the local distance-ladder measurement from SH0ES Collaboration [25]. Fig. 3.1 summarizes a selection of current H_0 determinations from different probes; we discuss the most relevant ones below.

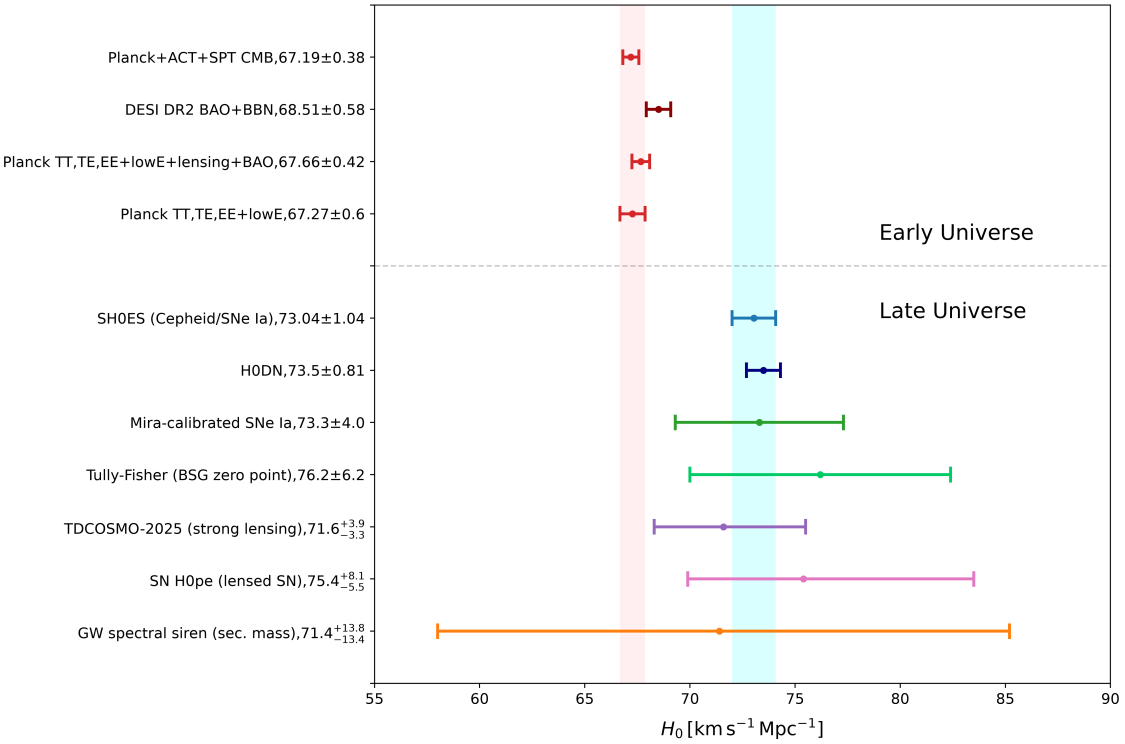

Figure 3.1: Various measurements and constraints of H_0 from different cosmological probes and their respective uncertainties.The values in the left column are obtained from early-Universe probes, while in column corresponds to late-Universe measurements obtained using different techniques, both distance-ladder and non-distance-ladder methods. Plot adapted from Ref. [53].

To infer the value of H_0 from CMB observations, one must assume a cosmological model. In contrast, late-time methods can provide more direct determinations of H_0, for example through the distance ladder. This method is a sequence of calibration steps, or rungs, in which each class of distance indicator is calibrated using the previous one, extending measurements to progressively larger scales. In its most common form, the SH0ES distance ladder consists of three rungs : geometric distances to nearby anchors, Cepheid period–luminosity relations calibrated against those anchors, and Type Ia supernovae calibrated by the Cepheids. Other realizations replace the intermediate rung with alternative indicators such as Mira variables or the Tully–Fisher relation.

It is important to note that the H_0 tension does not arise solely between Planck and SH0ES, but persists across a wide range of independent probes on both sides, as illustrated in Fig. 3.1. Unless stated otherwise, all values of H_0 reported in this section are quoted at 68% confidence level.

On the early-Universe side, model-dependent inferences of H_0 from CMB observations are remarkably consistent across experiments. The Planck 2018 analysis

yields $H_0 = 67.27 \pm 0.60$ km/s/Mpc from temperature data alone, tightening to $H_0 = 67.66 \pm 0.42$ km/s/Mpc when polarization, lensing, and BAO data are included [24].

Independent CMB measurements yield $H_0 = 70 \pm 2.2$ km/s/Mpc from WMAP [54] and $H_0 = 67.9 \pm 1.5$ km/s/Mpc from the Atacama Cosmology Telescope [55]. BAO measurements can also constrain H_0 independently of the CMB when combined with a sound-horizon calibration from Big Bang Nucleosynthesis (BBN); the DESI DR2 analysis yields $H_0 = 68.51 \pm 0.58$ km/s/Mpc [35].

The comoving sound horizon at a given redshift z is defined as:

$$r_s(z) = \int_z^\infty \frac{c_s(z')}{H(z')}\, dz', \qquad c_s(z) = \frac{c}{\sqrt{3[1+R(z)]}}, \qquad R(z) = \frac{3\rho_b(z)}{4\rho_\gamma(z)}. \tag{3.1}$$

Here, $c_s(z)$ is the sound speed of the photon–baryon fluid, c is the speed of light, and $R(z)$ baryon–to–photon momentum ratio. For BAO measurements, the relevant scale is the sound horizon evaluated at the baryon-drag epoch, $z_{\rm drag}$, such that $r_{\rm drag} \equiv r_s(z_{\rm drag})$. This characteristic comoving scale is imprinted in the matter distribution and acts as a standard ruler against which cosmological distances can be measured. This scale acts as a standard ruler, providing a known physical length imprinted in the galaxy distribution against which cosmological distances can be measured.

On the late-Universe side, the SH0ES collaboration reports $H_0 = 73.04 \pm 1.04$ km/s/Mpc from the Cepheid-calibrated Type Ia supernovae distance ladder [25]. Alternative calibrations yield consistent results : Mira-calibrated supernovae give $H_0 = 73.3 \pm 4.0$ km/s/Mpc [56], and the Tully–Fisher relation gives $H_0 = 76 \pm 2.54$ km/s/Mpc [57].

Crucially, several late-time probes that are entirely independent of the distance ladder also favor higher values of H_0. Gravitational-wave spectral sirens, using the secondary-mass feature, yield $H_0 = 71.4^{+13.8}_{-13.4}$ km/s/Mpc [58], while the multiply-imaged supernova SN H0pe provides $H_0 = 75.4^{+8.1}_{-5.5}$ km/s/Mpc [59]. While these measurements individually carry larger uncertainties, their consistency with the distance-ladder results strengthens the case that the tension is not driven by a single systematic in any one method.

Although some of these inconsistencies could arise from unidentified data errors or systematics, at present no single systematic effect has been shown to account for the full difference between the early- and late-Universe determinations.

These discrepancies have motivated a wide range of theoretical investigations beyond ΛCDM, from modifications of the dark sector to alternative theories of gravity. Before reviewing these proposals, however, it is worth noting that the framing of the Hubble tension itself is not unique. Although the most common perspective presents it as a discrepancy between early- and late-time probes, alternative interpretations have been proposed. For instance, Ref. [60] argues that the tension may be more accurately described as a disagreement between distance-ladder measurements and sound-horizon-calibrated probes, rather than an early- versus late-time divide.This illustrates that the Hubble tension is a multifaceted problem in which even the definition of the discrepancy remains under discussion.

With this in mind, we now briefly review two main classes of ΛCDM extensions proposed to address the tension.

3.1.1 Early-time modifications of ΛCDM

Early-time solutions to the Hubble tension propose a modification of the expansion history before recombination. Common approaches include early dark energy (EDE) models [61], which introduce an additional component into to the energy density content of the Universe, and scenarios with extra relativistic degrees of freedom [26].

To to provide a viable solution, or at least alleviate the tension, such early-Universe modifications must remain consistent with constraints from probes such as the CMB and BAO. The connection between these observables and H_0 can be understood from the characteristic acoustic scale measured by the CMB.

Although the extraction of cosmological parameters from CMB observations requires a detailed statistical analysis, the underlying mechanism can be understood analytically. The acoustic oscillations of the photon–baryon fluid leave a characteristic scale imprinted in the CMB power spectrum, visible through the positions of the acoustic peaks. This scale is measured with very high precision through the angular size of the sound horizon:

$$\theta_{\rm cmb} = \frac{r_s(z_{\rm cmb})}{D_M(z_{\rm cmb})} = \frac{\displaystyle\int_{z_{\rm cmb}}^{\infty} \frac{c_s(z)}{H(z)}\, dz}{\displaystyle\int_0^{z_{\rm cmb}} \frac{1}{H(z)}\, dz} = \frac{H_0\, r_s(z_{\rm cmb})}{c\, d_M(z_{\rm cmb})}, \tag{3.2}$$

where $d_M(z_{\rm cmb})$ denotes the dimensionless version of the comoving distance defined in Eq. (2.37), evaluated at the surface of last scattering, while $r_s(z_{\rm cmb})$ is the comoving sound horizon at that epoch, is the comoving sound horizon evaluated at $z = z_{\rm cmb}$. The combination $H_0 r_s$ is also closely related to the degeneracy constrained by BAO observations.

These two quantities probe different epochs of the cosmic expansion : $d_M(z_{\rm cmb})$ is more sensitive to the expansion history after recombination, while $r_s(z_{\rm cmb})$ is determined by the physics before recombination. Therefore, if the measured angular scale $\theta_{\rm cmb}$ and the dimensionless distance $d_M(z_{\rm cmb})$ remain approximately unchanged, an increase in H_0 requires a corresponding reduction of the sound horizon r_s.

This constitutes the main mechanism through which early-time modifications can alleviate the Hubble tension.

3.1.2 Late-time modifications of ΛCDM

An alternative approach is to leave the pre-recombination physics essentially unchanged and modify the expansion history at later times. A wide variety of models have been proposed following this approach, including scenarios that modify the DE sector at low redshift, or that allow interactions between DE and DM [53].

In these scenarios, the sound horizon r_s remains approximately fixed, while cosmological distances are modified through changes in $H(z)$. A larger value of H_0 would, by itself, decrease cosmological distances. Therefore, in late-time solutions, $E(z) = H(z)/H_0$ must evolve in such a way that the distance integral compensates for the increase in H_0.

Among the wide range of late-time modifications, increasing attention has recently focused on scenarios in which the DE density changes sign during the late-time

evolution of the Universe. In these models, the effective DE sector evolves from a negative energy density at high redshift to the positive value observed today. Since the models studied in this thesis belong to this class, we briefly review sign-switching dark-energy models at the beginning of Chapter 5, before introducing our contribution.

Eu estou pronto para conversarmos a sério
E se eu tiver a errar o passo no terraço do prédio
Dá-me a verdade crua e cristalina
Dá-me a tua perspetiva, a chave
Eu já nasci com um olho fechado

— Slow J

4 Inference and Observational Cosmology

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4.1 Bayesian Statistics and Numerical Tools

Cosmology is particularly distinctive among the physical sciences in that it is primarily observational rather than experimental. We cannot repeat the evolution of the Universe under controlled initial conditions; instead, we observe a single cosmological realization. Statistical inference therefore provides the natural framework through which theoretical predictions can be confronted with cosmological observations. However, the concept of probability itself does not have a unique interpretation across statistics, and the choice of framework matters.

From a frequentist perspective, probability is associated with the relative frequency of an outcome in a large number of repeated trials N. In the limit of infinitely many repetitions, we obtain:

$$P = \frac{n}{N}, \tag{4.1}$$

where n denotes the number of occurrences of event A among N trials.

From a Bayesian perspective, probability instead quantifies the degree of belief assigned to a proposition, given the available information.

Suppose we have collected some data D and have a model $\mathcal{M}$ containing a set of hypotheses described by the parameters θ [1]. When performing parameter estimation, we aim to obtain estimates of the parameters, their uncertainties, or, ideally, the probability distribution of θ given the data. This can be described by the mathematical quantity $p(\theta|D, \mathcal{M})$, called the posterior Probability Distribution Function (PDF), which represents the probability of the model parameters given the data.

As a direct consequence of the axioms of probability theory, the probability of the model in light of the data can be expressed by Bayes's theorem:

$$p(\theta|D, \mathcal{M}) = \frac{p(D|\theta, \mathcal{M})\ p(\theta|\mathcal{M})}{p(D|\mathcal{M})}, \tag{4.2}$$

where $p(D|\theta, \mathcal{M})$ is the sampling distribution of the data, $p(\theta|\mathcal{M})$ is the prior and $p(D|\mathcal{M})$ is known as the evidence of the model. We now briefly explain each of these quantities and their role in statistical inference.

Prior

Our choices when performing a data analysis depend on our assumptions and beliefs . This is where the role of the prior becomes important. Also written as $\pi(\theta)$, the prior encodes everything mentioned earlier: the choices and assumptions associated with the model and the data. In the absence of strong external information, uniform priors (also referred to as a flat priors) take the form $\pi(\theta) \propto c$ with c being a constant.

Our prior choices are listed in Section 5.5.2.

Evidence

The denominator in the RHS is a constant called the Bayesian evidence:

$$p(D|\mathcal{M}) = \int ... \int p(D|\theta, \mathcal{M})\ p(\theta|\mathcal{M})\ \mathrm{d}\theta_1 ... \mathrm{d}\theta_n. \tag{4.3}$$

This quantity plays an important role for selecting model that best describes the data, a procedure known as model selection. An downside of computing such a quantity is that it can be computationally expansive. Nevertheless, this quantity can be ignored when exploring the parameter space of a simple model, since it does not depend on the model parameters.

Likelihood

The sampling distribution describes how the degree of plausibility of the parameters changes when new data are acquired. When our set of parameters $\theta = \{\theta_1, ..., \theta_n\}$ is unknown and variable for a set of data D known and fixed, we call it the likelihood function, employing the notation $\mathcal{L}(\theta) \equiv p(D|\theta, \mathcal{M})$.

As stated earlier, the denominator in Bayes's Theorem (Bayesian Evidence) is a normalization constant. Thus, the posterior distribution can be obtained from the likelihood and prior as:

[1] Models can have different motivations, ranging from data-driven approaches that reproduce specific features or general trends in the data to physically motivated models.

$$p(\theta|D,\mathcal{M}) \propto p(D|\theta,\mathcal{M})\ p(\theta|\mathcal{M}), \tag{4.4}$$

where by the additional assumption of a flat prior $\pi(\theta) \propto c$, the posterior is proportional to the likelihood:

$$p(\theta|D,\mathcal{M}) \propto \mathcal{L}. \tag{4.5}$$

Maximizing the likelihood yields the maximum-likelihood estimate of the model parameters. In the case of a uniform prior, this point also coincides with the maximum-a-posteriori estimate, provided that the maximum lies within the allowed prior range.

Parameter Estimation

In general, a model depends on more than one parameter, with the number of parameters defining the dimensionality of the parameter space. Among these parameters, some may cprrespond to be calibration factors, while others may simply not of direct interested. Thus, when we want to reduce the posterior probability to a lower dimensional subspace,keeping the parameters of interest $\{\theta_1, ..., \theta_j\}$ we marginalize over the nuisance parameters:

$$p(\theta_1, ..., \theta_j|D,\mathcal{M}) = \int ... \int p(\theta_1, ...\theta_n|D,\mathcal{M})\ \mathrm{d}\theta_{j+1}...\mathrm{d}\theta_n. \tag{4.6}$$

In typical scenarios, evaluating the previous quantities analytically is general not possible, which motivates the use of alternative numerical methods to approximate the posterior distribution.

Monte Carlo (MC) methods provide an approximation to the posterior distribution by drawing a set of random samples from it. In this way, the posterior distribution $p(\theta|D,\mathcal{M})$ can be approximated by a sum of N Dirac delta distributions:

$$p(\theta \mid D,\mathcal{M}) \simeq p_N(\theta \mid D,\mathcal{M}) = \frac{1}{N}\sum_{i=1}^{N} \delta^{(n)}(\theta - \theta_i). \tag{4.7}$$

A particularly useful method for sampling the posterior distribution, commonly used in cosmology, is Markov Chain Monte Carlo (MCMC). In this approach, a sequence of points $x_1, x_2, \ldots, x_n, \ldots$, called a Markov chain, is generated in parameter space. The probability of the next point x_{n+1} depends only on the current point x_n.

This dependence between consecutive samples allows the chain to explore the posterior distribution, spending more time in regions of higher probability density. The initial samples in the chain must be discarded because, before convergence, they are not representative of the posterior distribution. This initial phase is known as the burn-in period.

A simple algorithm that implements this procedure is the Metropolis–Hastings algorithm [62], which we describe below.

Algorithm 1: Metropolis-Hastings Algorithm

Step 1. Choose an initial point θ_0 in the allowed parameter space, which defines the starting position of the Markov chain.

Step 2. At the current point θ_i, propose a candidate θ^* from a symmetric proposal distribution $q(\theta^*|\theta_i)$, such as a multivariate Gaussian centred at the current position. Assuming a symmetric proposal, $q(\theta^*|\theta_i) = q(\theta_i|\theta^*)$. The posterior probability at the candidate point is then evaluated from the likelihood and the prior.

Step 3. Compute the ratio

$$r = \frac{p(\theta^*|D, \mathcal{M})}{p(\theta_i|D, \mathcal{M})}. \tag{4.8}$$

The candidate is accepted with probability $\min(1, r)$. Therefore, if $r \geq 1$, it is always accepted, while for $r < 1$ it is accepted with probability r. If the candidate is rejected, the chain remains at the current point, $\theta_{i+1} = \theta_i$.

Step 4. Repeat the procedure until the chain has sufficiently explored the posterior distribution. Convergence can be assessed by comparing independent chains started from different regions of the parameter space and applying a convergence diagnostic.

The main advantage of MCMC methods is that they allow posterior distributions to be explored without evaluating them over the entire parameter space.

Once the posterior has been properly sampled, the resulting chain can also be used to obtain marginalized distributions, parameter constraints and correlations between parameters. In this sense, MCMC provides a practical way of performing the integrals required in Bayesian inference.

These advantages come with limitations:

- The consecutive points of a Markov chain are correlated, meaning that the number of effectively independent samples is smaller than the total number of points generated.
- If the proposed steps size is too large, many points will be rejected, whereas a very small step size can produce a high acceptance rate but cause the chain to move slowly through parameter space.
- For posteriors that contains several modes, a chain may explore one region very well while rarely, or never, reaching another relevant region.

Once the posterior distribution has been sampled, constraints on the model parameters can be expressed through regions containing a given fraction of the posterior probability. Although in a Bayesian framework these are referred to as credible regions, we follow the common convention in the cosmology community and refer to them as confidence levels (CL).

It is often useful to present results through two-dimensional marginalized posterior distributions, since these provide a qualitative view of the relation between different parameters and may reveal correlations or degeneracies between them. After

marginalizing over all parameters except θ_i and θ_j, we define a region $R(r)$ such that it contains a fraction r of the corresponding joint marginalized posterior probability:

$$\iint_{R(r)} p(\theta_i, \theta_j \mid D, \mathcal{M}), d\theta_i, d\theta_j = r. \tag{4.9}$$

The boundary of $R(r)$ defines the contours commonly displayed in triangle plots. Typical choices are $r = 0.68$ and $r = 0.95$, corresponding to the 68% and 95% confidence regions, respectively. single parameter, the posterior is further marginalized over the remaining parameters, and the corresponding confidence interval is defined analogously by:

$$\int_{R(r)} p(\theta_i \mid D, \mathcal{M}), d\theta_i = r. \tag{4.10}$$

Information Criteria

To assess the performance of the different cosmological models in fitting the observational data, we make use of information criteria. In particular, we consider the Akaike Information Criterion (AIC), its corrected version ($\mathrm{AIC_c}$), and the Bayesian Information Criterion (BIC). These quantities allow models to be compared by balancing the quality of the fit against their number of free parameters. The resulting best-fit likelihood is used to compute the different information criteria.

The AIC is defined as:

$$\mathrm{AIC} = -2 \ln \mathcal{L}_{\max} + 2k, \tag{4.11}$$

where $\mathcal{L}_{\max}$ is the maximum likelihood and k is the number of free parameters. The first term accounts for the goodness of fit, while the second penalizes the introduction of additional parameters. For Gaussian likelihoods, $-2 \ln \mathcal{L}_{\max}$ differs from the minimum χ^2 only by an additive constant, which cancels when models are compared using the same data.

For a small number of data points, the corrected AIC is given by:

$$\mathrm{AIC_c} = \mathrm{AIC} + \frac{2k(k+1)}{N_p - k - 1}, \tag{4.12}$$

where N_p denotes the total number of data points entering the analysis. In the limit $N_p \gg k$, the correction becomes negligible and $\mathrm{AIC_c} \simeq \mathrm{AIC}$.

The BIC introduces a different penalty for model complexity and is defined as:

$$\mathrm{BIC} = -2 \ln \mathcal{L}_{\max} + k \ln N_p. \tag{4.13}$$

Compared with the AIC, the BIC penalizes additional parameters more strongly as the size of the data set increases.

To compare an extended model with the reference ΛCDM model, we define:

$$\Delta\mathrm{IC} = \mathrm{IC}_{\mathrm{model}} - \mathrm{IC}_{\Lambda\mathrm{CDM}}, \tag{4.14}$$

where IC stands for AIC, $\mathrm{AIC_c}$, or BIC. Negative values indicate a preference for the extended model according to the corresponding criterion, while positive values

favor ΛCDM. Therefore, an improvement in the maximum likelihood alone does not necessarily imply a preference for the extended model, since the introduction of additional free parameters is explicitly penalized.

A relevant limitation arises when some of the additional parameters are only weakly constrained by the data. Such parameters still contribute to the total number of free parameters k and are therefore penalized by the information criteria. For this reason, they should be interpreted with some care when significant parameter degeneracies are present.

To provide a qualitative interpretation of the resulting differences, we follow the convention adopted in Ref. [63], based on Jeffreys' scale. The corresponding thresholds used for ΔAIC and ΔBIC are summarized in Table 4.1.

ΔAIC/ΔBIC	Interpretation
> 10	Decisively disfavored
$5 \sim 10$	Strongly disfavored
$2 \sim 5$	Moderately disfavored
$-2 \sim 2$	Compatible
$-5 \sim -2$	Moderately favored
$-10 \sim -5$	Strongly favored
< -10	Decisively favored

Table 4.1: Differences in ΔAIC and ΔBIC and the respective qualitative interpretation according to Jeffreys's scale. Negative values of ΔAIC and ΔBIC indicate that, for the specific data, the model is preferred over ΛCDM, while positive values indicate the opposite .

Methodology

For the numerical implementation of the Bayesian analysis, we employ the Cobaya package [64]. The posterior distributions are sampled using its Metropolis–Hastings MCMC sampler. The MCMC runs are performed in single-chain mode, with convergence assessed using the generalized Gelman–Rubin $R-1$ statistic implemented in Cobaya, adopting the criterion $R-1 < 0.01$, following the methodology described in Ref. [65]. The posterior distributions are analyzed and visualized using the GetDist package [66]. For the model-selection analysis, the maximum-likelihood values are obtained with the minimizer implemented in Cobaya, using its default BOBYQA algorithm [67–69].

4.2 Cosmological Parameters

Our analysis is performed only at the background level, therefore, the relevant equation Friedmann Eq. (2.30). By assuming a spatially flat Universe and treating the present-day matter density $\Omega_{m,0}$ as a free parameter, we can use Eq. (2.34) to constrain the DE density parameter:

$$\Omega_{\mathrm{de},0} = 1 - \Omega_{\mathrm{m},0} - \Omega_{\mathrm{r},0}, \tag{4.15}$$

. Assuming fixed neutrino properties and the standard effective number of relativistic species, we compute $\Omega_{\mathrm{r},0}$ as a derived parameter through matter–radiation equality:

$$\Omega_{\mathrm{r},0} = \frac{\Omega_{\mathrm{m},0}}{1 + z_{\mathrm{eq}}}, \tag{4.16}$$

where $z_{\mathrm{eq}} = 2.5 \times 10^4 \Omega_{\mathrm{m},0} h^2 \left(\frac{T_{\mathrm{cmb}}}{2.7}\right)$, $T_{\mathrm{cmb}} = 2.7255$ [70, 71].

The remaining quantity in the Friedmann equation is the Hubble constant H_0, which we also consider a free parameter.

In our analysis, we explore the parameter space described by the vector:

$$\theta = \{H_0; \Omega_{\mathrm{m},0}; \Omega_{\mathrm{b},0}; \Xi\}, \tag{4.17}$$

where Ξ denotes the set of extended parameters relative to ΛCDM.

4.3 Probes and Data

Astronomical observations can serve many purposes; in this work, we use them to constrain the parameters of our cosmological models. A comprehensive review of the different cosmological measurements currently available can be found in Ref. [26].

The probes used in this work are complementary, with each providing a different type of information. The CMB probes the early Universe around the epoch of recombination. BAO measurements are obtained from the distribution of galaxies at low redshift, but their constraining power comes from the comoving sound horizon at the drag epoch:

which is set by pre-recombination physics. Type Ia supernovae trace the luminosity distance–redshift relation and therefore probe the late-time expansion history. Lastly, cosmic chronometers provide a model-independent measurement of the Hubble parameter, requiring only the assumption of an FLRW background. The specific data sets used for each probe, together with the corresponding methodology, are described in the following sections.

4.3.1 Cosmic Microwave Background

According to the Hot Big Bang model, the early Universe was in a hot and dense state. As it expanded and cooled, it became energetically favorable for free electrons to combine mainly with protons, forming neutral hydrogen atoms. This process is known as recombination. Before this epoch, photons and baryons were tightly coupled through Thomson scattering, behaving as a single baryon–photon fluid that underwent through acoustic oscillations driven by the interplay between the gravitational attraction and the photon pressure. These oscillations later became imprinted in the CMB.

As recombination proceeded, the density of free electrons decreased and photons decoupled from matter, after which they propagated almost freely through the Universe. The corresponding last-scattering surface is located at $z_{\mathrm{cmb}} \simeq 1090$. These photons

are observed today as the CMB, with a temperature $T_{\rm cmb} = 2.7255\,\mathrm{K}$ [71]. The temperature anisotropies measured across the sky therefore contain information about both the physical conditions of the early Universe and its subsequent expansion history. Fig. 2.3 shows the CMB temperature anisotropies measured by the *Planck* satellite.

Although the full temperature and polarization power spectra depend on a wide range of effects, their sensitivity to the background expansion history can be encoded using a small number of geometric quantities. In particular, distance priors based on the angular scale of the sound horizon at decoupling preserve much of the CMB constraining power on the background expansion, while remaining applicable to a wide class of cosmological models. We employ the compressed Planck 2018 distance priors [72]. The likelihood is described by three quantities : the physical baryon density $\omega_{\rm b,0} \equiv \Omega_{\rm b,0}h^2$, the shift parameter R, and the acoustic angular scale ℓ_a:

$$R = \sqrt{\Omega_{\rm m,0}H_0^2}\,\frac{D_M(z_{\rm cmb})}{c}, \tag{4.18}$$

$$\ell_a = \pi\frac{D_M(z_{\rm cmb})}{r_s(z_{\rm cmb})}, \tag{4.19}$$

where $D_M(z)$ is the angular comoving distance, defined in Eq. (2.37), and $r_s(z)$ is the comoving sound horizon [73].

Using $H(z) = H_0E(z)$, the sound horizon defined in Eq. (3.1) can be rewritten in a form that makes its dependence on H_0 explicit:

$$r_s(z) = \frac{1}{H_0}\int_z^\infty \frac{c\,dz'}{E(z')\sqrt{3\left[1+\bar{R}_b/(1+z')\right]}}, \qquad c_s(z) = \frac{c}{\sqrt{3\left[1+\bar{R}_b/(1+z)\right]}}. \tag{4.20}$$

with $\bar{R}_b = 31500\Omega_{\rm b,0}h^2(T_{\rm cmb}/2.7K)^{-4}$. The decoupling redshift is evaluated using the fitting formula [74]:

$$z_{\rm cmb} = 1048\left[1+0.00124\,\omega_{\rm b}^{-0.738}\right]\left[1+g_1\omega_{\rm m}^{g_2}\right], \tag{4.21}$$

with $\omega_{\rm m} = \Omega_{\rm m,0}h^2$ and:

$$g_1 = \frac{0.0783\,\omega_{\rm b}^{-0.238}}{1+39.5\,\omega_{\rm b}^{0.763}}, \qquad g_2 = \frac{0.56}{1+21.1\,\omega_{\rm b}^{1.81}}. \tag{4.22}$$

For the Planck 2018 distance priors, the residual vector is [72]:

$$\Delta X_{\rm cmb} = \begin{pmatrix} R-1.74963 \\ \ell_a - 301.80845 \\ \omega_{\rm b} - 0.02237 \end{pmatrix}, \tag{4.23}$$

with covariance matrix:

$$\mathcal{C}_{\rm cmb} = 10^{-8}\begin{pmatrix} 1598.9554 & 17112.007 & -36.311179 \\ 17112.007 & 811208.45 & -494.79813 \\ -36.311179 & -494.79813 & 2.1242182 \end{pmatrix}. \tag{4.24}$$

The corresponding contribution to the likelihood is then written as:

$$\chi^2_{\rm cmb} = \Delta X^T_{\rm cmb} \mathcal{C}^{-1}_{\rm cmb} \Delta X_{\rm cmb}. \tag{4.25}$$

4.3.2 Standard Rulers - BAO

The CMB, as discussed previously, provides a picture of the Universe at recombination ($z \approx 1090$), encoding the physics of the baryon–photon plasma. The acoustic oscillations visible in the CMB angular power spectrum arise from the interplay between gravity pulling matter inward and radiation pressure in the baryon–photon fluid pushing it outward. However, these oscillations did not leave their imprint only in the photon distribution. Shortly after last scattering, at the drag epoch ($z_{\rm drag} \approx 1060$), baryons finally decoupled from the photon field, and the same acoustic physics that shaped the CMB anisotropies also imprinted the spatial distribution of baryonic matter. The pressure that had been driving the sound waves vanished, "freezing" the acoustic shell at a characteristic radius of roughly 150 Mpc.

Because this comoving scale is set entirely by well-understood pre-recombination physics and can be computed from first principles, BAO serve as a standard ruler for cosmological distance measurements. To implement this standard ruler in our analysis, we use $r_{\rm drag} \equiv r_s(z_{\rm drag})$, with $r_s(z)$ defined in Eq. (3.1). Since the models considered in this thesis modify only the late-time expansion history, we assume standard early-time physics and evaluate $r_{\rm drag}$ using the fitting formula from Ref. [35]:

$$r_{\rm drag} = 147.05 \left(\frac{\Omega_{\rm b,0} h^2}{0.02236}\right)^{-0.13} \left(\frac{(\Omega_{\rm b,0} + \Omega_{\rm c,0}) h^2}{0.1432}\right)^{-0.23} \text{ Mpc}, \tag{4.26}$$

where $\Omega_{\rm c,0}$ represents the cold dark matter density. $r_{\rm drag}$ is scaled to the best-fit values obtained from Planck 2018 [24].

There are two distinct ways in which BAO measurements can provide cosmological constraints. First, the BAO scale transverse to the line of sight constrains the angular diameter distance through:

$$D_A(z) = \frac{r_{\rm drag}}{\Delta\beta (1+z)}, \tag{4.27}$$

where $\Delta\beta$ is the angle under which the BAO shell is observed. Equivalently, surveys report this measurement in terms of the comoving distance, so that the dimensionless observable is the ratio:

$$\frac{D_M(z)}{r_{\rm drag}} = \frac{c}{r_{\rm drag}\ H_0} \int_0^z \frac{{\rm d}z'}{E(z')}. \tag{4.28}$$

Second, parallel to the line of sight, the BAO scale is sensitive to the expansion rate of the Universe, and the corresponding dimensionless observable is:

$$\frac{D_H(z)}{r_{\rm drag}} = \frac{c}{H(z)\ r_{\rm drag}} = \frac{c}{H_0\ r_{\rm drag}} \frac{1}{E(z)}. \tag{4.29}$$

When the data at a given redshift bin do not have sufficient statistical power to separate the radial and transverse BAO features, the two measurements are combined into a single volume-averaged distance:

$$D_V(z) = \left[z D_M^2(z) D_H(z) \right]^{1/3} \tag{4.30}$$

For the BAO analysis, we use measurements from the second data release of the Dark Energy Spectroscopic Instrument (DESI DR2) [35], covering the effective-redshift range $0.295 \leq z_{\rm eff} \leq 2.330$. The data contain both isotropic and anisotropic BAO measurements and are expressed in terms of the previously mentioned dimensionless quantities $\frac{D_M(z)}{r_{\rm drag}}$, $\frac{D_H(z)}{r_{\rm drag}}$ and $\frac{D_V(z)}{r_{\rm drag}}$, depending on the corresponding tracer and redshift range.

We define goodness of fit to the BAO measurements using the chi-squared statistics:

$$\chi^2_{\rm BAO} = \Delta X^T_{\rm BAO} C^{-1}_{\rm BAO} \Delta X_{\rm BAO}, \tag{4.31}$$

where the residual vector is defined as:

$$\Delta X_{\rm BAO} \equiv X^{\rm obs}_{\rm BAO} - X^{\rm th}_{\rm BAO}, \tag{4.32}$$

and $C_{\rm BAO}$ denotes the covariance matrix supplied with the DESI DR2 measurements. The full covariance information is included in the analysis, accounting for correlations between the different BAO observables and redshift measurements.

BBN

BAO measurements alone do not directly determine directly H_0. Since the BAO observables are ratios of distances to $r_{\rm drag}$, and distances scale as c/H_0, the data constrain only the combination $r_{\rm drag} H_0$.

Additional information about the Universe is required to calibrate the sound horizon. That can be accomplished using information from the CMB or through Big Bang Nucleosynthesis (BBN). The latter relies on measurements of the primordial abundances of light elements, particularly helium and deuterium, which are sensitive to the baryon-to-photon ratio and, consequently, to the physical baryon density $\omega_{\rm b}$.

Since the models considered in this work modify only the late-time DE sector, we assume the standard BBN scenario and keep the early-time physics unchanged throughout the analysis.

We therefore introduce $\Omega_{\rm b,0}$ as an additional free parameter and include the corresponding BBN likelihood in our analysis. In particular, we adopt the constraint [75]:

$$\Omega_{\rm b,0} h^2 = 0.02233 \pm 0.00036, \tag{4.33}$$

4.3.3 Standard Candles - Type Ia supernovae

Type Ia supernovae are thermonuclear explosions of white dwarfs in binary systems. Although their progenitor channels remain under investigation, empirical light-curve and color corrections make their luminosities sufficiently uniform for them to be used as standardizable candles.

The cosmological information provided by SNe Ia is encoded in the luminosity distance D_L, defined in Eq. (2.43). The observed brightness of a supernova is described by its apparent magnitude m_B, while its intrinsic luminosity is related to the absolute magnitude M. These quantities are related through the distance modulus:

$$\mu_{\rm SN} = m_B - M. \tag{4.34}$$

We use the Pantheon+ compilation of Type Ia supernovae [76], which contains 1701 light curves from 1550 spectroscopically confirmed SNe Ia, spanning the redshift range $0.001 \leq z \leq 2.26$.

For a supernova in the Hubble flow, the theoretical distance modulus is:

$$\mu_{\rm SN}^{\rm th}(z_{\rm hel}, z_{\rm cmb}) = 5\log_{10}\left[\frac{D_L(z_{\rm hel}, z_{\rm cmb})}{\rm Mpc}\right] + 25, \tag{4.35}$$

where $z_{\rm hel}$ and $z_{\rm cmb}$ are the heliocentric and CMB-frame redshifts, respectively. The luminosity distance is given by:

$$D_L(z_{\rm hel}, z_{\rm cmb}) = \frac{c}{H_0}(1+z_{\rm hel})\int_0^{z_{\rm cmb}} \frac{dz'}{E(z')}. \tag{4.36}$$

In this thesis, we consider two versions of the supernova likelihood : Pantheon+ without an external calibration, abbreviated as PP, and Pantheon+ including the SH0ES Cepheid calibration, abbreviated as PPS.

The difference in in the implementation of the two likelihoods is that PP provides only relative distances, while the SH0ES calibration included in PPS fixes M.

Pantheon+ without SH0ES.

Without an external calibration, the absolute magnitude M is degenerate with H_0, and the two quantities cannot be determined independently from the supernova sample alone. It is therefore convenient to introduce the Hubble-free luminosity distance:

$$d_L(z_{\rm hel}, z_{\rm cmb}) \equiv \frac{H_0}{c} D_L(z_{\rm hel}, z_{\rm cmb}), \tag{4.37}$$

and absorb the absolute normalization into the nuisance parameter:

$$\widetilde{\mathcal{M}} \equiv M + 5\log_{10}\left(\frac{c/H_0}{\rm Mpc}\right) + 25. \tag{4.38}$$

The residual vector is:

$$\Delta\mu_{\rm PP}(\widetilde{\mathcal{M}}) = m_B - 5\log_{10}(d_L) - \widetilde{\mathcal{M}}. \tag{4.39}$$

We marginalize over $\widetilde{\mathcal{M}}$, obtaining [77]:

$$\widetilde{\chi}^2_{\rm PP} = A_{\rm PP} + \ln\left(\frac{E_{\rm PP}}{2\pi}\right) - \frac{B^2_{\rm PP}}{E_{\rm PP}}, \tag{4.40}$$

where $A_{\rm PP} = \Delta\widetilde{\mu}^T_{\rm PP}\cdot C^{-1}_{\rm PP}\cdot\Delta\widetilde{\mu}_{\rm PP}$, $B_{\rm PP} = \Delta\widetilde{\mu}^T_{\rm PP}\cdot C^{-1}_{\rm PP}\cdot\mathbf{1}$, $E_{\rm PP} = \mathbf{1}^T\cdot C^{-1}_{\rm PP}\cdot\mathbf{1}$, and $\Delta\widetilde{\mu}_{\rm PP} = \Delta\mu_{\rm PP}(\widetilde{\mathcal{M}} = 0)$.

Pantheon+ with SH0ES.

When the SH0ES Cepheid calibration is included, the distance ladder fixes the absolute luminosity scale. The supernova absolute-magnitude nuisance parameter is already included in the data treatment and is therefore not marginalized over in the likelihood.

Following the Pantheon+ & SH0ES treatment, the theoretical quantity entering the likelihood is written as:

$$\mu^{\rm th}_{{\rm PPS},i} = \begin{cases} \mu_i^{\rm ceph}, & \text{Cepheid-host SNe,} \\ 5\log_{10}\left[\frac{D_L(z_{{\rm hel},i}, z_{{\rm cmb},i})}{\rm Mpc}\right] + 25, & \text{Hubble-flow SNe.} \end{cases} \tag{4.41}$$

For Cepheid-host supernovae, the cosmological distance modulus is therefore replaced by the measured Cepheid-based host distance modulus, while for Hubble-flow supernovae we use the cosmological prediction defined above.

The residual vector is:

$$\Delta\mu_{\rm PPS} \equiv \mu^{\rm obs}_{\rm PPS} - \mu^{\rm th}_{\rm PPS}, \tag{4.42}$$

and the corresponding goodness of fit is:

$$\chi^2_{\rm PPS} = \Delta\mu^T_{\rm PPS} C^{-1}_{\rm PPS} \Delta\mu_{\rm PPS}. \tag{4.43}$$

The covariance matrix $C_{\rm PPS}$ provided with the Pantheon+ & SH0ES compilation includes the statistical and systematic uncertainties, correlations between supernovae, and the information associated with the Cepheid calibration.

4.3.4 Cosmic Chronometers

Cosmic chronometers (CC) provide a direct and cosmology-independent measurements of the Hubble parameter. The method, exploits the fact that, by assuming an FLRW background, the expansion rate $H(z)$ can be written as:

$$H(z) = -\frac{1}{1+z}\frac{\mathrm{dz}}{\mathrm{dt}}, \tag{4.44}$$

where dt is the differential ageing of the Universe across a redshift interval dz. This technique involves no further cosmological assumptions beyond the FLRW metric.

The practical challenge lies in measuring dt reliably. The most suitable tracers are very massive, passively evolving galaxies, which formed the bulk of their stellar mass in short bursts at high redshift and have evolved without significant further star formation.

Unlike SNe and BAO, which constrain distance quantities involving integrals of the expansion history, cosmic chronometers provide direct measurements of $H(z)$ at discrete redshifts. This makes them particularly useful for probing localized features in the late-time expansion history.

We use the compilation of $N_{\rm CC} = 33$ cosmic chronometer measurements presented in Ref. [78], spanning approximately $0.07 \lesssim z \lesssim 2.0$. Each data point is specified by (z_i, H_i, σ_{H_i}), where H_i is the measured expansion rate at redshift z_i and σ_{H_i} is the corresponding uncertainty. The theoretical prediction, $H_{\rm th}(z_i)$, is obtained from the background evolution of the model.

Assuming the measurements to be uncorrelated, the corresponding chi-squared statistic is:

$$\chi^2_{\rm CC} = \sum_{i=1}^{N_{\rm CC}} \frac{[H_i - H_{\rm th}(z_i)]^2}{\sigma^2_{H_i}}. \tag{4.45}$$

4.4 Dataset combinations

Having introduced the individual cosmological probes, we now summarize the dataset combinations used in the analysis and discussed in Section 5.5. We also report the total number of data points, N_p, used in the computation of the AIC_c and BIC criteria. The BBN constraint on $\Omega_{\mathrm{b},0}h^2$ is implemented as an additional Gaussian likelihood term and therefore contributes one data point to N_p.

Dataset combination	$\boldsymbol{N_p}$
CMB	4
BAO	14
BAO+CMB	17
PPS	1702
PPS+BAO+CMB	1718
PP+CC	1735
PP+CC+BAO+CMB	1751

Table 4.2: Dataset combinations used in the analysis and their corresponding number of data points N_p. The BBN constraint is included in all combinations.

The combinations are chosen to assess the impact of different cosmological probes on the model parameters. In particular, we distinguish between analyses that include the SH0ES calibration (PPS) and those that do not (PP), to evaluate the impact of the local H_0 anchor on the parameter constraints. Cosmic chronometers are included as a complementary probe of the expansion rate in the combinations that use PP. For the $\Lambda_{\rm OI}$CDM family, the full set of combinations is explored; for $\Lambda_{\rm osc}$CDM, we restrict the analysis to a representative subset.

Poesia visual como miragens. Desenho imagens, começo a escrever imagens e nem sequer noto a passagem.
Assim, tentei guardar o melhor para o fim da jornada em que, por fim, provei que eu não troco a ordem das palavras.

— Nerve

5 The models

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Among late-time proposals to address the tensions we currently observe, increasing attention has recently been directed towards DE models whose energy density can change sign during the cosmic evolution.

This possibility was first explored within the graduated dark energy (gDE) model [27], in which the cosmological constant Λ is replaced by a dynamical DE density described by a smooth function that can switch sign at a certain redshift, becoming positive at late times.

Motivated by gDE, the abrupt Λ_sCDM model was subsequently introduced as the simplest realization of a sign-switching cosmological constant [79]. In this model, the cosmological constant is replaced by an effective DE density that undergoes an abrupt transition from negative (AdS-like) to positive (dS-like) values at low redshifts .

Despite its simple phenomenological nature, the abrupt Λ_sCDM model has been shown to alleviate both the H_0 and S_8 tensions [28]. From a theoretical perspective, a negative cosmological constant is well motivated in high-energy theory, where AdS spacetimes play a central role in string theory and in the AdS/CFT correspondence [80]. By contrast, obtaining controlled de Sitter vacua in string theory is generally more challenging and remains an active subject of research [81]. These considerations

provide additional motivation for exploring an AdS-like phase phenomenologically, rather than a fundamental derivation of the sign-switching scenario. Interestingly, recent model-independent reconstructions of the cosmic expansion have also reported hints of a sign-changing DE density, with a transition occurring near $z \sim 2$ [82].

The abrupt Λ_sCDM model can be regarded as the limiting case of a very rapid transition. For this reason, smooth realizations of AdS-to-dS-like transitions have also been studied in the literature [30, 34], replacing the instantaneous sign change with a continuous evolution of the DE density.

Additionally, recent analyses employing wavelet extensions of ΛCDM have shown that the effective DE density oscillates by construction and can become negative at redshifts $z \gtrsim 2$ in response to BAO data [83].

Motivated by these developments, we first briefly review abrupt Λ_sCDM and then introduce two new families of smooth sign-switching models, Λ_{osc}CDM and Λ_{OI}CDM.

Following the convention introduced in Section 2.2.4, we write the DE density as:

$$\rho_{de}(z) = \rho_{de,0}\, \varrho_{de}(z), \tag{5.1}$$

where $\varrho_{de}(z)$ denotes its dimensionless normalized evolution, with $\varrho_{de}(0) = 1$.

5.1 Abrupt sign-switching Λ_sCDM model

We start by reviewing th abrupt Λ_sCDM model. This model,introduces an additional parameter, $z_\dagger$, which determines the transition from a negative cosmological constant to a positive one. The model is characterized by a DE component that evolves as [79]:

$$\rho_s(z) = \rho_{s,0}\, \mathrm{sgn}(z_\dagger - z), \qquad \text{where} \quad \mathrm{sgn}(u) = \begin{cases} -1 & \text{for } u < 0, \\ 0 & \text{for } u = 0, \\ 1 & \text{for } u > 0. \end{cases} \tag{5.2}$$

As previously mentioned, the DE density behaves as a negative cosmological constant before undergoing an abrupt sign-switch at $z_\dagger$. After the transition, the model reduces to ΛCDM.

Using Eq. (2.32) under the same assumptions, we obtain the corresponding EoS parameter:

$$w_{de,s}(z) = -1 - \frac{2(1+z)}{3} \frac{\delta(z_\dagger - z)}{\mathrm{sgn}(z_\dagger - z)}. \tag{5.3}$$

Here, $\delta(z_\dagger - z)$ denotes the Dirac delta function. From the previous expression, one can see that the EoS is well defined at every redshift except at the transition point $z_\dagger$.

In Appendix A, we examine the singularities that arise in sign-switching models. Following the classification of Refs. [84, 85], we show that smoothing the sign change of the cosmological constant removes the Type-II (sudden) singularity present in Λ_sCDM, replacing it with the milder Type-V, or w-singularity.

5.2 Oscillatory sign-switching $\Lambda_{\rm osc}$CDM

These models contain one additional parameter relative to $\Lambda_{\rm s}$CDM, namely K, which controls both the smoothness of the transition and the oscillatory behavior. Although describing both features with a single parameter is advantageous from the point of view of model simplicity, it also introduces a degeneracy between them : as K increases, the transition becomes faster and the number of oscillations increases.

For the same value of K, the models exhibit oscillatory behavior around $\Lambda(z) = -1$ before the transition, followed by a monotonic evolution near $z_\dagger$, and subsequently oscillate around $\Lambda(z) = 1$. The main purpose of these models is to investigate the effect of the oscillations on our parameters. To this end, we introduce three models that, for the same value of K, exhibit a similar transition rate around $z_\dagger$ but differ in the number of oscillations after the transition.

For convenience, we define the arguments of the sine Integral for the three models as:

$$u_z(z) = A_1 K(z_\dagger - z), \tag{5.4}$$

$$u_a(z) = A_2 K\left(a_\dagger - \frac{1}{1+z}\right), \tag{5.5}$$

$$u_x(z) = A_3 K\left(x_\dagger + \ln(1+z)\right), \tag{5.6}$$

where A_1, A_2, and A_3 are fixed such that the three models exhibit a similar transition sharpness for the same value of K.

$\Lambda^{\mathbf{z}}_{\mathbf{osc}}$CDM

Originally defined in redshift space, the DE density of this model is described by the sine Integral [1]:

$$\rho^z_{\rm osc}(z) = \frac{\rho^z_{\rm osc,0}}{N}\mathrm{Si}\Big(u_z(z)\Big). \tag{5.7}$$

Here, $A_1 = 0.64$, while N is a normalization factor, $N = \mathrm{Si}\Big(A_1 K z_\dagger\Big)$. Imposing Eq. (2.32), one obtains[2]:

$$w^z_{\rm osc}(z) = -1 - \frac{A_1 K(1+z)}{3}\frac{\mathrm{sinc}\Big(u_z(z)\Big)}{\mathrm{Si}\Big(u_z(z)\Big)}. \tag{5.8}$$

Due to the form of the sine integral entering the DE density, the EoS exhibits oscillatory behavior around $w = -1$ throughout most of the cosmological evolution, except near the sign-switching point $z = z_\dagger$, where the DE density vanishes and the EoS of the DE becomes ill-defined.

As K increases, the transition becomes progressively sharper and approaches the abrupt $\Lambda_{\rm s}$CDM sign-switch. However, the oscillations close to the transition do not disappear completely. Instead, they become confined to an increasingly narrow region around $z = z_\dagger$, while the first overshoot approaches a finite value of approximately 9%.

$\Lambda^{\mathbf{a}}_{\mathbf{osc}}$CDM

[1] The sine Integral is defined as $\mathrm{Si}(u) = \int_0^u \frac{\sin(y)}{y}\,dy$.

[2] The sinc function is defined as $\mathrm{sinc}(u) = \frac{\sin u}{u}$, with $\lim_{u\to 0}\mathrm{sinc}(u) = 1$.

Originally defined in terms of the scale factor, the energy density of this model can be written as:

$$\rho^a_{\rm osc}(z) = \frac{\rho^a_{\rm osc,0}}{N}\mathrm{Si}\Big(u_a(z)\Big), \tag{5.9}$$

Here, $a_\dagger = \frac{1}{1+z_\dagger}$ and $A_2 = 10$. Imposing Eq. (2.32), one obtains:

$$w^a_{\rm osc}(z) = -1 + \frac{A_2 K}{3(1+z)}\frac{\mathrm{sinc}\Big(u_a(z)\Big)}{\mathrm{Si}\Big(u_a(z)\Big)}. \tag{5.10}$$

Among the three models, $\Lambda^a_{\rm osc}$CDM exhibits the highest number of oscillations for the same value of K, with its energy density oscillating rapidly after the transition.

As before, for $K \to \infty$ the model approaches an abrupt $\Lambda_{\rm s}$CDM-like behavior.

The same does not apply when only the limit $z_\dagger \to \infty \implies a_\dagger \to 0$ is considered. In this case, the energy density takes the form:

$$\rho^a_{\rm osc}(z) = \rho^a_{\rm osc,0}\frac{\mathrm{Si}\Big(\frac{A_2 K}{1+z}\Big)}{\mathrm{Si}\Big(A_2 K\Big)}, \tag{5.11}$$

and the ΛCDM behavior is recovered only when the additional limit $K \to \infty$ is also imposed.

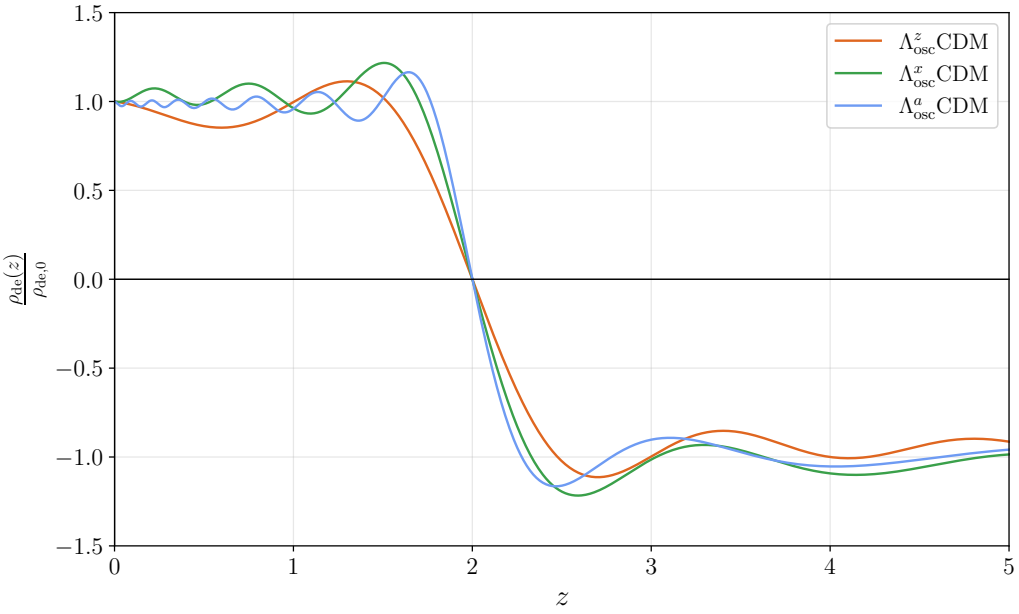


Figure 5.1: Evolution of the normalized DE density for the models of the $\Lambda_{\rm osc}$CDM family, assuming $K = 7$ and $z_\dagger = 2$.

$\Lambda^{\rm x}_{\rm osc}$CDM

Originally defined in terms of $x = \ln(a) = -\ln(1+z)$, the energy density of this model can be written as:

$$\rho^x_{\rm osc}(z) = \frac{\rho^x_{\rm osc,0}}{N}\mathrm{Si}\Big(u_x(z)\Big). \tag{5.12}$$

Here, $x_\dagger = -\ln(1+z_\dagger)$ and $A_3 = 2.5$. Imposing Eq. (2.32), one obtains:

$$w^x_{\rm osc}(z) = -1 + \frac{A_3 K}{3}\frac{\mathrm{sinc}\Big(u_x(z)\Big)}{\mathrm{Si}\Big(u_x(z)\Big)}. \tag{5.13}$$

Compared with $\Lambda^z_{\rm osc}$CDM, the energy density of $\Lambda^x_{\rm osc}$CDM oscillates more rapidly after the transition. Here, $z_\dagger \to \infty$, corresponds to $x_\dagger \to -\infty$, and the ΛCDM behavior is recovered, while for $K \to \infty$ the model approaches an abrupt $\Lambda_{\rm s}$CDM-like behavior.

Thus, while keeping the same number of free parameters, the choices of A_1, A_2, and A_3 allow us to compare the effect of increasing the number of oscillations in sign-switching DE models.

Figure 5.1 illustrates the different behaviors of the three models for the same parameter values, $K = 7$ and $z_\dagger = 2$.

5.3 Oscillatory sign-switching $\Lambda_{\rm OI}$CDM

The second family of models is complementary to the one introduced above. For intermediate values of K, instead of exhibiting oscillatory behavior mainly before and after the transition from $\Lambda(z) = -1$ to $\Lambda(z) = 1$, these models exhibit oscillations during the transition itself.

These models are written as the sum of a monotonically evolving term, given by the Erf function, and an additional term responsible for the oscillatory behavior. Whenever possible, the expressions will be written in terms of the variable $u = K(x_\dagger - x)$, although the precise meaning of $x_\dagger$ differs between the models.

Semi-Wave DE model ($\Lambda^{\rm SW}$CDM)

The first model has an overall monotonic evolution, while the additional term makes the transition slower than that obtained from the error function [3] (Erf) alone.

Its energy density can be written as:

$$\rho_{\rm sw}(x) = \frac{\rho_{\rm sw,0}}{N}\left(\mathrm{Erf}(u) - \frac{\sin(u)}{(1+u^2)^2}\right), \qquad u = K(x_\dagger - x). \tag{5.14}$$

Here, $x_\dagger$ corresponds to the crossing point at which the DE density changes from negative to positive. Imposing Eq. (2.32), one obtains:

$$w_{\rm sw}(z) = -1 - \frac{K\left(\frac{\cos(u)}{(u^2+1)^2} - \frac{4u\sin(u)}{(u^2+1)^3} - \frac{2e^{-u^2}}{\sqrt{\pi}}\right)}{3\left(\mathrm{Erf}(u) - \frac{\sin(u)}{(1+u^2)^2}\right)}. \tag{5.15}$$

Triple-Transition DE model ($\Lambda^{\rm TT}$CDM)

[3] The error function, is defined as $\mathrm{Erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-y^2}\,dy$.

In this model, the DE density crosses $\Lambda(x) = 0$ three times during its oscillatory evolution. The corresponding DE density can be written as:

$$\rho_{\rm tt}(x) = \frac{\rho_{\rm tt,0}}{N}\left(\mathrm{Erf}(u) - \frac{\sin(2u)}{(1+u^2)^4}\right), \qquad u = K(x_{2\dagger} - x). \tag{5.16}$$

Here, $x_{2\dagger}$ corresponds to the second transition, where $\Lambda(x) = 0$. Imposing Eq. (2.32), one obtains:

$$w_{\rm tt}(z) = -1 - \frac{2K\left(\frac{\cos(2u)}{(u^2+1)^4} - \frac{4u\sin(2u)}{(u^2+1)^5} - \frac{e^{-u^2}}{\sqrt{\pi}}\right)}{3\left(\mathrm{Erf}(u) - \frac{\sin(2u)}{(1+u^4)^2}\right)}. \tag{5.17}$$

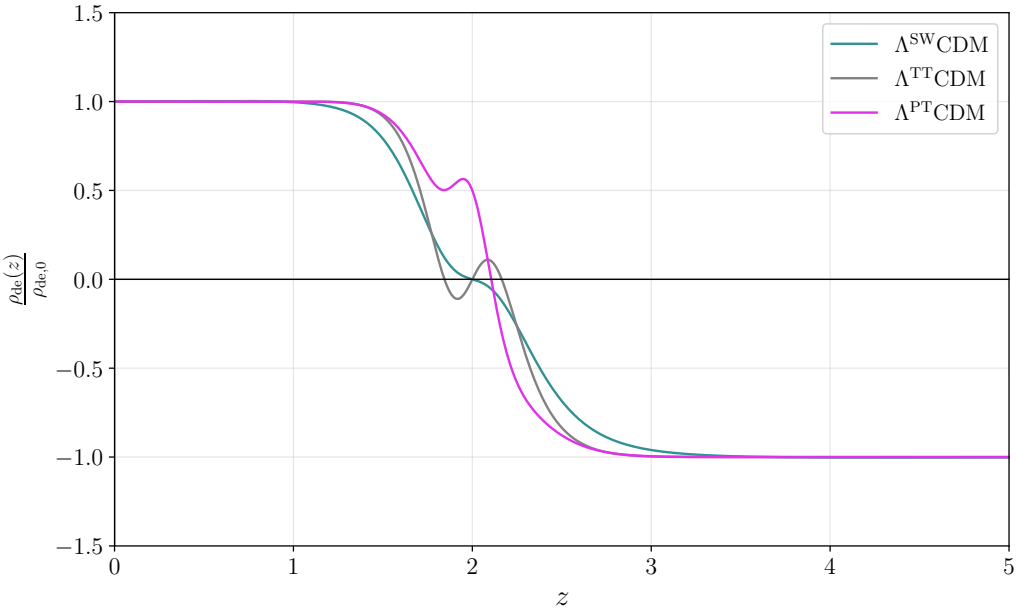


Figure 5.2: Evolution of the normalized DE density for the models of the $\Lambda_{\rm OI}$CDM family, assuming $K = 7$ and $z_\dagger = 2$.

Post-Transition DE model ($\Lambda^{\rm PT}$CDM)

For this model, the overall DE density increases monotonically, while the oscillations appear once the energy density becomes positive, motivating the name Post-Transition DE. The corresponding energy density can be written as:

$$\rho_{\rm pt}(x) = \frac{\rho_{\rm pt,0}}{N}\left(\mathrm{Erf}(u) - \frac{A_4\cos(3u)}{(1+u^2)^6}\right), \qquad u = K(x_\star - x). \tag{5.18}$$

Here, the meaning of $x_\star$ is slightly different from that of the sign-switching parameter $x_\dagger$. At $u = 0$,the $\cos(u)$ term does not vanish, and therefore $x_\star$ does not coincide exactly with $x_\dagger$ for small values of K. As a result, the transition occurs earlier than the value indicated by $x_\dagger$. However, as K increases, $x_\star \to x_\dagger$, revealing a degeneracy between $x_\star$ and K in determining the effective transition point. Imposing Eq. (2.32), one obtains:

$$w_{\rm pt}(z) = -1 + \frac{K\left(\frac{2e^{-u^2}}{\sqrt{\pi}} + \frac{3A_4\sin(3u)}{(1+u^2)^6} + \frac{12A_4 u\cos(3u)}{(1+u^2)^7}\right)}{3\left(\mathrm{Erf}(u) - \frac{A_4\cos(3u)}{(1+u^2)^6}\right)}. \tag{5.19}$$

Throughout our analysis, whenever the distinction is not relevant, we will denote both $z_\star$ and $z_{2\dagger}$ by $z_\dagger$ in order to avoid overloading the notation. In Appendix A, where the precise location of the different transition points becomes important, the original notation will be retained. We choose $A_4 = 0.5$

Figure 5.2 illustrates the different behaviors of the proposed models for identical parameter choices.

Even though $w_{\rm de}$ is ill-defined at the transition point for all the models introduced above, the corresponding total EoS parameter defined in Eq. (2.76) remains well defined. In Fig. 5.3, we show the evolution of the total EoS for the two model families, neglecting the radiation contribution.

For the $\Lambda_{\rm osc}$CDM family, the total EoS exhibits a strongly oscillatory behavior due to the form of the Si(u) function and its derivative. As K increases, the derivative term becomes increasingly important and eventually dominates the evolution, giving rise to the strong oscillatory behavior of the total EoS.

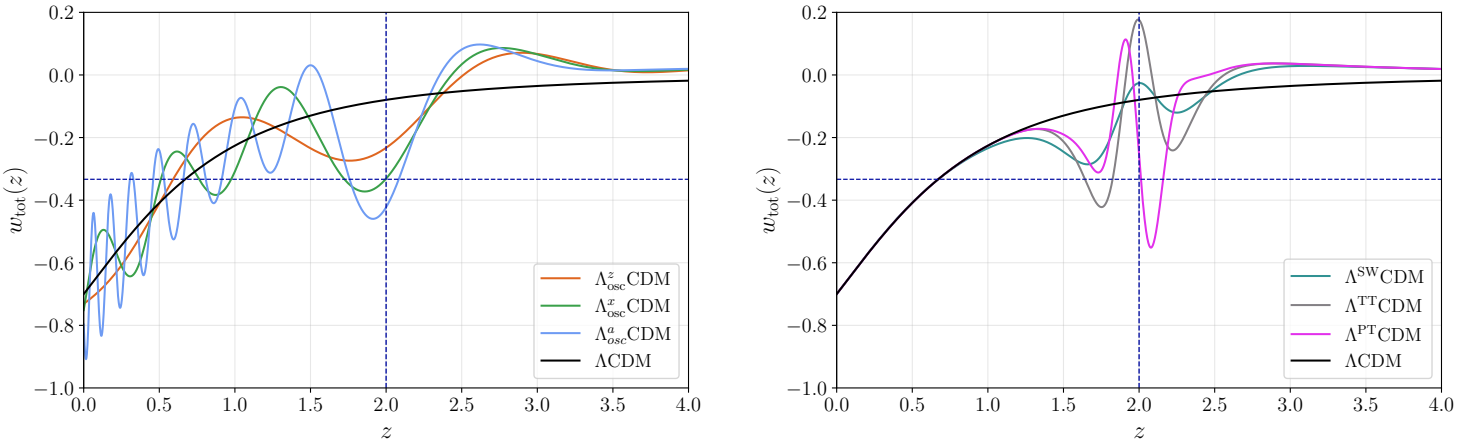


Figure 5.3: Evolution of the total EoS parameter, $w_{\rm tot}$, as a function of redshift. The vertical line denotes the transition redshift, $z_\dagger = 2$, while the horizontal line corresponds to $w_{\rm tot} = -1/3$.

5.4 Background Universe in Sign-Switching models

Some definitions commonly used in GR require particular care once the DE density, $\rho_{\rm de}(z)$, is allowed to become negative. In this section, we discuss how this possibility affects quantities commonly used to characterize the behavior of different cosmological components .

Three assumptions should be stated, as they restrict our analysis but are consistent with the conditions under which our models are defined. We restrict the analysis to a spatially flat background Universe. Furthermore, since our models

describe late-time modifications, the contribution from radiation, $\Omega_{\rm r}(z)$, is neglected throughout this section.

As seen in the models introduced above, the corresponding EoS parameter can become ill-defined. This raises the question of how phantom-like and quintessence-like behavior should be characterized when the DE density is allowed to change sign. We can rewrite Eq. (2.32) in terms of the scale factor as:

$$\frac{\mathrm{d}\rho_{\rm de}(a)}{\mathrm{d}\ln a} = -3\left(\rho_{\rm de} + p_{\rm de}\right) = -3(1+w_{\rm de})\rho_{\rm de}. \tag{5.20}$$

Once sign-changing energy densities are allowed, the usual boundary $w_{\rm de} = -1$ is no longer, by itself, a reliable criterion for distinguishing between phantom-like and quintessence-like regimes [86].

From the first equality in Eq. (5.20), we see that an evolution in which $\rho_{\rm de}$ increases as the Universe expands, corresponding to phantom-like behavior, is characterized by:

$$\rho_{\rm de} + p_{\rm de} < 0. \tag{5.21}$$

Since:

$$\rho_{\rm de} + p_{\rm de} = (1+w_{\rm de})\rho_{\rm de}, \tag{5.22}$$

this condition implies:

$$\rho_{\rm de} + p_{\rm de} < 0 \quad \Longleftrightarrow \quad \begin{cases} w_{\rm de} < -1, & \rho_{\rm de} > 0, \\ w_{\rm de} > -1, & \rho_{\rm de} < 0. \end{cases} \tag{5.23}$$

A direct consequence is that quantities whose interpretation usually relies on the EoS parameter must be reconsidered once the energy density is allowed to change sign. Let us now consider a single fluid. The Raychaudhuri equation, Eq. (2.19), reads:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p). \tag{5.24}$$

If the energy density is assumed to be positive and $p = w\rho$, one recovers the usual condition $w < -1/3$ for accelerated expansion. However, this criterion cannot be applied directly when the energy density is allowed to become negative. Instead, for a general fluid f, we introduce the quantity:

$$\mathcal{A}_f(z) = \rho_f(z) + 3p_f(z), \tag{5.25}$$

also known as the active gravitational mass density [87]. This quantity determines whether a given fluid contributes attractively or repulsively to the Raychaudhuri equation: when $\mathcal{A}_f > 0$, the fluid produces an attractive contribution, whereas when $\mathcal{A}_f < 0$ it produces a repulsive one.

Even if an individual fluid contributes repulsively, this does not necessarily imply that the system as a whole behaves repulsively. The contributions from all fluids must be taken into account through the total active gravitational mass density, $\mathcal{A}_{\rm tot} = \sum_f (\rho_f + 3p_f)$. Neglecting the contribution from radiation, $\mathcal{A}_{\rm tot}$ reduces to:

$$\mathcal{A}_{\rm tot} = \rho_{\rm m}(z) + \mathcal{A}_{\rm de}(z). \tag{5.26}$$

This quantity serves as a diagnostic of cosmic acceleration. For the models considered here, the deceleration parameter $q(z)$ introduced in Eq. (2.49), the total equation-of-state parameter $w_{\rm tot}(z)$, and $\mathcal{A}_{\rm tot}$ all yield the same qualitative description of the accelerating behavior.

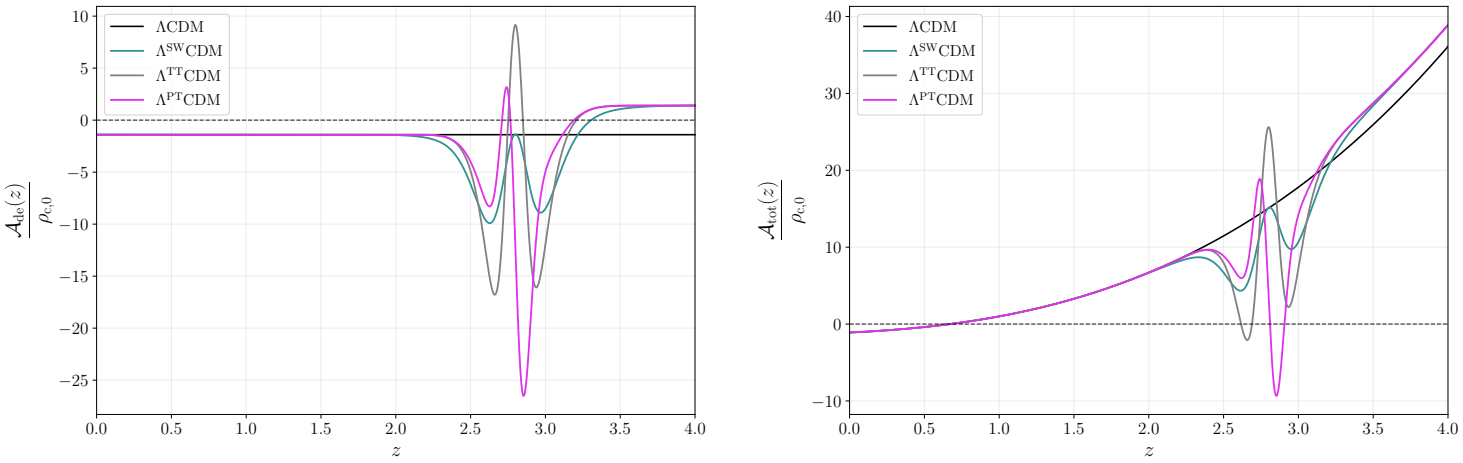


Figure 5.4: $\mathcal{A}_{\rm de}$ and $\mathcal{A}_{\rm tot}$ expressed as functions of redshift. When $\mathcal{A}_{\rm de} < 0$, the DE sector contributes repulsively; however, the Universe is in an accelerated phase only when $\mathcal{A}_{\rm tot} < 0$.

Fig.5.4 show the evolution of both $\mathcal{A}_{\rm de}$ and $\mathcal{A}_{\rm tot}$ for a Universe with the current energy density values $\Omega_{\rm m,0} = 0.3$, $\Omega_{\rm de,0} = 0.7$ and the parameter values $K = 15$ and $z_\dagger = 2.8$. These specific parameter values will be useful for interpreting the results shown in Section 5.7.

We notice that, for both $\Lambda^{\rm PT}$CDM and $\Lambda^{\rm TT}$CDM models, the background evolution undergoes an additional phase of accelerated expansion in the past. In contrast, the $\Lambda^{\rm TT}$CDM model enters an accelerated phase only when the other two models begin their second phase of accelerated expansion.

5.5 Observational Constraints

We now present the observational constraints on the sign-switching DE models and compare them with ΛCDM.

Priors

As stated in Section 4.2 we will explore the parameter space:

$$\alpha = \{H_0; \Omega_{\rm m,0}; \Omega_{\rm b,0}; \Xi\}, \tag{5.27}$$

where ΛCDM contains the baseline cosmological parameters, $\Lambda_{\rm s}$CDM contains the additional parameter $\Xi = \{z_\dagger\}$ and the remaining oscillatory models contain the two additional parameters $\Xi = \{z_\dagger, K\}$ parameters. The respective prior are summarized in Table 5.1, where we assumed a flat prior in all the parameters.

Parameter	Prior
H_0	$[40, 100]$
$\Omega_{\mathrm{b},0}$	$[0.01, 0.09]$
$\Omega_{\mathrm{m},0}$	$[0.1, 0.9]$
$z_\dagger$	$[1, 4]$
K	$[0.8, 30]$

Table 5.1: Prior ranges and distributions for the sampled parameters.

We adopt standard flat priors on the baseline cosmological parameters $\theta = \{H_0; \Omega_{\mathrm{m},0}; \Omega_{\mathrm{b},0}\}$ and now turn to the prior specifications for the extended parameters $\Xi = \{z_\dagger, K\}$.

For the transition redshift parameter $z_\dagger$, we consider the interval $z_\dagger \in [1, 4]$, allowing the posterior to be explored and reducing possible boundary effects. We have checked that the main qualitative conclusions remain stable under reasonable changes of the prior range.

We further note that the preferred values of $z_\dagger$ lie above the effective redshift of the Lyman-α BAO measurement, $z_{\mathrm{eff}} \simeq 2.33$. A more complete assessment of this possible tension would require a perturbative treatment of structure formation, which lies beyond the scope of the present background-level analysis and is left for future work.

For the oscillation/transition sharpness parameter K, we adopt the interval $K \in [0.8, 30]$ for all models. By construction,the constants A_1, A_2, and A_3, ensure that all models exhibit a similar transition sharpness for the same value of K, allowing a direct comparison of their oscillatory behavior.

At the lower bound, all models exhibit a smooth transition and, in the case of the Λ_{osc}CDM family, nearly no oscillatory behavior is displayed.

At the upper bound, the Λ_{osc}CDM models approach, but do not fully recover, the abrupt Λ_{s}CDM-like behavior, while the Λ_{OI}CDM family also evolves progressively near the abrupt Λ_{s}CDM limit as K increases. As will become clear from the analysis, extending the prior to higher values of K does not provide additional constraining power of the data considered here.

Although the posterior distributions[4] carry more information than the mean values and uncertainties alone, we show only a selected set in this chapter. For the Λ_{osc}CDM family, we present the posteriors for combinations VII and VIII in Fig. 5.5, corresponding to PP+CC+BAO+CMB and PPS+BAO+CMB, respectively. For the Λ_{OI}CDM family, we show only combination VIII in Fig. 5.6. The remaining posteriors are collected in Appendix B. The mean values and standard deviations of the cosmological parameters for each family are listed separately in Tables 5.3 and 5.2. The ΛCDM and Λ_{s}CDM entries are identical in both tables and are reproduced for the reader's convenience.

[4]To avoid overloading the notation and to facilitate the comparison between the posterior distributions, the transition parameters $x_{2\dagger}$ and $x_\star$ are both denoted by $z_\dagger$ in the plots.

Table 5.2: Mean values and standard deviations of the cosmological parameters obtained for each model.

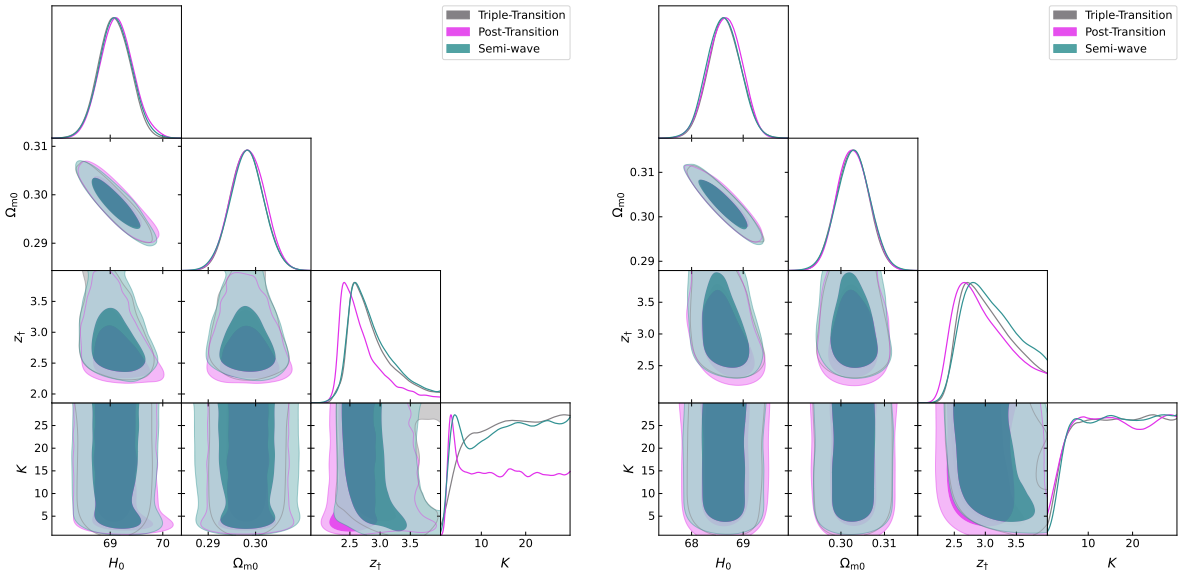
Model/Dataset	H_0	$\Omega_{\mathrm{m},0}$	$10^2\Omega_{\mathrm{b},0}h^2$	$z_\dagger$	K	r_{drag}
ΛCDM						
CMB	$67.41^{+0.50}_{-0.55}$	0.3146 ± 0.0073	2.2362 ± 0.014			
BAO	68.72 ± 0.45	0.2972 ± 0.0084	2.233 ± 0.035			147.8 ± 1.1
BAO+CMB	68.41 ± 0.31	0.3010 ± 0.0039	2.251 ± 0.012			147.48 ± 0.18
PP+CC	67.53 ± 1.70	$0.3314^{+0.0164}_{-0.0178}$	$2.233^{+0.035}_{-0.037}$			
PPS	73.52 ± 0.98	0.3322 ± 0.0187	2.234 ± 0.034			
PP+CC+BAO+CMB	68.31 ± 0.29	0.3024 ± 0.0037	2.251 ± 0.012			147.43 ± 0.18
PPS+BAO+CMB	68.71 ± 0.29	0.2974 ± 0.0036	2.260 ± 0.012			147.51 ± 0.18
$\Lambda_{\mathbf{s}}$CDM						
CMB	$70.08^{+0.26}_{-2.87}$	$0.2928^{+0.0253}_{-0.0061}$	2.236 ± 0.014	> 1.33		
BAO	68.72 ± 0.42	$0.2979^{+0.0081}_{-0.0090}$	2.235 ± 0.035	> 2.45		147.7 ± 1.2
BAO+CMB	68.74 ± 0.33	0.3015 ± 0.0038	2.241 ± 0.013	$3.03^{+0.31}_{-0.61}$		147.18 ± 0.21
PP+CC	$67.78^{+1.61}_{-1.74}$	0.3319 ± 0.0174	2.232 ± 0.035	> 1.44		
PPS	73.55 ± 0.98	0.3321 ± 0.0180	2.233 ± 0.036	unconstrained		
PP+CC+BAO+CMB	68.66 ± 0.32	0.3026 ± 0.0037	$2.239^{+0.013}_{-0.012}$	$3.00^{+0.26}_{-0.60}$		147.16 ± 0.20
PPS+BAO+CMB	69.10 ± 0.30	0.2982 ± 0.0035	2.246 ± 0.012	$2.78^{+0.12}_{-0.44}$		$147.16^{+0.20}_{-0.21}$
$\Lambda^{\mathbf{SW}}$CDM						
CMB	$70.13^{+0.49}_{-2.93}$	$0.2923^{+0.0259}_{-0.0078}$	2.236 ± 0.013	> 1.38	unconstrained	
BAO	$69.07^{+0.39}_{-0.85}$	$0.3067^{+0.0065}_{-0.0199}$	2.233 ± 0.036	> 1.48	> 2.63	$146.43^{+2.79}_{-0.93}$
BAO+CMB	68.74 ± 0.33	0.3015 ± 0.0039	2.241 ± 0.013	$3.08^{+0.30}_{-0.59}$	> 4.21	147.18 ± 0.20
PP+CC	67.66 ± 1.68	0.3333 ± 0.0185	2.231 ± 0.037	> 1.40	unconstrained	
PPS	73.51 ± 0.99	$0.3348^{+0.0177}_{-0.0202}$	2.232 ± 0.035	unconstrained	unconstrained	
PP+CC+BAO+CMB	68.62 ± 0.32	0.3029 ± 0.0038	2.239 ± 0.013	$3.10^{+0.33}_{-0.56}$	> 4.33	147.16 ± 0.19
PPS+BAO+CMB	69.10 ± 0.31	0.2982 ± 0.0035	2.246 ± 0.013	$2.88^{+0.20}_{-0.46}$	> 3.24	147.17 ± 0.20
$\Lambda^{\mathbf{TT}}$CDM						
CMB	$69.58^{+0.56}_{-2.28}$	$0.2964^{+0.0205}_{-0.0081}$	2.236 ± 0.014	> 1.51	unconstrained	
BAO	$68.99^{+0.36}_{-0.78}$	$0.3054^{+0.0061}_{-0.0185}$	2.234 ± 0.036	> 1.34	unconstrained	$146.65^{+2.54}_{-0.84}$
BAO+CMB	68.73 ± 0.31	0.3016 ± 0.0037	2.241 ± 0.012	$3.06^{+0.31}_{-0.55}$	> 4.01	147.19 ± 0.20
PP+CC	67.52 ± 1.68	0.3332 ± 0.0180	2.234 ± 0.035	> 1.42	unconstrained	
PPS	73.53 ± 1.00	$0.3340^{+0.0174}_{-0.0187}$	2.233 ± 0.036	unconstrained	unconstrained	
PP+CC+BAO+CMB	68.64 ± 0.31	0.3027 ± 0.0037	2.239 ± 0.012	$3.05^{+0.28}_{-0.55}$	> 3.96	147.16 ± 0.20
PPS+BAO+CMB	69.08 ± 0.30	0.2982 ± 0.0034	2.246 ± 0.013	$2.87^{+0.18}_{-0.45}$	> 4.22	147.17 ± 0.20
$\Lambda^{\mathbf{PT}}$CDM						
CMB	$70.01^{+0.45}_{-2.81}$	$0.2931^{+0.0249}_{-0.0074}$	2.236 ± 0.014	> 1.36	unconstrained	
BAO	$69.02^{+0.34}_{-0.81}$	$0.3056^{+0.0058}_{-0.0178}$	$2.232^{+0.037}_{-0.034}$	> 1.35	> 2.31	$146.61^{+2.51}_{-0.83}$
BAO+CMB	68.74 ± 0.34	0.3015 ± 0.0040	2.240 ± 0.013	$3.00^{+0.30}_{-0.60}$	> 4.13	147.19 ± 0.21
PP+CC	67.63 ± 1.74	0.3328 ± 0.0179	2.234 ± 0.035	> 1.41	unconstrained	
PPS	$73.52^{+0.95}_{-1.09}$	$0.3323^{+0.0186}_{-0.0172}$	2.236 ± 0.036	unconstrained	unconstrained	
PP+CC+BAO+CMB	68.65 ± 0.31	$0.3028^{+0.0035}_{-0.0038}$	2.238 ± 0.012	$2.97^{+0.29}_{-0.57}$	unconstrained	147.15 ± 0.20
PPS+BAO+CMB	69.13 ± 0.31	0.2984 ± 0.0035	2.245 ± 0.012	$2.71^{+0.12}_{-0.43}$	> 2.78	$147.13^{+0.20}_{-0.22}$

5.5.1 Parameter Constraints

Since our analysis is restricted to background-level observables and does not assume a specific perturbative description of the sign-switching DE models, we employ the compressed Planck 2018 likelihood based on CMB shift parameters, which provides an effective description of the geometric constraints encoded in the CMB.

This approach, however, carries less constraining power than the full CMB likelihood, particularly for parameters correlated with H_0 and for models with non-trivial late-time deviations from ΛCDM. Therefore, the constraints obtained from the compressed CMB likelihood should be interpreted with some caution, while an analysis including perturbations and the full CMB likelihood is left for future work.

The comoving sound horizon at the drag epoch also deserves attention. According to the fitting formula in Eq. (4.26), $r_{\rm drag}$ depends on the physical baryon density and on the total matter density. In our analysis, $\Omega_{\rm b,0}$ is treated as a free parameter, while a BBN prior on $\Omega_{\rm b,0}h^2$ is included in all dataset combinations. For the BAO-only combination, the constraining power on $r_{\rm drag}$ is weaker than for combinations including CMB information. This is expected, since the CMB distance priors provide an additional direct constraint on $\Omega_{\rm b,0}h^2$, bringing the inferred value of $r_{\rm drag}$ closer to the corresponding ΛCDM constraint obtained from the same dataset combination.



(a) PPS+BAO+CMB **(b)** PP+CC+CMB+BAO

Figure 5.5: 68% and 95% C.L. 2D contours and 1D marginalized posterior distributions for the $\Lambda_{\rm OI}$CDM family using PPS+BAO+CMB and PP+CC+CMB+BAO.

SN and CC combinations

We begin with dataset combinations that rely solely on late-time probes. These datasets have limited constraining power on the additional parameters, leaving the oscillation/transition parameter K essentially unconstrained.

In contrast to PPS, the PP+CC combination shows a lower bound $z_\dagger > 1.4$ at 95% C.L. This result was supported in the literature , because CC measurements offer a direct probe of the Hubble parameter $H(z)$, instead of other quantities based

on integrals like the luminosity density, presented in SN probes. Consequently, CC measurements can be more sensitive to parameters that modify the evolution of $H(z)$.

Apart from the additional model parameters, the remaining cosmological parameters show similar mean values and uncertainties across the different models.

BAO and CMB combinations

For the BAO-only dataset, both H_0 and $\Omega_{\mathrm{m},0}$ tend to increase across all models relative to ΛCDM, while their corresponding uncertainties also become broader.

For the abrupt Λ_{s}CDM case, the transition redshift is pushed beyond the redshift range directly probed by DESI [35]. While the largest effective redshift is $z_{\mathrm{eff}} = 2.33$, we find the lower bound $z_\dagger > 2.45$ at 95% C.L. This suggests that, within the Λ_sCDM model, the DESI DR2 data tend to disfavor a negative DE density over the observed redshift range.

Regarding the smooth extensions of Λ_{s}CDM, one can see a different behavior, which can be split in respect to the families:

- Λ_{OI}CDM : For the BAO data, the posterior disfavors the lowest values of the transition parameter for Λ^{SW}CDM, with the lower bound $K > 2.63$ at 95% C.L. In contrast Λ^{TT}CDM does not show a corresponding lower bound. This can be understood from the fact that, for the same value of K, the transition in Λ^{TT}CDM occurs more rapidly than in Λ^{SW}CDM, as shown in Fig. 5.2. The same argument cannot be directly applied to Λ^{PT}CDM because of the role of the parameter $x_\star$. As discussed when introducing this model, for small values of K, $x_\star$ does not coincide exactly with the effective transition point. Instead, the position of the transition depends on both $x_\star$ and K. Only as K increases we have $x_\star \to x_\dagger$. Consequently, comparing the models at a fixed value of K does not necessarily correspond to comparing transitions occurring at the same position.

- Λ_{osc}CDM: With BAO data and all the corresponding combinations, we notice that the parameter K exhibits a strong multimodal distribution, a feature that was not present in the previous family. In contrast to the Λ_{OI}CDM models, these combinations do not constrain K, with the exception of the Λ^a_{osc}CDM model.

For the CMB-only combination, the sign-switching models allow larger values of H_0 than ΛCDM. This behavior is mainly associated with an extended right-hand tail of the marginalized H_0 posterior, rather than with a strong shift of its peak. The two-dimensional distributions (Appendix B) show that larger values of H_0 are correlated with lower values of $\Omega_{\mathrm{m},0}$ and, in particular, with smaller values of the transition redshift $z_\dagger$. The parameter $z_\dagger$ remains weakly constrained, with all models favoring a sign-switching redshift of $z_\dagger \gtrsim 1.26$ at 95% C.L. The Λ^x_{osc}CDM model shows a stronger constraint, with $z\dagger > 2.07$ at 95% C.L. The parameter K remains unconstrained for this dataset. These weak constraints on the additional parameters may be related to the fact that we do not use the full CMB likelihood.

For the combined BAO+CMB dataset, we again find differences in the Hubble constant H_0 and the matter density parameter $\Omega_{\mathrm{m},0}$, with all sign-switching models tending to favor slightly higher values than those obtained within ΛCDM. In contrast,

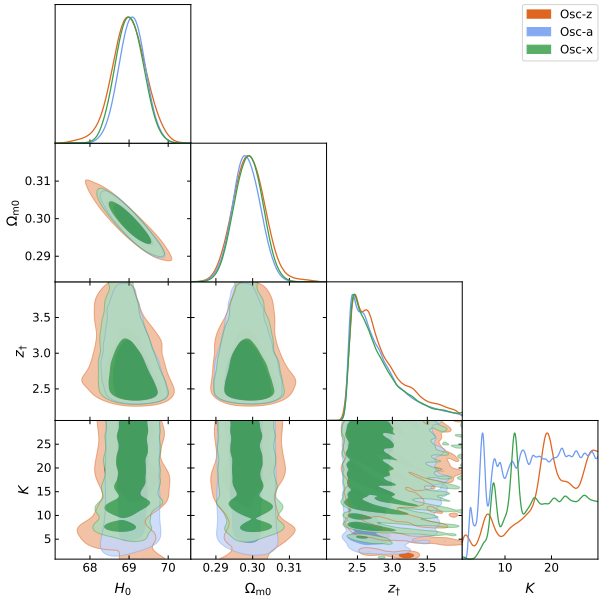


Figure 5.6: 68% and 95% C.L. 2D contours and 1D marginalized posterior distributions for the $\Lambda_{\rm osc}$CDM family using PPS+BAO+CMB .

the baryon density $\Omega_{\rm b,0}h^2$ is generally inferred to take slightly lower values. Regarding the sign-switching models, all oscillatory models show values of the transition parameter $z_\dagger$ comparable to those of $\Lambda_{\rm s}$CDM, yielding $z_\dagger \sim 3.0$. In the $\Lambda_{\rm OI}$CDM family, K does not have an upper limit, provided that the transition is sufficiently fast.

For the $\Lambda_{\rm osc}$CDM family, the posterior distributions of K are strongly multimodal and exhibit no upper bound, reflecting a degeneracy between K and $z_\dagger$. In this case, interpreting the constraints through mean values and standard deviations can be misleading, as these summary statistics do not capture the multimodal structure of the posterior.

PPS+BAO+CMB and PP+CC+BAO+CMB combinations

We now turn to the PPS+BAO+CMB combination.

All sign-switching models favor higher values of H_0 and $\Omega_{\rm m,0}$ than ΛCDM, with the transition redshift converging to a consistent mean value of $z_\dagger \sim 2.80$ across all models.

For the $\Lambda_{\rm osc}$CDM family, the parameter K exhibits the same multimodal posterior distribution across all three models. However, the individual models display distinct posterior shapes, indicating that as the number of oscillations increases, the multimodality in K becomes more pronounced, a trend that persists across the BAO, BAO+CMB, and PPS+BAO+CMB combinations. It should be noted that standard MCMC samplers do not efficiently explore multimodal distributions, and the results for these models should therefore be interpreted with care.

For the $\Lambda_{\rm OI}$CDM family, the constraints generally favor sufficiently fast transitions, while no upper bound on K is obtained. The data therefore remain compatible with very rapid transitions, approaching the abrupt behavior of the $\Lambda_{\rm s}$CDM model.

Nevertheless, slower transitions should not be dismissed. In principle, they could provide a greater alleviation of the Hubble tension, particularly if the transition occurred at lower redshifts than those favored by our analysis. However, the preference

for sign-switching at higher redshifts appears to be mainly driven by the DESI DR2 dataset, which disfavors a negative DE density between its data points.

For the PP+CC+BAO+CMB combination, all sign-switching models again favor higher values of H_0 and $\Omega_{\mathrm{m},0}$, although the shifts relative to ΛCDM are smaller than in the PPS+BAO+CMB case. The transition redshift $z_\dagger$ also admits a higher mean central value than in the PPS+BAO+CMB combination.

Table 5.3: Mean values and standard deviations of the cosmological parameters obtained for each model.

Model/Dataset	H_0	$\Omega_{m,0}$	$10^2\Omega_{b0}h^2$	$z_\dagger$	K	r_{drag}
ΛCDM						
CMB	$67.41^{+0.50}_{-0.55}$	0.3146 ± 0.0073	2.2362 ± 0.014			
BAO	68.72 ± 0.45	0.2972 ± 0.0084	2.233 ± 0.035			147.8 ± 1.1
BAO+CMB	68.41 ± 0.31	0.3010 ± 0.0039	2.251 ± 0.012			147.48 ± 0.18
PPS	73.52 ± 0.98	0.3322 ± 0.0187	2.234 ± 0.034			
PPS+BAO+CMB	68.71 ± 0.29	0.2974 ± 0.0036	2.260 ± 0.012			147.51 ± 0.18
$\Lambda_{\mathbf{s}}$CDM						
CMB	$70.08^{+0.26}_{-2.87}$	$0.2928^{+0.0253}_{-0.0061}$	2.236 ± 0.014	> 1.33		
BAO	68.72 ± 0.42	$0.2979^{+0.0081}_{-0.0090}$	2.235 ± 0.035	> 2.45		147.7 ± 1.2
BAO+CMB	68.74 ± 0.33	0.3015 ± 0.0038	2.241 ± 0.013	$3.03^{+0.31}_{-0.61}$		147.18 ± 0.21
PPS	73.55 ± 0.98	0.3321 ± 0.0180	2.233 ± 0.036	unconstrained		
PPS+BAO+CMB	69.10 ± 0.30	0.2982 ± 0.0035	2.246 ± 0.012	$2.78^{+0.12}_{-0.44}$		$147.16^{+0.20}_{-0.21}$
$\Lambda^{\mathbf{z}}_{\mathbf{osc}}$CDM						
CMB	$70.59^{+0.36}_{-3.70}$	$0.2895^{+0.0311}_{-0.0075}$	2.236 ± 0.014	> 1.26	unconstrained	
BAO	68.65 ± 0.78	$0.3031^{+0.0077}_{-0.0169}$	2.234 ± 0.036	> 1.63	unconstrained	$147.2^{+2.0}_{-1.0}$
BAO+CMB	$68.56^{+0.71}_{-0.55}$	$0.3032^{+0.0053}_{-0.0069}$	2.240 ± 0.013	$3.03^{+0.26}_{-0.60}$	unconstrained	147.19 ± 0.20
PPS	73.44 ± 1.04	0.3363 ± 0.0275	2.232 ± 0.036	unconstrained	unconstrained	$\pm$
PPS+BAO+CMB	68.99 ± 0.44	0.2991 ± 0.0045	$2.246^{+0.012}_{-0.013}$	$2.83^{+0.14}_{-0.47}$	> 4.94	$147.17^{+0.19}_{-0.21}$
$\Lambda^{\mathbf{a}}_{\mathbf{osc}}$CDM						
CMB	$70.20^{+0.25}_{-3.07}$	$0.2921^{+0.0267}_{-0.0061}$	2.236 ± 0.014	> 1.31	unconstrained	
BAO	68.75 ± 0.48	$0.2979^{+0.0085}_{-0.0100}$	2.234 ± 0.036	> 2.44	> 3.29	147.7 ± 1.3
BAO+CMB	68.72 ± 0.38	0.3016 ± 0.0042	2.241 ± 0.012	$3.04^{+0.28}_{-0.60}$	> 2.80	147.19 ± 0.20
PPS	73.52 ± 1.06	0.3329 ± 0.0188	$2.235^{+0.037}_{-0.034}$	unconstrained	unconstrained	$\pm$
PPS+BAO+CMB	69.08 ± 0.34	0.2984 ± 0.0038	2.246 ± 0.012	$2.79^{+0.11}_{-0.43}$	> 4.66	147.17 ± 0.20
$\Lambda^{\mathbf{x}}_{\mathbf{osc}}$CDM						
CMB	$70.31^{+0.33}_{-3.35}$	$0.2915^{+0.0289}_{-0.0069}$	2.236 ± 0.013	> 2.07	unconstrained	
BAO	$68.72^{+0.63}_{-0.56}$	$0.2988^{+0.0090}_{-0.0114}$	$2.235^{+0.035}_{-0.034}$	> 2.46	unconstrained	147.6 ± 1.3
BAO+CMB	$68.77^{+0.42}_{-0.50}$	0.3012 ± 0.0050	2.241 ± 0.013	$3.03^{+0.26}_{-0.59}$	> 3.97	147.19 ± 0.20
PPS	73.40 ± 0.90	0.3361 ± 0.0270	2.233 ± 0.036	unconstrained	unconstrained	
SN+BAO+CMB	69.01 ± 0.36	0.2989 ± 0.0039	2.246 ± 0.013	$2.80^{+0.12}_{-0.44}$	> 7.21	147.17 ± 0.20

5.5.2 Statistical Model Selection - Λ_{OI}CDM

Given the strongly multimodal structure found for the Λ_{osc}CDM models, we cannot ensure that the global maximum likelihood is reliably identified with the present minimization procedure. Since the AIC and BIC depend directly on this quantity, we therefore do not quote information criteria for this family.

PPS, PP, CC, BAO and CMB combinations				
Model	PPS	BAO	CMB+BAO	PPS+BAO+CMB
	$\chi^2_{\rm min}$ above; (ΔAIC, ΔAIC$_c$, ΔBIC) below			
ΛCDM	1452.017 (0, 0, 0)	10.283 (0, 0, 0)	16.525 (0, 0, 0)	1500.684 (0, 0, 0)
Λ_sCDM	1451.703 (1.686, 1.695, 7.125)	10.283 (2.000, 4.044, 2.639)	10.613 (−3.912, −2.425, −3.079)	1489.285 (−9.399, −9.390, −3.950)
$\Lambda^{\mathbf{SW}}$CDM	1450.616 (2.599, 2.621, 13.478)	6.890 (0.606, 5.706, 1.885)	10.364 (−2.161, 1.447, −0.495)	1489.046 (−7.638, −7.617, 3.260)
$\Lambda^{\mathbf{TT}}$CDM	1450.179 (2.162, 2.183, 13.041)	5.328 (−0.955, 4.145, 0.323)	10.368 (−2.157, 1.451, −0.491)	1488.973 (−7.711, −7.690, 3.187)
$\Lambda^{\mathbf{PT}}$CDM	1451.275 (3.258, 3.279, 14.137)	5.564 (−0.719, 4.381, 0.559)	10.239 (−2.286, 1.323, −0.619)	1486.961 (−9.723, −9.702, 1.175)

Model	PP+CC	PP+CC+BAO+CMB
ΛCDM	1417.533 (0, 0, 0)	1437.006 (0, 0, 0)
Λ_sCDM	1417.533 (2.000, 2.009, 7.459)	1430.860 (−4.146, −4.137, 1.322)
$\Lambda^{\mathbf{SW}}$CDM	1416.266 (2.734, 2.754, 13.651)	1430.595 (−2.411, −2.390, 8.525)
$\Lambda^{\mathbf{TT}}$CDM	1415.882 (2.349, 2.370, 13.267)	1430.595 (−2.411, −2.390, 8.525)
$\Lambda^{\mathbf{PT}}$CDM	1416.942 (3.409, 3.430, 14.327)	1430.667 (−2.339, −2.318, 8.597)

Table 5.4: Model comparison for the different dataset combinations. In each entry, the first value corresponds to $\chi^2_{\rm min}$, while the quantities in parentheses denote (ΔAIC, ΔAIC$_c$, ΔBIC). All differences are computed relative to ΛCDM.

The information criteria in Table 5.4 show that the statistical preference for the sign-switching models depends strongly on the dataset combination. For the PPS and PP+CC combinations, the improvement in $\chi^2_{\rm min}$ is generally too small to compensate for the additional model parameters, particularly for the two-parameter extensions. As a result, the BIC in particular shows a strong penalization for the oscillatory models. The BAO combination, although not disfavored, also does not show an improvement with respect to ΛCDM.

For BAO+CMB, all models show compatibility with ΛCDM, although the penalization from AIC$_{\rm c}$ is stronger than from the other criteria.

We now turn to the larger dataset combinations. Although the inclusion of BAO and CMB improves the relative performance of the extended models compared with the late-time PP+CC combination, the BIC still favors the simpler description, with $\Lambda_{\rm s}$CDM being less penalized than the oscillatory models due to its smaller number of free parameters.

The PPS+BAO+CMB combination shows a different behavior. Here, the AIC and AIC$_{\rm c}$ favor the extended models, reflecting the larger improvement in $\chi^2_{\rm min}$. The BIC

remains more conservative : Λ_sCDM is favored, whereas the smooth sign-switching models are less supported. Among the latter, PT exhibits the smallest positive ΔBIC.

For the PP+CC+BAO+CMB combination, the AIC and AIC_c also favor the extended models, but the BIC penalization of the smooth sign-switching models is stronger.

5.6 Cosmography

Using Eq. (2.49) we can plot the cosmographic parameters with respect to z, were we wil ignore radiation. Since we are interested in compare and study the oscillatory behavior, we fix $z_\dagger = 2$ and study two regimes :

- $K = 2$, corresponding to a slower transition/ weaker oscillatory behavior.
- $K = 10$, corresponding to a sharper transition and oscillatory behavior.

We first consider the $\Lambda_{\rm OI}$CDM family. Figs. 5.8 and 5.7 show the evolution of the deceleration parameter, jerk, snap and lerk for $K = 2$ and $K = 10$, respectively. The corresponding DE densities are also shown below the deceleration parameter in order to make the following discussion clearer.

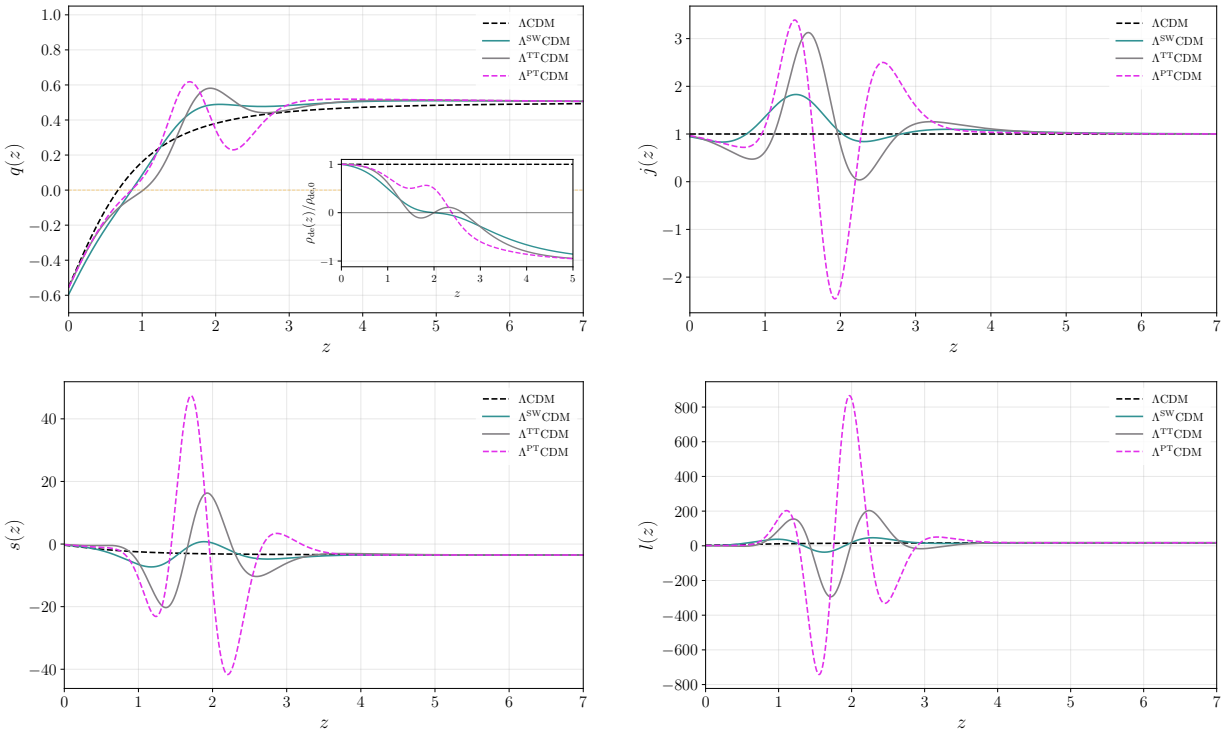


Figure 5.7: Cosmographic parameters for the $\Lambda_{\rm OI}$CDM and ΛCDM, assuming $K = 2$ and $z_\dagger = 2$. The corresponding energy densities are shown below $q(z)$.

Starting with the deceleration parameter, all models approach the ΛCDM behavior at sufficiently high redshift, as expected, since the matter contribution dominates over DE in this regime.

As we approach $z = z_\dagger$, the models begin to deviate more significantly from ΛCDM. For $K = 2$, $q(z)$ tends to increase around the transition, while for $K = 10$ we notice the opposite effect, with the models crossing $q(z) = 0$, thus entering an acceleration phase during the transition. The Λ^{SW}CDM and Λ^{PT}CDM models exhibit a single additional acceleration phase around the transition, whereas Λ^{TT}CDM enters two acceleration phases before the models converge again towards the ΛCDM behavior.

A comment should be made regarding Λ^{PT}CDM. Its deceleration parameter falls below $q(z) = -1$ near the transition. Since :

$$q = -1 - \frac{\dot{H}}{H^2} \quad , \tag{5.28}$$

the condition $q(z) < -1$ implies $\dot{H} > 0$, corresponding to a phase of super-acceleration, during which the Hubble parameter increases with cosmic time. In GR, such a phase implies a violation of the null energy condition (NEC). This type of behavior can also arise in smooth realizations of Λ_{s}CDM [30].

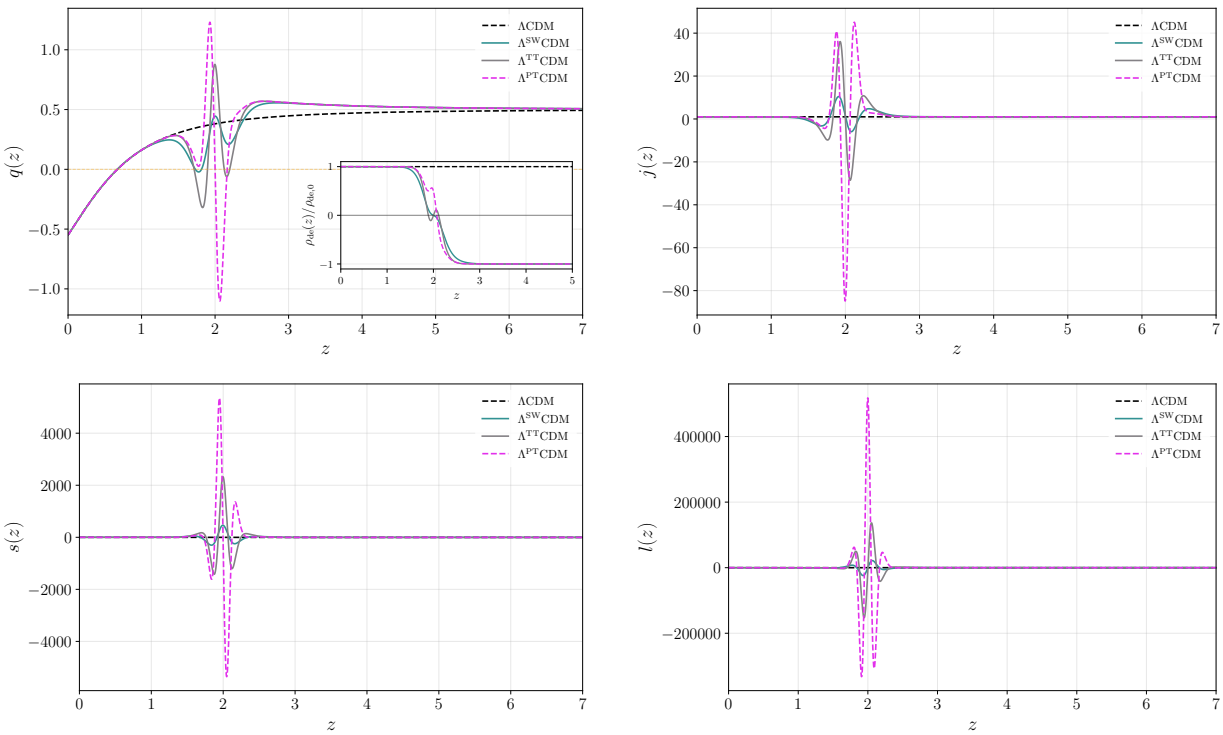


Figure 5.8: Cosmographic parameters for the Λ_{OI}CDM models and ΛCDM, assuming $K = 10$ and $z_\dagger = 2$. The corresponding energy densities are shown below $q(z)$.

The second cosmographic parameter, the jerk $j(z)$, remains fixed at $j(z) = 1$ for ΛCDM when radiation is neglected. For both the slower and faster transitions, we observe pronounced deviations from the standard cosmological model, with the jerk oscillating around $j(z) = 1$. Here, increasing K has two main effects : the deviations are confined to a narrower redshift interval, while their amplitude increases. This behavior follows directly from the expression for the jerk parameter, Eq. (2.50), since a rapid variation in $q(z)$ produces a larger variation in $j(z)$. The same effect becomes even more pronounced for the higher-order cosmographic parameters, the snap and lerk.

These higher-order cosmographic parameters may therefore provide a sensitive way of distinguishing these models from ΛCDM. If future observations constrain these quantities sufficiently well, the large values and oscillatory features predicted around the transition could provide an additional test of the models.

We now turn to the $\Lambda^{x}_{\rm osc}$CDM family. For these models, we restrict the analysis to the deceleration parameter, since the oscillatory behavior of the DE density produces large variations in its derivatives. These variations are progressively amplified in the higher-order cosmographic parameters, leading to extremely large values.

Fig. 5.9 shows $q(z)$ for $K = 2$ and $K = 10$. As expected, the oscillatory behavior becomes more pronounced as K increases, particularly for the $\Lambda^{x}_{\rm osc}$CDM and $\Lambda^{a}_{\rm osc}$CDM models. Unlike the $\Lambda_{\rm OI}$CDM family, the DE density in these models continues to oscillate around $\rho_{\rm de}(z)/\rho_{\rm de,0} = 1$ after the sign transition. Since $q(z)$ depends on the derivative of $H(z)$ and the Hubble parameter depends in the contribution from the DE component,these variations are also reflected in $q(z)$.

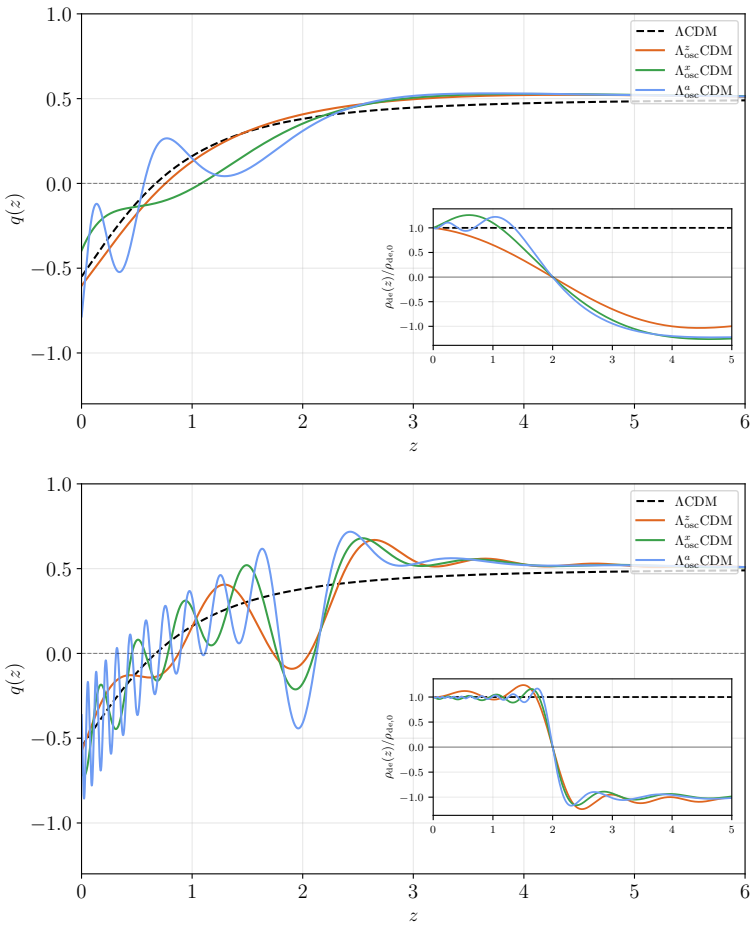


Figure 5.9: Deceleration parameter $q(z)$ for the $\Lambda_{\rm osc}$CDM models, assuming $z_{\dagger} = 2$, for $K = 2$ and $K = 10$.

The increase in the number of oscillations can be understood from the different functional forms of the arguments used in this family. For the $\Lambda^{z}_{\rm osc}$CDM, $\Lambda^{a}_{\rm osc}$CDM, and $\Lambda^{x}_{\rm osc}$CDM models, differentiating Eq. (5.4) with respect to redshift gives:

$$\frac{\mathrm{d}u_z}{\mathrm{d}z} = -0.64K, \tag{5.29}$$

$$\frac{\mathrm{d}u_a}{\mathrm{d}z} = \frac{10K}{(1+z)^2}, \tag{5.30}$$

$$\frac{\mathrm{d}u_x}{\mathrm{d}z} = \frac{2.5K}{1+z}. \tag{5.31}$$

Their dependence on redshift is different in each case. In particular, while $u_z(z)$ varies linearly with z, the arguments $u_x(z)$ and $u_a(z)$ have a non-linear dependence on redshift. Since these arguments enter the Sine Integral function, their different redshift dependence affects both the number of oscillations and how rapidly they occur. This effect becomes stronger as K increases.

Fig. 5.10 shows the evolution of $q(z)$ for the $\Lambda^z_{\rm osc}$CDM model for $K = 10$, $K = 20$, and $K = 50$. In contrast to the other models in this family, increasing K does not produce a strong growth in the amplitude of the oscillations at redshifts below $z = 1$. Among the $\Lambda_{\rm osc}$CDM models, the $\Lambda^z_{\rm osc}$CDM model exhibits the simplest behavior in the deceleration parameter.

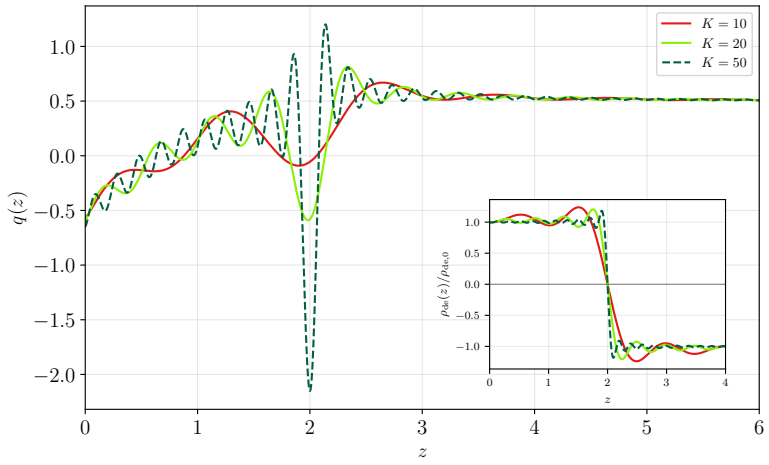


Figure 5.10: Deceleration parameter $q(z)$ for the $\Lambda^z_{\rm osc}$CDM model, assuming $z_\dagger = 2$, for $K = 10$, $K = 20$, and $K = 50$.

To conclude the cosmographic analysis of the $\Lambda_{\rm osc}$CDM family, the introduction of a varying number of oscillations in the DE density leads to a non-trivial behavior of the deceleration parameter. Although all models undergo the same negative-to-positive energy density transition, the way the oscillations are introduced has a clear impact on their kinematic evolution.

Evolution of Background Quantities

Using samples from the MCMC chains, we now study the evolution of different background quantities predicted by the models.

To avoid a redundant discussion, we only show the evolution of the background quantities for the $\Lambda^{\rm SW}$CDM model, together with $\Lambda_{\rm s}$CDM and ΛCDM. The qualitative behavior of the remaining $\Lambda_{\rm OI}$CDM models is similar.

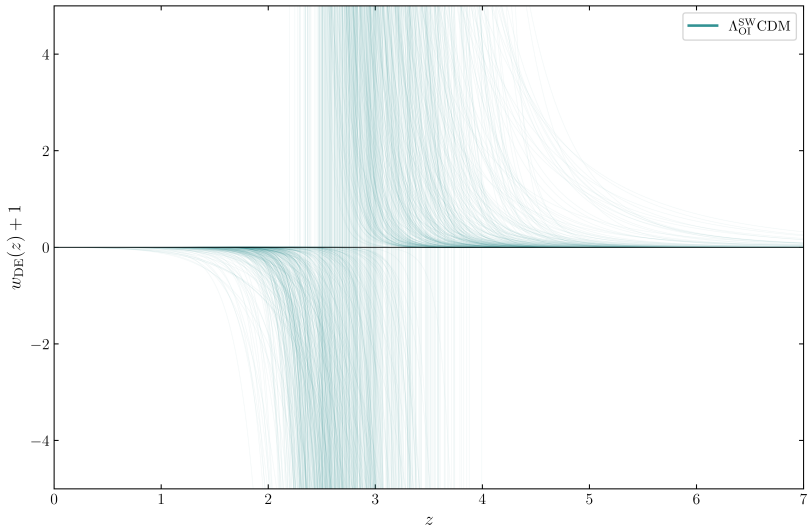


Figure 5.11: Evolution of the DE EoS plus one, $w_{\rm de}(z)+1$, for the $\Lambda^{\rm SW}$CDM model as a function of redshift for the PPS+BAO+CMB combination with 400 samples.

Figs. 5.11 and 5.12 show the corresponding evolution using 400 random samples from the PPS+BAO+CMB chains presented in the previous observational analysis.

Starting with Fig. 5.11, we show the evolution of $w_{\rm de}+1$, which, as previously discussed, diverges at the redshift where the DE density crosses zero. As the transition is approached from higher redshifts, $z \to z_\dagger^+$, we have $w_{\rm de}(z) \to +\infty$. Immediately after the transition, $z \to z_\dagger^-$, we have $w_{\rm de}(z) \to -\infty$.

Regarding the evolution of the fractional densities $\Omega_f(z)$, we notice a characteristic feature of phenomenological AdS-like to dS-like transition models.

During the epoch in which the DE fractional density is negative, the matter density parameter can become larger than one. Neglecting radiation and assuming a spatially flat Universe, we have:

$$\Omega_{\rm m}(z) = 1 - \Omega_{\rm de}(z), \tag{5.32}$$

and therefore:

$$\Omega_{\rm de}(z) < 0 \implies \Omega_{\rm m}(z) > 1. \tag{5.33}$$

This behavior is also present in $\Lambda_{\rm s}$CDM. However, in our models the change occurs continuously rather than abruptly.

Lastly, the right panel shows the evolution of $H(z)/(1+z)$. After the transition, i.e. for $z < z_\dagger$, the sign-switching models tend to predict a higher Hubble parameter than ΛCDM. The abrupt transition in $\Lambda_{\rm s}$CDM produces a correspondingly sharp change in $H(z)/(1+z)$, while the $\Lambda^{\rm SW}$CDM model smooths this behavior across the transition.

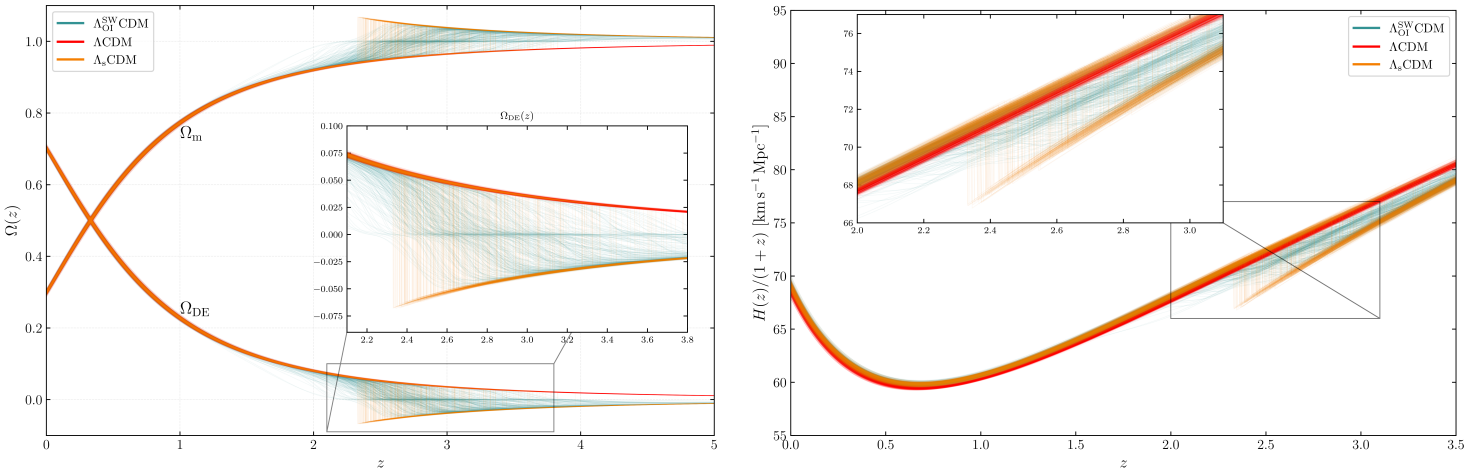


Figure 5.12: Evolution of background cosmological quantities for the $\Lambda^{\rm SW}$CDM model as a function of redshift for the PPS+BAO+CMB combination with 400 samples. Left: fractional energy densities, $\Omega_f(z)$. Right: comoving Hubble parameter, $H(z)/(1+z)$. In both panels, the red lines correspond to ΛCDM and the orange lines to $\Lambda_{\rm s}$CDM.

5.7 Linear Perturbation Analysis

We now present the results of the cosmological perturbation analysis. The usual fluid description, given by Eqs. (2.86), provides a convenient way of describing an effective DE fluid. For the oscillatory sign-switching models considered here, however, this formulation can become ill-defined at particular stages of the evolution. In particular, divergences may arise in the adiabatic sound speed, $c_a^2 = p'_{\rm de}/\rho'_{\rm de}$, at turning points of the DE density, while $w_{\rm de} = p_{\rm de}/\rho_{\rm de}$ diverges when $\rho_{\rm de}$ crosses zero. In the present analysis, we therefore neglect DE perturbations and account for their influence only through the background expansion.

We restrict the perturbation analysis to the $\Lambda_{\rm OI}$ family. Unlike these models, the $\Lambda_{\rm osc}$CDM family does not exactly recover the $\Lambda_{\rm s}$CDM behavior, even in the limit $K \to 0$. These models show repeated crossings of $w_{\rm de} = -1$ and do not exhibit exactly the step function behavior due to the Gibbs phenomenon for larger values of K. Consequently, neglecting DE perturbations is less well justified for this family of models.

For the $\Lambda_{\rm OI}$CDM models, on the other hand, sufficiently rapid transitions confine the main variation of the DE density to a short interval of cosmic evolution. We therefore expect the associated DE perturbations to have a limited impact. The perturbative analysis is thus restricted to matter and radiation, while dark energy affects their evolution through the background dynamics.

The colors corresponding to the different Fourier modes are represented by the color bar. The left and right dashed vertical lines indicate radiation–matter equality and matter–DE equality, respectively. The solid vertical line corresponds to $x_\dagger$, $x_{2\dagger}$, and $x_\star$ for the respective models.

For the perturbation analysis, we adopt the best-fit values of H_0, $\Omega_{\rm m,0}$, and $z_\dagger$ obtained from the PPS+BAO+CMB combination (see Table 5.2). For K, the observational analysis yields only lower bounds, indicating that the data prefer sharper

transitions without constraining this parameter from above. We therefore set $K = 15$, which lies in that regime and provides a sufficiently rapid transition to approximate the sharp sign-switching limit while still resolving the oscillatory features of each model.

For each model, we compute the numerically evolution of the matter and radiation perturbations δ_m, v_m, δ_r, and v_r, after expressing the gravitational potential Ψ and its derivative in terms of the perturbation variables. The integration starts deep in the radiation-dominated era, at $z \simeq 10^6$, with the initial conditions derived above, and is continued until the far future, $z \simeq -0.99$. The calculation is repeated for Fourier modes in the range $3.33 \times 10^{-4} \leq k \leq 0.1\, h\,\mathrm{Mpc}^{-1}$. The former coincides to the current Hubble horizon, while the latter is the maximum mode before the linear approximations breaks and non-linear contribution become important.

Matter perturbations

In Fig. 5.13 we see the evolution of the fractional energy density perturbations in matter δ_m from radiation-dominated era $x = -15$, to a far future $x = 12$. The dashed lines correspond to our models while the dotted lines correspond to the ΛCDM model. For each quantity, we present the results for six different wave numbers : $k = 3.33 \times 10^{-4}\, h\,\mathrm{Mpc}^{-1}$ (dark blue), $k = 1.04 \times 10^{-3}\, h\,\mathrm{Mpc}^{-1}$ (purple), $k = 3.27 \times 10^{-3}\, h\,\mathrm{Mpc}^{-1}$ (magenta), $k = 1.02 \times 10^{-2}\, h\,\mathrm{Mpc}^{-1}$ (red), $k = 3.19 \times 10^{-2}\, h\,\mathrm{Mpc}^{-1}$ (orange), and $k = 0.1\, h\,\mathrm{Mpc}^{-1}$ (yellow). We also plot the same correspondence of colors for the wave numbers in Fig. 5.14 and Fig. 5.15.

The sign-switching models behave almost indistinguishably to all the modes in comparison to ΛCDM with only the lower modes that entered the horizon around the sign-switching of dark energy showing a deviation.

After entering the horizon, matter perturbations begin to grow through gravitational instability. During the matter-dominated era, their evolution is approximately given by:

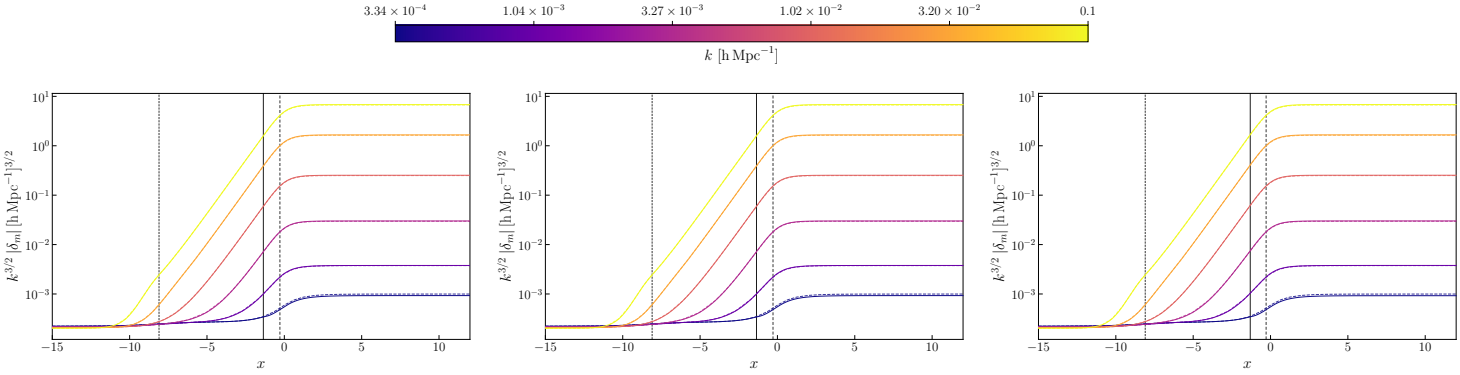
$$\delta_{\mathrm{m}} \propto a \quad \xrightarrow{x=\ln a} \quad \delta_{\mathrm{m}} \propto e^{x},$$

corresponding to an exponential dependence on x. Once dark energy starts to dominate, the growth is progressively suppressed and δ_{m} approaches an approximately constant value for each Fourier mode. Modes with larger wavenumber k enter the horizon earlier and therefore have more time to evolve, typically reaching larger values of δ_{m} over the course of cosmic evolution.

Gravitational potential

Fig. 5.14 shows the evolution of the gravitational potential Ψ, normalized to its initial value Ψ_{ini}, for the different Fourier modes considered. At both very early times and in the far future, the evolution remains close to that predicted by ΛCDM. The largest differences arise in the end of the matter-dominated epoch, when the DE contribution becomes dynamically relevant, and extend into the early stages of DE domination. The behavior of the gravitational potential can be separated according to the range of wave numbers :

For large k, these modes enter the horizon during the radiation-dominated era and consequently experience an early decay of the gravitational potential, together with small oscillatory features that are progressively damped before radiation–matter



(a) Λ^{SW}CDM **(b)** Λ^{TT}CDM **(c)** Λ^{PT}CDM

Figure 5.13: Evolution of the matter perturbations δ_m for the three models compared with the standard ΛCDM scenario.

equality. During matter domination, the gravitational potential approaches an approximately constant value.

Around the sign-switching epoch, a small enhancement can be observed, followed by a second period of decay close to matter–DE equality. At sufficiently late times, the potential asymptotically approaches a very small value.

For intermediate k wave numbers, the gravitational potential remains nearly constant during most of the radiation-dominated epoch and starts to decay close to radiation–matter equality. It subsequently settles into a second approximately constant regime during matter domination. The magnitude of this decay depends on the scale, with larger wave numbers being more strongly affected. Around the sign-switching epoch, a more pronounced feature appears than for the large-k modes, after which the evolution follows a similar late-time behavior.

For small k wave numbers remain approximately constant throughout the radiation-dominated epoch, showing then a mild decay around radiation–matter equality, followed by a second plateau during matter domination. These small modes display the most distinctive behavior around near the sign-switch, where the gravitational potential undergoes a rapid variation. After the transition, their evolution again approaches the behavior observed for the remaining modes.

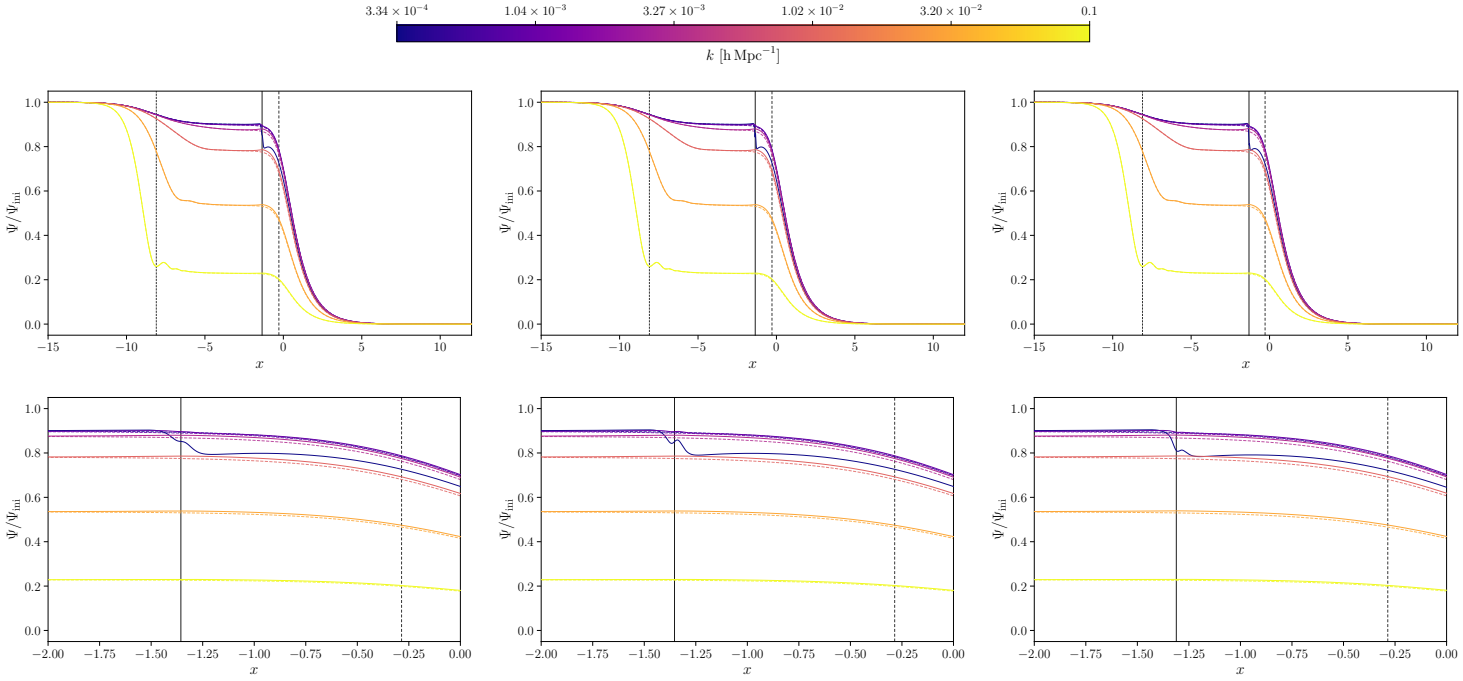
In the amplified panels we see that for the Λ^{SW}CDM model the gravitational potential decays during the transition rather than exhibiting an enhancement. This behavior is not observed in the same way for the other models. The qualitative differences can be understood by considering the third equation in Eqs. (2.92).

Since we are neglecting DE perturbations, the total pressure perturbation is given only by the radiation contribution, $\delta p = \delta p_{\mathrm{r}}$. For radiation:

$$\delta p_{\mathrm{r}} = \frac{1}{3}\rho_{\mathrm{r}}\delta_{\mathrm{r}}, \tag{5.34}$$

and therefore :

$$\frac{3}{2}\frac{\delta p}{\rho_{\mathrm{tot}}} = \frac{1}{2}\Omega_{\mathrm{r}}(x)\delta_{\mathrm{r}}. \tag{5.35}$$



(a) Λ^{SW}CDM **(b)** Λ^{TT}CDM **(c)** Λ^{PT}CDM

Figure 5.14: Normalized gravitational potential Ψ/Ψ_{ini} for the three models compared with ΛCDM. The lower panels show the behavior around the sign-switching epoch.

Around the transition epoch, which occurs at low redshift, the radiation fraction is of order $\Omega_{\mathrm{r}}(x) \sim 10^{-3}$. Thus, the source term on the RHS is therefore strongly suppressed by the small radiation abundance, and we neglect it in the following discussion. The equation then becomes:

$$\Psi_{xx} + \left[3 - \frac{1}{2}\left(1 + 3w_{\mathrm{tot}}\right)\right]\Psi_x - 3w_{\mathrm{tot}}\Psi \approx 0. \tag{5.36}$$

Using Eq. (5.26), this can be rewritten as:

$$\Psi_{xx} + \left(3 - \frac{\mathcal{A}_{\mathrm{tot}}}{2\rho_{\mathrm{tot}}}\right)\Psi_x + \left(1 - \frac{\mathcal{A}_{\mathrm{tot}}}{\rho_{\mathrm{tot}}}\right)\Psi \approx 0. \tag{5.37}$$

Eq. (5.37) shows that the dynamics of the gravitational potential depend explicitly on $\mathcal{A}_{\mathrm{tot}}$.

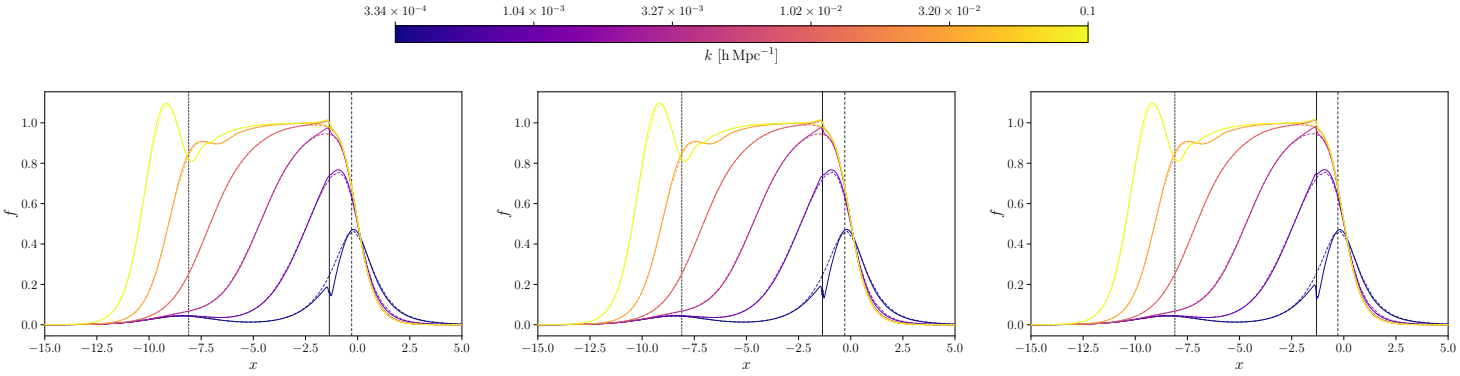
From Fig. 5.4, where the same value of K and a similar value of $z_\dagger$ were used, we see that for Λ^{SW}CDM, $\mathcal{A}_{\mathrm{tot}}$ remains positive throughout the transition, whereas for both Λ^{TT}CDM and Λ^{PT}CDM it becomes negative around the transition epoch. This suggests that whether the background Universe enters an accelerated phase during the transition can influence the evolution of Ψ and can contribute to the different behavior observed around the sign switch.

Growth rate

In Fig.5.15, we show the evolution of the growth rate for the different models considered. At early times, all models exhibit an approximately constant behavior

while the corresponding modes remain super-Hubble. Modes with larger values of k enter the horizon earlier, after which the growth rate increases and approaches the usual sub-Hubble evolution. In this regime, the dependence on the Fourier scale becomes progressively weaker, allowing a more direct comparison between the different models and ΛCDM.

At high redshift, the predictions remain very similar to those of ΛCDM, while the main differences appear once the DE component becomes dynamically relevant. In the presence of a negative DE density, the modified expansion history can favor a larger growth rate prior to the sign-switching epoch. After the transition to positive dark energy, the subsequent accelerated expansion suppresses the growth of matter perturbations, and the evolution gradually approaches the behavior expected in ΛCDM.



(a) Λ^{SW}CDM **(b)** Λ^{TT}CDM **(c)** Λ^{PT}CDM

Figure 5.15: Growth rate f for the three models compared with ΛCDM.

The largest-scale modes show a somewhat different evolution. Since these modes enter the Hubble radius only at relatively late times, they have had less time to develop significant matter perturbations, resulting in a smaller growth rate compared with modes that entered the horizon earlier. At sufficiently late times, once dark energy dominates the background evolution, the growth rate decreases for all modes.

For models in which the DE density changes, a mild enhancement of the growth rate may occur around the sign-switching epoch, particularly when the DE density becomes negative enough to significantly affect the background dynamics.

Such a negative-energy phase can reduce the expansion-induced damping of matter perturbations and therefore temporarily favor structure formation.

Matter Power spectrum

Before interpreting the matter power spectrum, it is important to stress that these results are preliminary, since a full cosmological fit may modify the trends discussed below.

Fig. 5.16 shows the matter power spectrum for wave numbers in the range $k \in [0.000334, 0.2]\, h\,\mathrm{Mpc}^{-1}$. As already observed in the growth-rate analysis, modes with larger k, corresponding to smaller physical scales, tend to show an enhancement of structure formation, whereas modes with smaller k appear comparatively suppressed. Overall, however, the models display an enhancement of structure growth relative to the ΛCDM scenario.

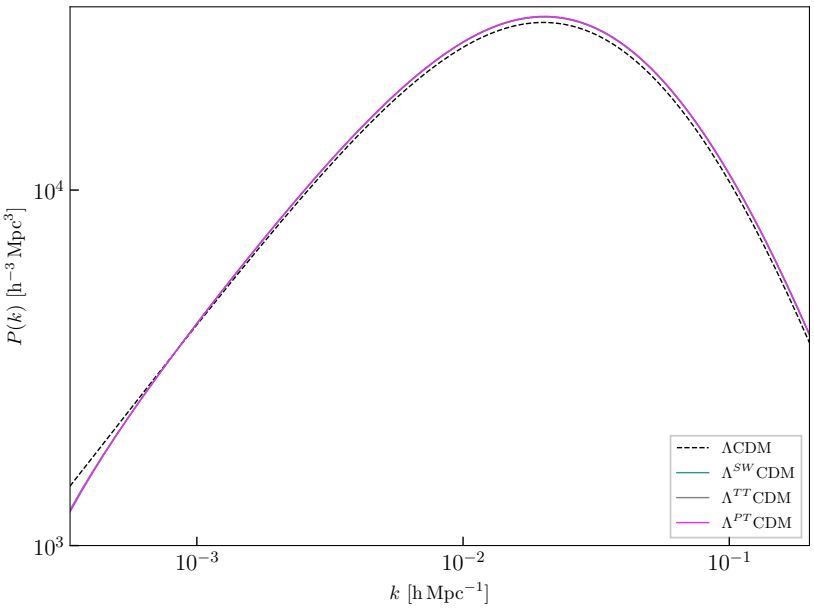


Figure 5.16: Matter power spectrum as a function of wavenumber k. The dashed black line corresponds to ΛCDM, while the remaining lines represent the three models.

fσ_8 distribution

Fig. 5.17 shows the evolution of $f\sigma_8$ for the three models over the redshift range $z \in (0, 2)$, together with the corresponding ΛCDM prediction and observational measurements.

Since the sign-switching transition occurs at $z_\dagger \approx 2.8$ for the parameter values considered, the main modifications to the expansion history take place at redshifts above the range currently probed by data. By $z \approx 2$, the DE density in all three models has already settled close to its ΛCDM value, and the subsequent evolution of $f\sigma_8$ is therefore nearly indistinguishable from the standard prediction.

A more robust assessment of the impact of these models on structure formation would require fitting the perturbative predictions directly to growth-rate measurements, which is left for future work.

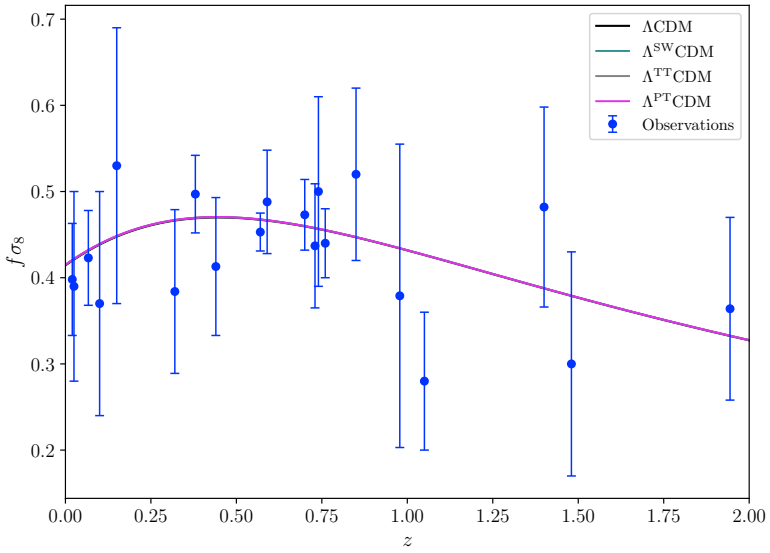


Figure 5.17: Evolution of $f\sigma_8$ over the redshift range $0 \leq z \leq 2$ for ΛCDM and the three Λ_{OI}CDM models, compared with the observational measurements from Ref. [88]. The blue points represent the observational data with their corresponding uncertainties.

There's no such thing as a painless lesson, they just don't exist. Sacrifices are necessary. You can't gain anything without losing something first. Although, if you can endure that pain and walk away from it, you'll find that you now have a heart strong enough to overcome any obstacle. Yeah... a heart made fullmetal.

— Edward Elric

6 Conclusions

In this thesis, motivated by the $\Lambda_{\rm s}$CDM model, we introduced a new class of smooth, oscillatory sign-switching DE models, organized into two families: $\Lambda_{\rm osc}$CDM and $\Lambda_{\rm OI}$CDM.

We undertook a detailed analysis of the background evolution of these models and discussed the consequences of allowing the DE density to cross zero continuously. In particular, we showed that a smooth sign change removes the Type-II sudden singularity present in $\Lambda_{\rm s}$CDM, replacing it with the milder Type-V (w-singularity).

We confronted our models with several dataset combinations, and the analysis showed that sign-switching models can provide a viable alternative to the standard cosmological model when sufficient constraining power is available. In particular, for the PPS+BAO+CMB combination, the AIC and $\text{AIC}_{\rm c}$ favor the sign-switching models over ΛCDM. Although $\Lambda_{\rm s}$CDM is often statistically preferred, mainly due to its smaller number of free parameters, our models exhibit a richer phenomenology. Future observations will reveal whether this additional phenomenology is supported by the data.

Regarding the cosmographic analysis, both families can exhibit an additional phase of acceleration around the sign-switching transition, depending on their extended parameters. The $\Lambda_{\rm OI}$CDM family shows a comparatively simpler behavior, whereas $\Lambda_{\rm osc}$CDM displays strongly oscillatory cosmographic parameters, particularly in the $\Lambda^{a}_{\rm osc}$CDM and $\Lambda^{x}_{\rm osc}$CDM models, where the amplitudes grow rapidly with K and may become physically unrealistic. We also found that higher-order cosmographic quantities amplify features that are barely visible in the deceleration parameter, suggesting that they could serve as discriminating signatures between models.

Regarding the perturbation analysis, matter perturbations are nearly indistinguishable from ΛCDM for most Fourier modes, with deviations appearing only for the lowest modes that enter the horizon near the sign-switching epoch. The matter power spectrum shows an overall enhancement of structure growth relative to ΛCDM, with the effect being more pronounced at smaller scales. The evolution of $f\sigma_8$ remains broadly consistent with ΛCDM and compatible with current observational data, since the sign-switching transition at $z_\dagger \sim 2.8$ occurs above the redshift range probed

by existing growth-rate measurements.

The work developed in this thesis is expected to lead to three publications. The first, *Oscillating Sign-Switching Dark Energy Models* [89], is currently in preparation and presents the models together with selected results from the different analyses. Two complementary works are also in progress: one focusing on the cosmographic analysis [90], and another on the observational constraints and perturbation analysis [91].

In future work, a full analysis including DE perturbations and the complete CMB likelihood would provide significantly tighter constraints and allow for a more robust comparison with ΛCDM.

Second, the strongly multimodal posterior distributions found for the $\Lambda_{\rm osc}$CDM family motivate the use of a nested sampler such as PolyChord, which is better suited to exploring complex, multimodal parameter spaces. We have already begun testing this approach, while a more complete analysis is left for future work. Finally, confronting the cosmographic predictions with observational data could provide an independent test of these models.

Appendices

A

Singularities in Sign-Switching Models

Contents

A.1 $\Lambda_{\rm Osc}$CDM family **77**
A.2 $\Lambda_{\rm QI}$CDM family **78**

In this appendix, we examine the types of singularities that may arise in our models due to the sign change of the DE density.

As already shown in the literature [29], the abrupt $\Lambda_{\rm s}$CDM model exhibits a Type-II (sudden) singularity at the transition point, characterized by

$$\begin{aligned} t = t_\dagger, \qquad & a = a_\dagger < \infty, \\ \rho_{\rm tot}(a_\dagger) < \infty, \qquad & |p_{\rm tot}(a_\dagger)| \to \infty. \end{aligned} \tag{A.1}$$

We now extend this analysis to our smooth sign-switching models. Using the variable $x = \ln a$, the total energy density and pressure can be written as

$$\rho_{\rm tot}(x) = \rho_{\rm r,0} e^{-4x} + \rho_{\rm m,0} e^{-3x} + \rho_{\rm de,0}\, \varrho_{\rm de}(x), \tag{A.2}$$

$$p_{\rm tot}(x) = \frac{1}{3}\rho_{\rm r,0} e^{-4x} - \rho_{\rm de,0} \left[\varrho_{\rm de}(x) + \frac{1}{3}\frac{{\rm d}\varrho_{\rm de}(x)}{{\rm d}x} \right], \tag{A.3}$$

where $\varrho_{\rm de}(x)$ denotes the dimensionless DE density for each model.

A.1 $\Lambda_{\rm Osc}$CDM family

Here, we discuss only the $\Lambda^x_{\rm osc}$CDM model, since the other two models exhibit similar singularity behavior. For this model:

$$\varrho^x_{\rm osc}(x) = \frac{1}{N}{\rm Si}\left[A_3 K (x_\dagger - x)\right], \qquad N = {\rm Si}(A_3 K x_\dagger). \tag{A.4}$$

The total energy density and pressure are then:

$$\rho_{\rm tot}(x) = \rho_{\rm r,0}e^{-4x} + \rho_{\rm m,0}e^{-3x} + \frac{\rho_{\rm de,0}}{N}\mathrm{Si}\left[A_3K(x_\dagger - x)\right], \tag{A.5}$$

$$p_{\rm tot}(x) = \frac{1}{3}\rho_{\rm r,0}e^{-4x} - \frac{\rho_{\rm de,0}}{N}\mathrm{Si}\left[A_3K(x_\dagger - x)\right] + \frac{\rho_{\rm de,0}}{N}\frac{A_3K}{3}\mathrm{sinc}\left[A_3K(x_\dagger - x)\right]. \tag{A.6}$$

Evaluating at $x = x_\dagger$, we obtain:

$$\rho_{\rm tot}(x_\dagger) = \rho_{\rm r,0}e^{-4x_\dagger} + \rho_{\rm m,0}e^{-3x_\dagger}, \tag{A.7}$$

$$p_{\rm tot}(x_\dagger) = \frac{1}{3}\rho_{\rm r,0}e^{-4x_\dagger} + \frac{\rho_{\rm de,0}}{N}\frac{A_3K}{3}. \tag{A.8}$$

The total energy density at $x_\dagger$ does not depend on K and remains finite. The total pressure also remains finite for any finite value of K. However, in the limit $K \to \infty$:

$$\lim_{K\to\infty} |p_{\rm tot}(x_\dagger)| = \infty. \tag{A.9}$$

Thus, a Type-II (sudden) cosmological singularity occurs in the limit $K \to \infty$. For finite K, a w-singularity occurs instead at the transition point.

A.2 $\Lambda_{\rm OI}$CDM family

$\Lambda^{\rm SW}$ and $\Lambda^{\rm TT}$

Both models share the same structure, differing only in the constants that determine the oscillatory term. Their dimensionless DE density can be written as:

$$\varrho_{\rm sw/tt}(x) = \frac{1}{N}\left[\mathrm{Erf}(u) - \frac{\sin(n_1u)}{(1+u^2)^{n_2}}\right], \qquad u = K(x_\dagger - x), \tag{A.10}$$

where $(n_1, n_2) = (1, 2)$ corresponds to the $\Lambda^{\rm SW}$ model, while $(n_1, n_2) = (2, 4)$ corresponds to the $\Lambda^{\rm TT}$ model.

The total energy density and pressure are:

$$\rho_{\rm tot}(x) = \rho_{\rm r,0}e^{-4x} + \rho_{\rm m,0}e^{-3x} + \frac{\rho_{\rm de,0}}{N}\left[\mathrm{Erf}(u) - \frac{\sin(n_1u)}{(1+u^2)^{n_2}}\right], \tag{A.11}$$

$$p_{\rm tot}(x) = \frac{1}{3}\rho_{\rm r,0}e^{-4x} - \frac{\rho_{\rm de,0}}{N}\left[\mathrm{Erf}(u) - \frac{\sin(n_1u)}{(1+u^2)^{n_2}}\right] + \frac{K\rho_{\rm de,0}}{3N}\left[\frac{2e^{-u^2}}{\sqrt{\pi}} - \frac{n_1\cos(n_1u)}{(1+u^2)^{n_2}} + \frac{2n_2u\sin(n_1u)}{(1+u^2)^{n_2+1}}\right]. \tag{A.12}$$

At $u = 0$, corresponding to $x = x_\dagger$, these reduce to:

$$\rho_{\rm tot}(x_\dagger) = \rho_{\rm r,0}e^{-4x_\dagger} + \rho_{\rm m,0}e^{-3x_\dagger}, \tag{A.13}$$

$$p_{\rm tot}(x_\dagger) = \frac{1}{3}\rho_{\rm r,0}e^{-4x_\dagger} + \frac{K\rho_{\rm de,0}}{3N}\left[\frac{2}{\sqrt{\pi}} - n_1\right]. \tag{A.14}$$

For the $\Lambda^{\rm SW}$ model, the DE density crosses $\rho_{\rm de}(x) = 0$ only once. As in the $\Lambda^{a}_{\rm osc}$CDM model, the total energy density at $x_\dagger$ does not depend on K and remains finite. The total pressure also remains finite for finite K. In the limit $K \to \infty$, however, the model exhibits a sudden cosmological singularity.

The situation differs for the $\Lambda^{\rm TT}$ model, where $(n_1, n_2) = (2, 4)$. Although $x = x_{2\dagger}$ remains one of the zeros of the DE density, the oscillatory correction introduces two additional zero crossings. Solving numerically:

$$\mathrm{Erf}(u) - \frac{\sin(2u)}{(1+u^2)^4} = 0, \tag{A.15}$$

we obtain:

$$u_i = \{-u_0,\, 0,\, +u_0\}, \qquad u_0 \simeq 0.37. \tag{A.16}$$

Since $u = K(x_{2\dagger} - x)$, where we adopt the original notation $x_{2\dagger}$ to distinguish the central transition from the other crossing points, their positions are:

$$x_i = x_{2\dagger} - \frac{u_i}{K}. \tag{A.17}$$

At each of these points, the DE density vanishes while the pressure remains finite and non-zero for finite K. Consequently, the $\Lambda^{\rm TT}$ model exhibits three w-singularities.

The separation between the additional crossing points and the central one scales as $1/K$. Thus:

$$\lim_{K\to\infty} x_i = x_{2\dagger}. \tag{A.18}$$

As $K \to \infty$, the three w-singularities converge to the same point. In the abrupt limit, the model reduces to $\Lambda_{\rm s}$CDM and exhibits a sudden singularity.

$\Lambda^{\rm pt}$

Although similar to the $\Lambda^{\rm SW}$ model, the $\Lambda^{\rm pt}$ transition does not occur exactly at $x = x_\star$, except in the limit $K \to \infty$. The position of the actual transition depends on K. The dimensionless DE density can be written as:

$$\varrho_{\rm pt}(x) = \frac{1}{N}\left[\mathrm{Erf}(u) - \frac{A_4\cos(3u)}{(1+u^2)^6}\right], \qquad u = K(x_\star - x), \tag{A.19}$$

and the total energy density and pressure are given by:

$$\rho_{\rm tot}(x) = \rho_{\rm r,0}e^{-4x} + \rho_{\rm m,0}e^{-3x} + \frac{\rho_{\rm de,0}}{N}\left[\mathrm{Erf}(u) - \frac{A_4\cos(3u)}{(1+u^2)^6}\right], \tag{A.20}$$

$$p_{\rm tot}(x) = \frac{1}{3}\rho_{\rm r,0}e^{-4x} - \frac{\rho_{\rm de,0}}{N}\left[\mathrm{Erf}(u) - \frac{A_4\cos(3u)}{(1+u^2)^6}\right] + \frac{K\rho_{\rm de,0}}{3N}\left[\frac{2e^{-u^2}}{\sqrt{\pi}} + \frac{3A_4\sin(3u)}{(1+u^2)^6} + \frac{12A_4u\cos(3u)}{(1+u^2)^7}\right]. \tag{A.21}$$

As for the $\Lambda^{\rm TT}$ model, the corresponding transition point can be determined numerically, giving approximately :

$$x_\dagger = x_\star - \frac{0.241}{K}. \tag{A.22}$$

Evaluating at the transition point, we obtain:

$$\rho_{\rm tot}(x_\dagger) = \rho_{\rm r,0} e^{-4\left(x_\star - \frac{0.241}{K}\right)} + \rho_{\rm m,0} e^{-3\left(x_\star - \frac{0.241}{K}\right)}, \tag{A.23}$$

$$p_{\rm tot}(x_\dagger) = \frac{1}{3}\rho_{\rm r,0} e^{-4\left(x_\star - \frac{0.241}{K}\right)} + \frac{2.502\, K\, \rho_{\rm de,0}}{3N}. \tag{A.24}$$

Unlike the previous models, the total energy density at $x_\dagger$ depends on K for finite values; this dependence vanishes in the limit $K \to \infty$, recovering Eq. (A.7). The total pressure remains finite for finite K, yielding a w-singularity at the transition point. In the limit $K \to \infty$, the pressure diverges:

$$\lim_{K\to\infty} |p_{\rm tot}(x_\dagger)| = \infty, \tag{A.25}$$

and a Type-II (sudden) cosmological singularity occurs.

B

Supplementary Posterior Distributions

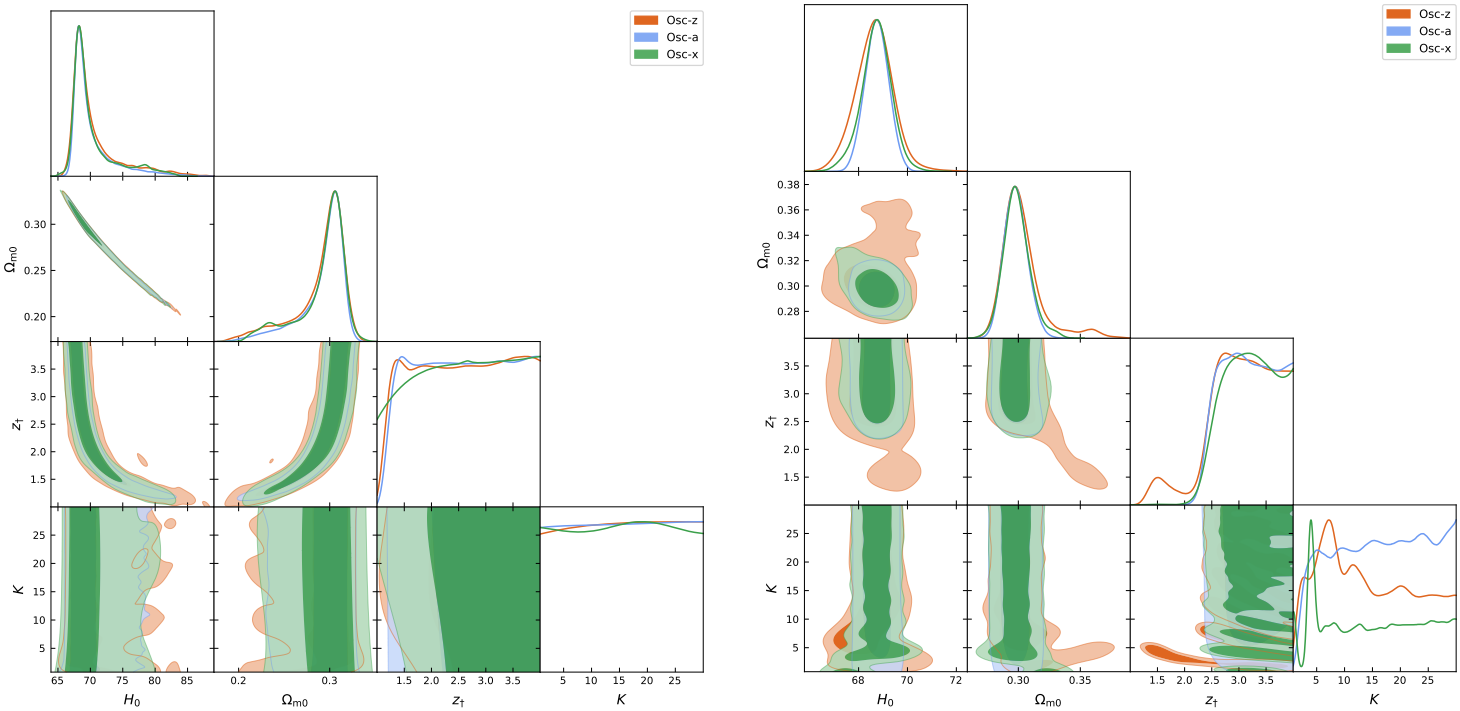
Contents

B.1 Λ_{osc}CDM 81
B.2 Λ_{OI}CDM 81

This appendix presents the remaining posterior distributions obtained for the Λ_{osc}CDM and Λ_{OI}CDM families, complementing the selected combinations discussed in Section 5.5. The contours denote the 68% and 95% confidence levels (C.L.).

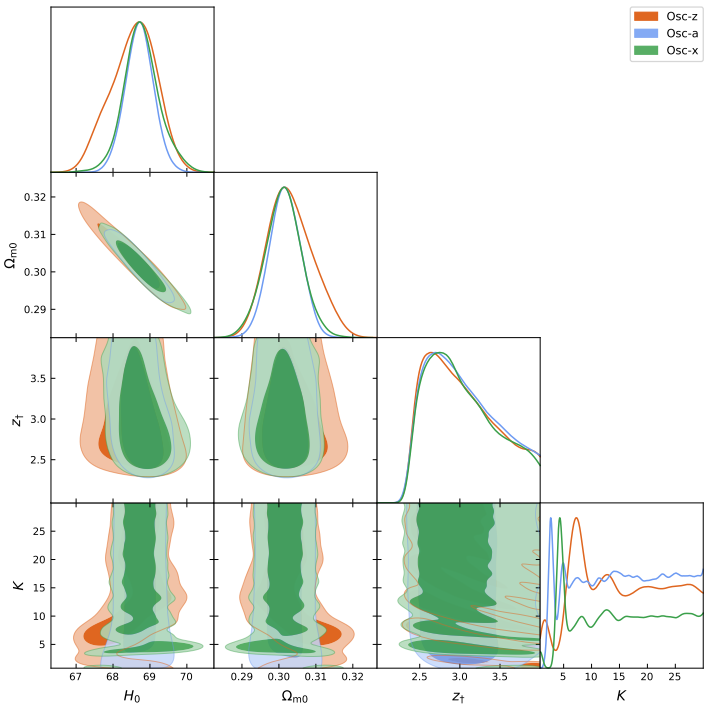
B.1 Λ_{osc}CDM

B.2 Λ_{OI}CDM

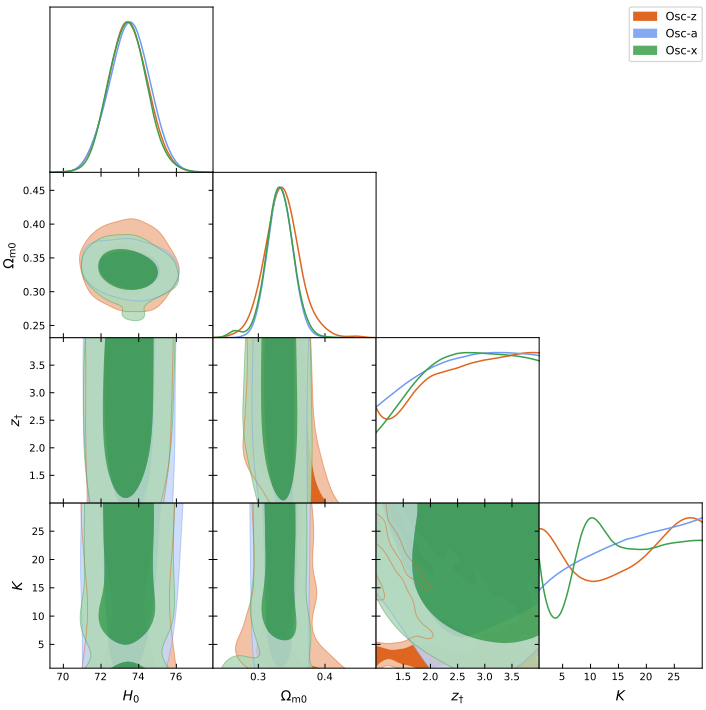


CMB

BAO

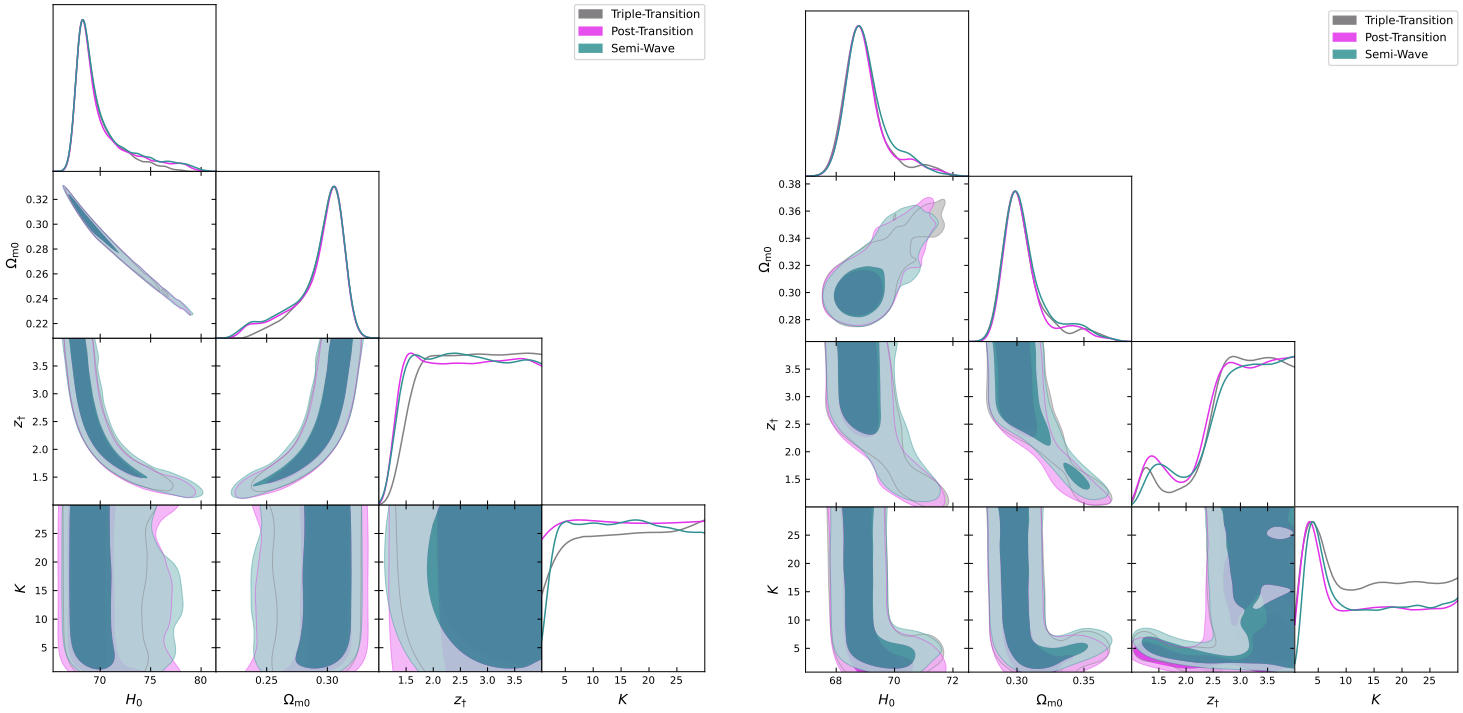


BAO+CMB

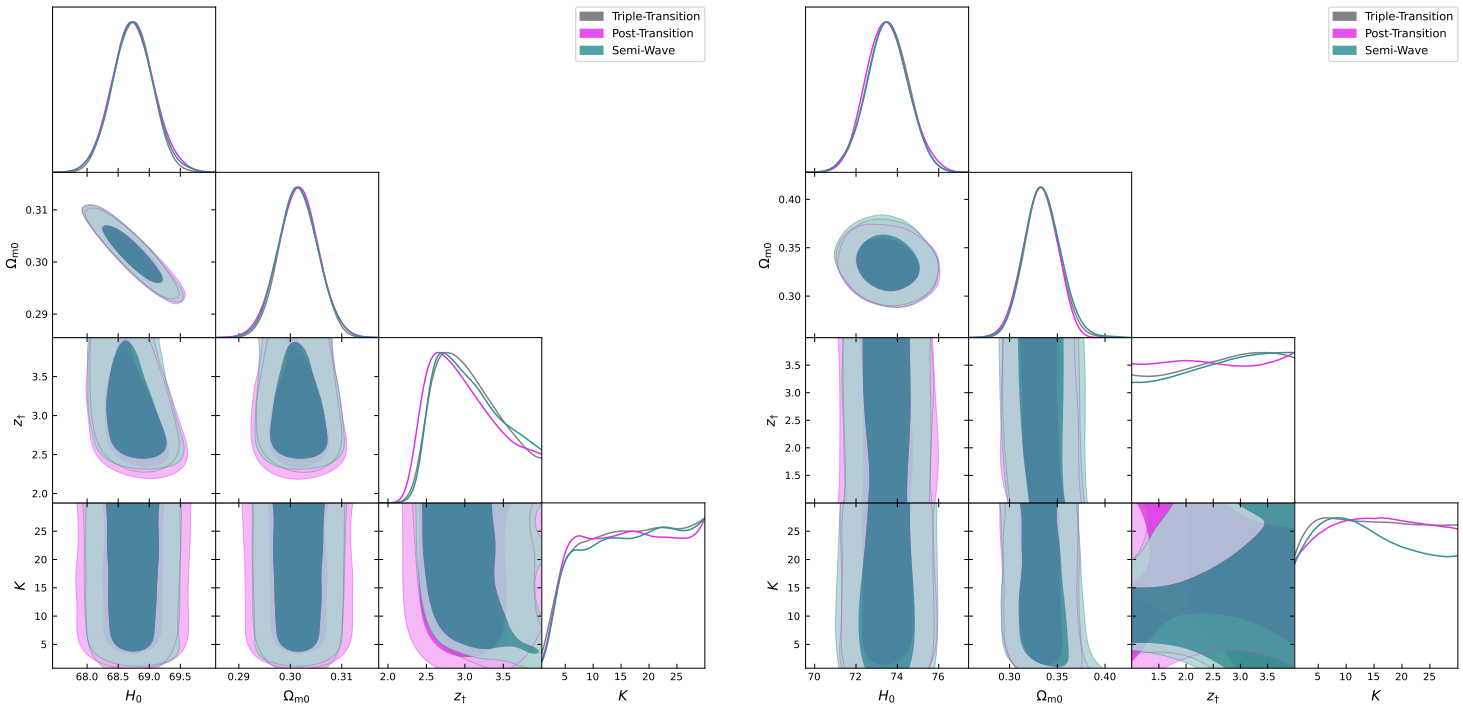


PPS

Figure B.1: Two-dimensional posterior distributions for the $\Lambda_{\rm osc}$CDM family, using the remaining dataset combinations not presented in Section 5.5. Contours denote the 68% and 95% confidence levels (C.L.).

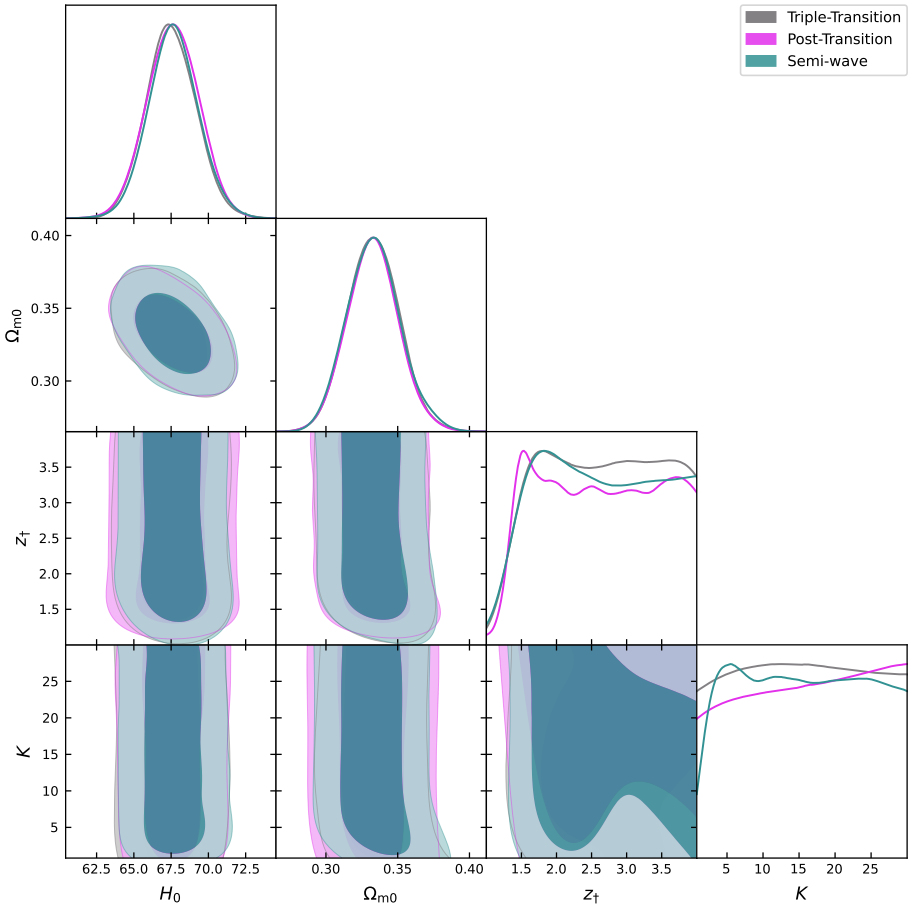


CMB BAO



BAO+CMB PPS

Figure B.2: Two-dimensional posterior distributions for the $\Lambda_{\rm OI}$CDM family for the CMB, BAO, BAO+CMB, and PPS dataset combinations. Contours denote the 68% and 95% confidence levels (C.L.).



PP+CC

Figure B.3: Two-dimensional posterior distributions for the $\Lambda_{\rm OI}$CDM family using the PP+CC dataset combination. Contours denote the 68% and 95% confidence levels (C.L.).

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